THE CHAUTAUQUAN

A Monthly Magazine Devoted to the Promotion of True Culture. Organ of the Chautauqua Literary and Scientific Circle.

Vol. V
April, 1885.
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Theodore L. Flood, D.D., Editor
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A MONTHLY MAGAZINE DEVOTED TO THE PROMOTION OF TRUE CULTURE. ORGAN OF THE CHAUTAUQUA LITERARY AND SCIENTIFIC CIRCLE.

Vol. V. APRIL, 1885. No. 7.

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REQUIRED READING FOR APRIL.

ARISTOTLE.

BY WILLIAM C. WILKINSON.

[The "College Greek Course in English" did not, for a reason alluded to in the following paper, include Aristotle among the authors represented. The readers of The Chautauquan will be glad to get some acquaintance with so great an ancient name through this supplementary chapter from Prof. Wilkinson's pen.]

Philosopher, though he by eminence is ranked, Aristotle was, too, something of an encyclopedist. He traversed almost the whole circle of the sciences, as that circle existed for the ancient world. But he was not simply first a learner, and then a teacher, of what others had found out before him. He was also an explorer and discoverer. Inventor also he was, if between discovery and invention we are to make a difference. He was a great methodizer and systematizer of knowledge. He bore to Plato the personal relation of pupil.

The history of Aristotle's intellectual influence is remarkable. That influence has suffered several phases of wax and wane, several alternate occultations and renewals of brightness. During a certain period of time, covering several hundred years, he was, perhaps beyond the fortune of any other man that ever lived, the lord of human thought. We mean the time of the schoolmen[1] so called. From near the close of the thirteenth century, until the era of the Reformation, Aristotle reigned supreme in the schools of Christian theology, which is the same thing as to say that he was acknowledged universal monarch of the European mind. The business of the schoolmen may be said to have been to state the dogmas of the church in the forms of the Aristotelian logic, and then to reconcile those dogmas so stated, with the teachings of the Aristotelian philosophy.

Curiously enough, the introduction of Aristotle to the doctors of the church was through the Mohammedan Arabs. These men had, during a term of centuries, been the continuers of the intellectual life of the race. While through the long night of those ages of darkness the Christian mind slept, the Arabian mind, waking, gave itself largely to the study of Aristotle. The Greek philosopher was posthumously naturalized a barbarian; for Aristotle's writings were now translated from their original tongue into Arabic. In this Arabic version, the celebrated Ibn Roshd (chiefly famous under his latinized name A-verˈroës) knew Aristotle and commented on him. The Arabic commentaries of Averroes were translated into Latin, and the thought of Aristotle thus became once more accessible to European students. Averroes (A. D. 1149-1198) himself was of the Moors of Spain.

For centuries previous to the time when the son and successor of good Haroun al Raschid,[2] known at least by name to the readers and lovers of Tennyson, collected at Bagdad all the scattered volumes of Greek letters that his agents could find in Armenia, Syria and Egypt—for centuries, we say, previous to this, Aristotle suffered an almost complete arrest and suspension of intellectual influence. That man would have been a bold prophet who should then have predicted what a resurrection to power awaited the slumbering philosopher.

Still earlier, however, than this, that is, during the interval between the third Christian century and the sixth, Aristotle enjoyed a great vogue. He was studied and commented on as if all human wisdom was summed up in him. The spirit of independent and original philosophy had perished, and whatever philosophic aptitude survived was well content to exhaust itself in expounding Aristotle. Aristotle's works became a kind of common Bible to the universal mind of the Roman empire. This was the period of the Greek scholiasts, so-called—in more ordinary language, commentators.

Taking the reverse or regressive direction of history, we have thus run back to a point of time some six or seven centuries subsequent to the personal life and activity of Aristotle. During the latter half of these centuries, Aristotle's fame was gradually growing, from total obscurity to its great culmination in splendor under the scholiasts.

Before that growth began, the productions of Aristotle had experienced a fortune that is one of the romances of literary history. The great pupil of Plato had himself no great pupil to continue after his death the illustrious succession of Grecian philosophy. His writings, unduplicated manuscripts they seem to have been, fell into the hands of a disciple, who, dying, bequeathed them to a disciple of his own, residing in the Troad. To the Troad accordingly they went. Here, with a view to save them from the grasp of a ruthless royal collector of valuable parchments, the family having these works in possession hid them in an underground vault, in which they lay moldering and forgotten one hundred and fifty years! It was thus in all nearly two hundred years that Aristotle's thoughts were lost to the world. When at last it was deemed safe, the precious documents were brought out and sold to a rich and cultivated Athenian. This gentleman, let us name him for honor, it was A-pelˈli-con, had unawares purchased his prize for a rapacious Roman collector. Sylla seized it, on his capture of Athens, and sent it to Rome. At Rome it had the good fortune to be appreciated. One An-dro-niˈcus edited the collection, and gave to the world that, probably, which is now the accepted text of Aristotle.

But, romantic as has been the succession of vicissitudes befalling his productions and his fame, Aristotle is, in his extant writings, anything but a romantic author. A less adorned, a less succulent style, than the style in which the Stagirite (he was of Stagˈi-rus, in Macedonia) wrote, it would be difficult to find. Still it is a style invested at least with the charm of evident severe intentness, in the writer, on his chosen aim. Cicero, it is true, speaks of Aristotle's style in language of praise that would well befit a characterization of Plato. But Cicero must have had in view works of the philosopher other than those which we possess, works written perhaps in the author's more florid youth. With this conjecture agrees the fact that a list of Aristotle's works, made by the authorities of the renowned Alexandrian library, contains numerous titles not appearing in the writings that remain to us attributed to Aristotle.

Aristotle was not, as Plato was, properly a man of letters. Or, if he did bear this character, the evidence of it has perished. What we possess of his intellectual productions exhibits the author in the perfectly dry and colorless light of a man of science. Even in those treatises of his in which he comes nearest to the confines of pure and proper literature, his interest is rather scientific than literary. He discusses in two separate books the art of rhetoric and the art of poetry; but he conducts his discussion without enthusiasm, without imagination, in the severely strict spirit of the analyst and philosopher. The text of the two treatises now referred to survives in a state of great imperfection. Indeed, the same is the case generally with Aristotle’s works. Critics have even surmised that, in some instances, notes of lectures, taken by pupils while the master according to his wont was walking about and extemporizing discourse, have done duty in place of authentic autograph originals supplied by the hand of Aristotle himself. The title “Peripatetic” (walk-about), given to the Aristotelian philosophy, was suggested by the great teacher’s habit, thus alluded to, of doing his work as teacher under the stimulus of exercise on his feet in the open air.

The non-literary character of Aristotle’s works has to a great extent excluded him from the course of Greek reading adopted by colleges—this, and moreover the fact that he occupies a position at the extreme hither limit, if not quite outside the extreme limit, of the Greek classic age. Still he is now and then read in college; and at any rate he is too redoubtable a name among those names which in their motions were

“Full-welling fountain-heads of change,”

not to be an interesting object of knowledge to the readers of THE CHAUTAUQUAN.

The productions of Aristotle are numerous. The Alexandrian bibliography of him gives one hundred and forty-six titles of his works. Of the books thus catalogued not a vestige remains, except in an occasional quotation from them at the hands of some other ancient writer. The works commonly printed as Aristotle’s form an entirely different list. We give a few of the leading titles or subjects: “Organon,” a collective name for various writings that made up a system of logic; “Rhetoric,” “Po-etˈics” (art of poetry), “Ethics,” “Politics,” “Natural Philosophy,” “Biology,” “Metaphysics.” [This last word, which has acquired in modern use a very distinct meaning of its own, was originally a mere meaningless designation of certain investigations or discussions entered into by Aristotle *after* his physical researches. The preposition *meta* (after), and *physica* (physics), give the etymology of the term.] The comprehensive or, as we before said, encyclopædic range of Aristotle’s intellectual activity will to the observant reader be sufficiently indicated by this list of titles.

For his work in natural history, Aristotle was powerfully supported by one of the most resplendent military geniuses that the world has ever seen, Alexander the Great. To this prince and warrior, when he was a lad, the philosopher had discharged the office of private teacher. It would appear that either Aristotle was courtier enough, or young Alexander was man enough, to make this relation a pleasant one to the boy. For, in later years, the conqueror of the world presented to his former teacher a round million of dollars to make himself comfortable withal. But who can tell which it was, gratitude for benefit received, or remorse for trouble occasioned, that prompted the *ex post facto*[3] royal munificence? Perhaps it was both—a tardy gratitude quickened by a generous remorse.

The chief glory of Aristotle is to have at once invented and finished the science of logic. For this is an achievement which may justly be credited to the philosopher of Stagirus. It would generally be conceded that, since Aristotle's day, little or nothing substantial has been added to the results of his labor in the field of pure logic. The name Orˈga-non (instrument) is not Aristotle's word, but that of some ancient editor of his works. It is a noteworthy name, as having dictated to Bacon the title to his epoch-making work, the *Novum Organum* (the new method or instrument).

It would not be easy to give an exhaustive account of Aristotle's productions, and make the account attractive reading. We shall not undertake so impracticable a task. Let our readers accept our word for it that Aristotle, though a justly renowned name in the history of thought, is not fitted to be a popular author.

From his "History of Animals" we present a specimen extract that will perhaps with some readers go far toward confuting what we just now said. There are, we confess, some things in this treatise that read almost as if they might belong to that truly fascinating book, "Goldsmith's Animated Nature:"

> "The cuckoo is said by some persons to be a changed hawk, because the hawk which it resembles disappears when the cuckoo comes, and indeed very few hawks of any sort can be seen during the period in which the cuckoo is singing, except for a few days. The cuckoo is seen for a short time in the summer,

> and disappears in winter. But the hawk has crooked talons, which the cuckoo has not, nor does it resemble the hawk in the form of its head, but in both these respects is more like the pigeon than the hawk, which it resembles in nothing but its color; the markings, however, upon the hawk are like lines, while the cuckoo is spotted.
>
> "Its size and manner of flight is like that of the smallest kind of hawk, which generally disappears during the season in which the cuckoo is seen. But they have both been seen at the same time, and the cuckoo was being devoured by the hawk, though this is never done by birds of the same kind. They say that no one has ever seen the young of the cuckoo. It does, however, lay eggs, but it makes no nest, but sometimes it lays its eggs in the nests of small birds, and devours their eggs, especially in the nests of the pigeon (when it has eaten their eggs). Sometimes it lays two, but usually only one egg; it lays also in the nest of the hypolais,[4] which hatches and brings it up. At this season it is particularly fat and sweet-fleshed; the flesh also of young hawks is very sweet and fat. There is also a kind of them which builds a nest in precipitous cliffs."

This morsel, our readers must consider, is not a very characteristic specimen of the feast that, take all his works together, Aristotle spreads for his students. But it is as toothsome as any we could offer. If it makes our readers wish for more, that is as friendly a feeling as we could possibly hope to inspire in them toward Aristotle. We shall now let them, in that mood, bid the great philosopher farewell.

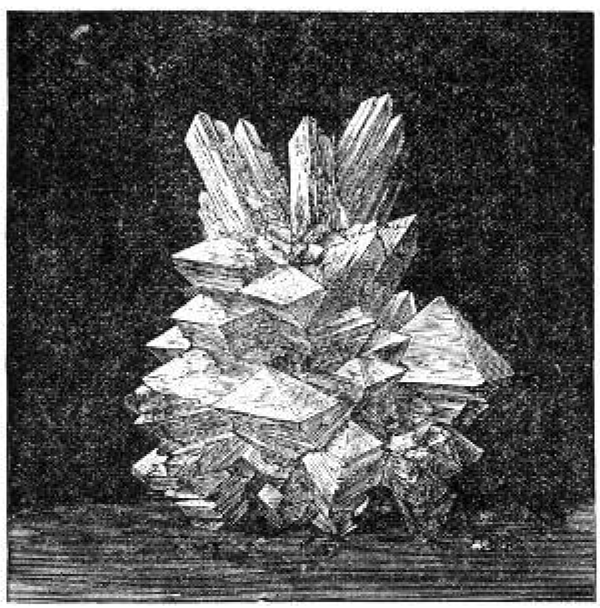
HOME STUDIES IN CHEMISTRY AND PHYSICS.

BY PROF. J. T. EDWARDS, D.D.
Director of the Chautauqua School of Experimental Science.

CHEMISTRY OF EARTH.

John B. Gough declares that a few kind words spoken to him, in a crisis of his life, saved him from ruin. He afterward carefully educated the orphan daughters of the gentleman who uttered those words.

“Why,” you say, “it was a little thing.” “Yes, *little* for him, but a big thing for me.”



CRYSTALS OF ALUM.

The importance of *many* things depends upon the point of observation. To a hypothetical astronomer on a distant star, this world would be too minute for observation. In that shining pathway of the heavens, called the “milky way,” there have been discovered eighteen millions of stars, each hundreds of times larger than our earth; yet *our* atom in immensity is, just now, of marvelous interest to us. Indeed, it must be of interest to the highest intelligences, for such are the harmonies of God’s universe that the minutest planet is in many of its forces and laws representative of the whole. So that our world is, in a sense, both a microcosm and a cosmo.

Let us briefly consider some characteristics of the earth, from the standpoint of the chemist.

All substances have been divided into two great classes, the inorganic and organic. The latter contains two subdivisions—the vegetable and animal world. Nature thus comprises three great sub-kingdoms, the mineral, vegetable and animal.

A mineral is an inorganic body (that is, one in which no parts are formed for special purposes), possessed of a definite chemical composition, and usually of a regular geometric form. It may seem at first glance that the last part of this definition is not correct, but there is reason to believe that all mineral substances may, under favorable circumstances, assume crystalline forms. Water and air are minerals. Other liquids and gases are included in the term, but as we have had already something to say of these latter substances, we shall, for the purposes of this article, use the word earth in the popular sense; namely, inorganic matter, which at ordinary temperature is solid. All materials are classified into

ELEMENTS AND COMPOUNDS.

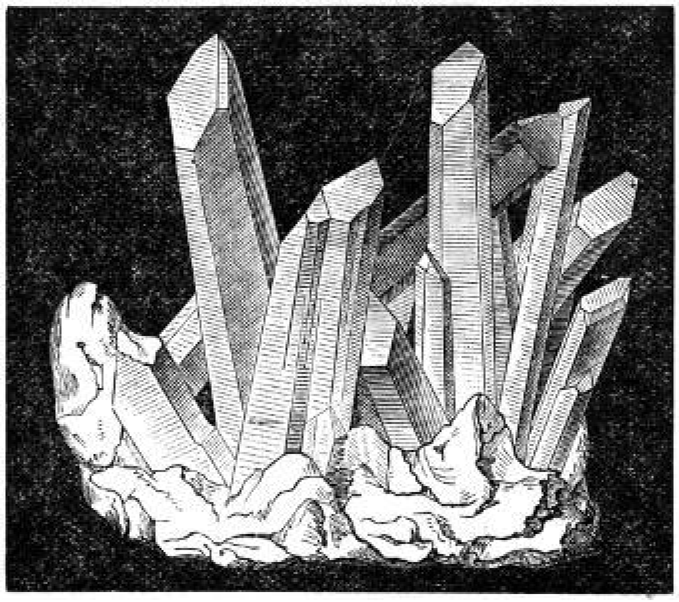
By an element is meant a substance which has never been resolved into parts, and conversely, one that can not be produced by the union of two or more substances. There is some difference of opinion as to their number. It is usually given as sixty-four. There are a great many compounds. Nature seems to delight in surprising us by the simplicity of the means employed in producing marvelous results. As the mind of Milton combined the twenty-six letters of our alphabet to form “Paradise Lost,” so the Infinite arranged and re-arranged the elements to form the sublime poem of creation. Fifty-one of the elements are metals, and thirteen metalloids; gold is a familiar example of the former, and sulphur of the latter. A few, like hydrogen and oxygen, are gases; two are liquids; quicksilver and bromine: the greater number exist as solids. But few of them are found native, *i. e.*, chemically uncombined with other substances. In the fierce heat of former ages they were mixed as in a mighty crucible, and few escaped the power of affinities thus engendered. Gold and copper are sometimes found pure, but even they, more frequently than otherwise, exist fused with other substances.

Compounds are of three classes—acids, bases and salts. Sand is a specimen of the first, lime of the second, and clay of the third. *Fixedness* is a characteristic of mineral compounds, yet they are by no means incapable of change; certain influences come in to promote it, of which the following are the most important—heat, solution, friction and percussion.

Two gases, oxygen and hydrogen, may remain side by side for years uncombined, but a single spark will cause them to rush together with terrific energy.

If the contents of the blue and white papers in a Seidlitz powder are mixed, no chemical action follows, but if dissolved separately in glasses of water, and then poured together, a violent effervescence takes place. If a small amount of potassium chlorate and a *little* piece of sulphur be put together in a mortar, and then pressed by the pestle, sharp detonations follow. Dynamite, which is nitro-glycerine mixed with infusorial earth, sugar or sawdust, is quite harmless when free from acid, unless struck. The above instances illustrate the various influences that stimulate chemical combination. Almost all the crust of the earth is formed of three substances—quartz, lime, and alumina. Wherever we stand on the round globe, it is safe to say that one or all of these are beneath our feet.

QUARTZ.

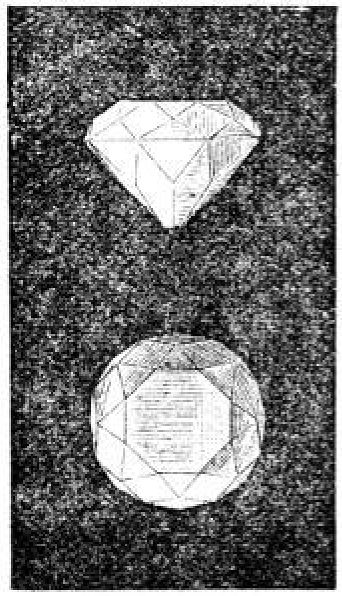


QUARTZ CRYSTALS.

This mineral comprises about one half the earth’s crust. Its symbol is SiO_2, being a compound of silicon and oxygen, in the proportions indicated. It is very hard, easily scratching glass, of which it forms an important constituent, is acted upon by only one acid—hydrofluoric; this attacks it eagerly, as may be shown by the following interesting experiment: Take a little lead saucer, or in the absence of this, spread lead foil carefully over the inside of an ordinary saucer, and in this place some powdered fluor spar. This mineral is quite abundant in nature, and is always to be obtained, in the form of a powder, from dealers in chemicals. Have a pane of glass covered by a thin film of wax. Now trace upon this surface with a sharp point, anything you may desire, verse or picture. Pour into the saucer containing the fluor spar, sufficient sulphuric acid to make a paste. Place over this the plate of glass, with the waxed side down, and let it remain for twenty-four hours. Remove the wax by heating, and on the glass you will find a perfect etching, the HF having removed the silica.

The same effect may be produced in a few moments by applying to the bottom of the saucer a moderate heat. Care should be taken not to inhale the fumes, as they are highly corrosive.

Quartz can be melted at a high temperature, and may be dissolved in certain hot solutions. It is still a question in dispute, whether the numerous quartz veins found in rocks were introduced there in melted form or in solution. Probably, sometimes in one state and sometimes in the other. Any visitor to a glass manufactory can see how easily glass in a melted state is manipulated; and travelers often bring from the geysers[1] fine specimens of silica called geyserite, derived from the material held in solution in the hot water, and deposited on the edge of the "basin."



SIDE AND TOP VIEW OF THE REGENT OR PITT DIAMOND (REDUCED IN SIZE)—CUT IN THE FORM OF THE "BRILLIANT."

Quartz may be classified under two varieties—the common and the rare. Sand, pebbles, many conglomerates, all sandstone rocks come under the former head. The old red sandstone described by Hugh Miller,[2] in which fossil fish are so abundant, and the new red sandstone of the Connecticut valley, famous for its bird or reptile tracks, brought to light through the labors of Dr. Hitchcock,[3] were formed of sand cemented together under pressure by the peroxide of iron. There are many beautiful varieties of the rarer forms of quartz. Not a few of these were known to the ancients, as may be seen by reading the twenty-first chapter of Revelations, where a number are mentioned in the description of the heavenly city. "The wall of it was of jasper, and the foundations of the wall of the city were garnished with all manner of precious stones. The first foundation was jasper; the second, sapphire; the third, a chalcedony; the fourth, an emerald; the fifth, a sardonyx; the sixth, sardius; the seventh, chrysolite; the eighth, beryl; the ninth, a topaz; the tenth, a chrysoprasus; the eleventh, a jacinth; the twelfth, an amethyst."

All of these excepting the sapphire, which is crystallized alumina, are either pure or mixed varieties of quartz, colored with some metallic oxide. One of the most beautiful forms of these precious stones is the agate, especially that kind called the onyx, which consists of a succession of opaque and transparent layers. When carved into gems, this is called the cameo. A wonderful carved cameo was in the Tiffany exhibit at the Centennial Exposition, valued at four thousand dollars. The several layers were so cut as to represent a man looking through the bars of his prison.

LIME.

Another very plentiful substance in the earth is lime. It is chiefly found in the form of three salts, the carbonate, sulphate and phosphate ($CaCO_3$) ($CaSO_4$) ($Ca_3(PO_4)_2$), respectively. The first is familiarly known as limestone. When crystallized, it appears as marble. The shades of marble are due to the tinting of metallic oxides, and sometimes to the presence of fossils. The most beautiful marble is obtained from Carrara, Italy, which has long been famous for furnishing the material used for statues. It is pure white. Pure black marble is found in some ancient Roman sculptures. Sienna marble is yellow. Italy furnishes one kind that is red. Verd-antique is a mixture of green serpentine and white limestone, while our beautiful Tennessee marble, used so profusely in the new Capitol at Washington, is a blended red and white.

Common limestone is almost entirely the product of minute animals[4] which lived in early geologic times. Ages before the Romans drove piles into the Thames, or the first hut was erected on the banks of the Seine, these little creatures laid the foundations which underlie London and Paris. They built the rocky barriers which gave to England the name Albion, derived from the white cliffs along her shore. It is a suggestive crumb of comfort for little folk, that the great tasks in the building of our earth have been performed by the smallest creatures.

The wide distribution of limestone is shown from the fact that it is found to be an ingredient in almost all waters. It is readily dissolved, as is seen in the numerous caves which are found in limestone regions.

When limestone is heated, the carbonic anhydride[5] is expelled, leaving quicklime. All are familiar with the manifold uses of this material. United with sand, it forms a silicate of lime, called mortar, which becomes harder with age. In the old stone mill[6] at Newport, R. I., which is of unknown antiquity, the mortar in some places actually protrudes beyond the stones, showing it to be more durable than the rock itself. The catacombs of Rome were excavated in a very soft kind of limestone, called calcareous tufa.

Sulphate of lime, also known as gypsum and plaster of Paris, is widely distributed. One beautiful variety is called satin spar, and another alabaster.

Great quantities of sulphate of lime are quarried for use in the arts and for agricultural purposes. Dr. Franklin was one of the first to discover its value in connection with crops, and is said to have sown it with grain on a side hill, so that when the wheat sprang up, observers were surprised to see written in gigantic green letters, “Effects of Gypsum!” I suspect he got the hint from Dr. Beattie, who sowed seeds so that their flowers formed the name of his son, to prove to the boy the existence of a God, from evidences of design in nature.

ALUMINA—Al_2O_3.

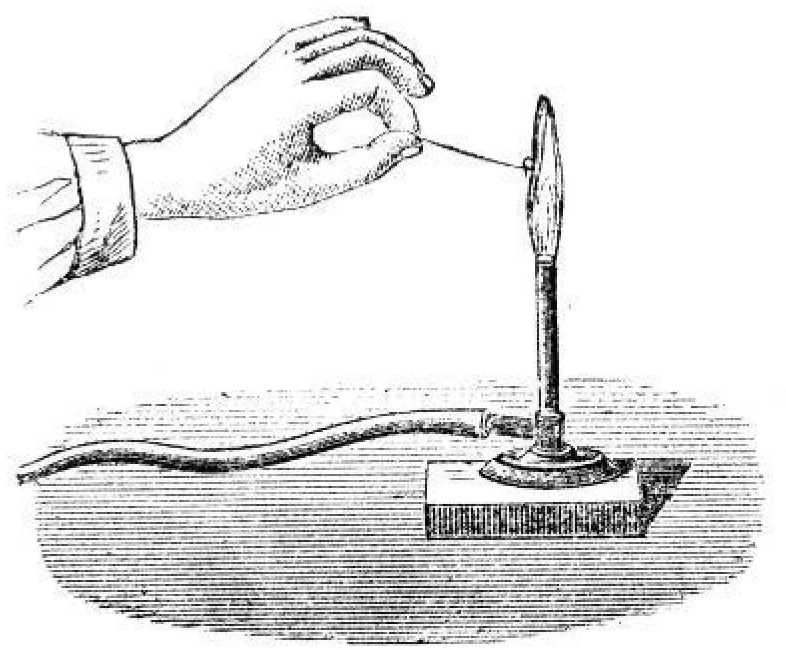
This material is found both alone and in combination with silica. It forms an important ingredient in alum. Crystallized, it furnishes some of our most rare and beautiful gems, the color of which depends upon the metal combined with them.

The ruby is red, the emerald green, the topaz yellow, the sapphire blue.

Slate rocks consist largely of this material, and clay is a compound of alumina with siliceous anhydride. Among the first earthy substances utilized by man was clay. We find remains of pottery even as far back as the stone age[7]. The ingenuity of man seems to have been displayed constantly and successfully in the ceramic[8] art, the art of making pottery. Note the accounts given by Prescott, in his “Conquest of Peru and Mexico,” and the Cesnola collection of Cypriote remains[9] exhibited in the Metropolitan Museum in New York City.

History is repeating itself by renewing the ancient enthusiasm for decoration of china and earthen ware. Bricks made from clay are found to rival granite in durability, and surpass it in resistance to heat, as was proven in the great fires of Boston and Chicago. It will be observed from the symbol of alumina that it is largely composed of the metal aluminum. If this could be readily liberated from the oxygen with which it is combined, the world would be immensely enriched.

Every clay bank or clayey soil contains it in great quantities. Next to oxygen and silicon, it is the most abundant element in the earth. Note its valuable properties. It is but two and one-half times heavier than water, as bright and non-oxidizable as silver, malleable, ductile, tenacious, and can be welded and cast. Who will lay the world under obligation by doing with alumina what has been done with iron ores, cheaply liberate the oxygen?



TESTING FOR IRON WITH A BORAX BEAD.—THE COMPOUNDS OF IRON WITH BORAX GIVE A BOTTLE GREEN COLOR.

In this brief enumeration of earth materials, we have intentionally omitted the forms of carbon. They constitute no insignificant portion of the earth’s crust, but belong to the class of organic substances. We introduce, however, an illustration showing one of the shapes in which is cut the diamond—that most costly of all forms of matter,—crystallized carbon.

THE COMMON METALS.

First in importance is iron. The fact already mentioned that its oxide is the most common coloring matter in the mineral world will also indicate its wide dissemination.

Trap rock, gneiss, even granite, sands, clays and other rocks all borrow tints from this source. Iron is never found native except in meteors. It exists most abundantly in the form of three ores, the composition of which is as follows:

Black or magnetic oxide (Fe_3O_4), red oxide (Fe_2O_3), hydrated sesquioxide ($Fe_2O_33H_2O$). From all of these the oxygen is removed in a blast furnace, by the use of some form of carbon. As thus prepared, it is called cast-iron. Two other varieties are employed in the arts, wrought iron and steel. The last differs from the first in having less carbon, and from the second in having more. The general properties of this material are too well known to require description here. A single property of this substance alone has marvelously affected the commerce of the world; that is, the power first discovered in magnetic iron ore, of attracting iron, and pointing northward. The first compass, it is said, consisted of a piece of this metal placed on a cork floating on water.

Copper seems to have been one of the few metals known to barbarous peoples. It is found pure, and in combination. Specimens obtained from the Lake Superior region, in mines worked by the mound builders,[10] have led some to believe that they possessed the art of hardening copper. Malachite is a carbonate of copper, of a beautiful mottled green color, and is made into elegant ornaments. Some magnificent specimens were in the Russian exhibit at the Philadelphia Exposition. It is found in great perfection in the Ural mountains.

Tin is obtained from its binoxide (SnO_2). It was known to the ancients. Some historians claim that the Phœnicians procured it long before the time of Christ, from the mines of Cornwall, England. Until recently our country has seemed to be destitute of this valuable metal. Reports now indicate that Dakota is destined to supply this deficiency. It is a handsome metal, but little affected by oxygen, and capable of being rolled into thin sheets.

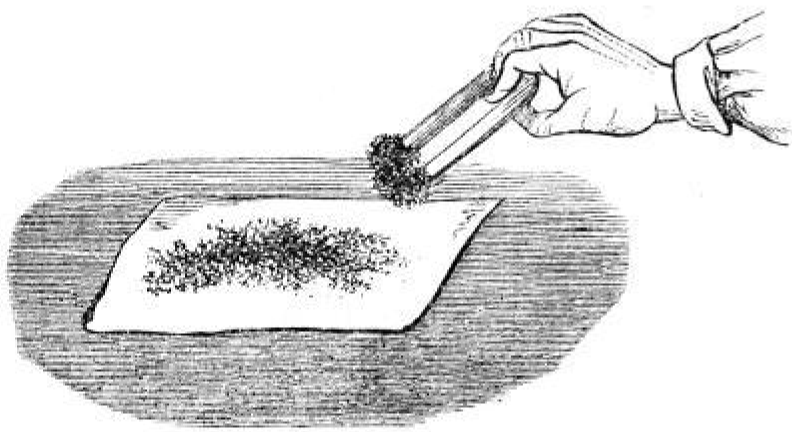
Zinc is found in two different ores: red oxide (ZnO) and zinc blende (ZnS), from which it can be separated by smelting, in much the same manner as we obtain iron.

Lead constitutes the fifth of the common metals which are preëminently useful. It is found in the sulphide of lead (PbS), the sulphide being expelled by roasting the ore. It forms numerous compounds, some of which are of great value. For example, lead carbonate ($PbCO_3$), the white lead which furnishes the most valuable ingredient of all paints.

NOBLE METALS.

These are so called because they retain their brilliancy and are not easily affected by other substances. Three of them are specially important: gold, silver and platinum. Gold is mentioned in the second chapter of Genesis: “and the gold of that land is good.” Although constituting an inconsiderable part of the earth, it is much more widely distributed than many suppose, but often exists in such small quantities that its production is not profitable.

Australia and California are the gold lands. It is found principally in three situations: in sands which have been washed from the mountains, in little pockets in the rocks, and in veins of quartz. From the first it is separated by simply washing away the lighter materials, from the last situation it is procured by quarrying the rock, crushing it with stamping machines, then washing with water to remove the pulverized quartz, and gathering up the powdered gold with quicksilver. The mercury is removed by vaporizing. Gold is nineteen times heavier than water, extremely ductile, and the most malleable of all substances. Silver is abundant in the mountains of the west. It is usually found in the form of black sulphide (Ag_2S) or horn-silver ($AgCl$). When unpolished it is perfectly white, and is called dead or frosted silver. All are familiar with the properties of this attractive metal. Just now its producers in Colorado seem to fear its displacement from its important position in the coinage of the country. In nitrate of silver ($AgNO_3$) we have a material that perpetuates the faces of our friends, many a goodly landscape, and marvelous picture.



MAGNET GATHERING IRON FILINGS.—A MAGNET WILL ALSO ATTRACT NICKEL FILINGS.

Platinum stands at the extreme limit of the elementary world in point of weight, being twenty-one and fifty-three hundredths times heavier than water. Russia has almost a monopoly of the production of this metal. It is about the value of gold, and to the chemist is of immense importance, on account of its high point of fusibility, which is over 4,000°. It is so ductile that it can be drawn out into wire so fine as to be invisible to the naked eye. This microscopic wire is used for centering the field of view in the finest telescopes.

EARTH'S CRUST AND CENTER.

Our earth is called "terra firma;" it is regarded as the very embodiment of stability, but every waving outline, every hill and mountain peak, not less than the rumbling of the earthquake, and the bursting forth of volcanic fires, indicate that it has been, and may again be, the scene of mighty disturbances. Indeed, upon reflection, one wonders that we can live on it at all. The temperature of the earth increases one degree for every fifty feet as we approach the center. At this rate, at the depth of fifty miles the heat would be sufficient, according to Humboldt,[11] to melt the hardest rocks. Fifty miles is one one-hundred and sixtieth of the earth's diameter. It thus appears that if we should have a globe three feet in diameter full of molten liquid, surrounded by a covering of infusible material *one eighth of an inch* in thickness, that film of solid matter would represent the earth's crust. Think of it!

A “LEAD TREE.”

Ex.—Place in a glass a dilute solution of acetate of lead; suspend in it a strip of zinc. Some of the lead will be precipitated in crystals upon the zinc. This is caused by the zinc taking a portion of the acetic acid, and thus forming a new compound called zinc acetate, thereby liberating some of the lead.

Besides, that awful, fiery sea within is subject to tides, currents and convulsions that constantly threaten to disrupt and destroy this crust. It is supposed that masses of water percolate through cracks and fissures until they reach the internal fires and are suddenly converted into steam at an enormously high temperature, which gives it such tremendous expansive force as to shake the globe itself. This action, combined with the violent explosion of gases, creates the sublime and dreadful phenomena of

EARTHQUAKES AND VOLCANOES.

The destruction of Lisbon and many other cities is matter of history. But last year a charming city in the Mediterranean was destroyed in a few seconds, and the stricken inhabitants of Spain are still trembling with horror at the recent shocks that have desolated their fair country.

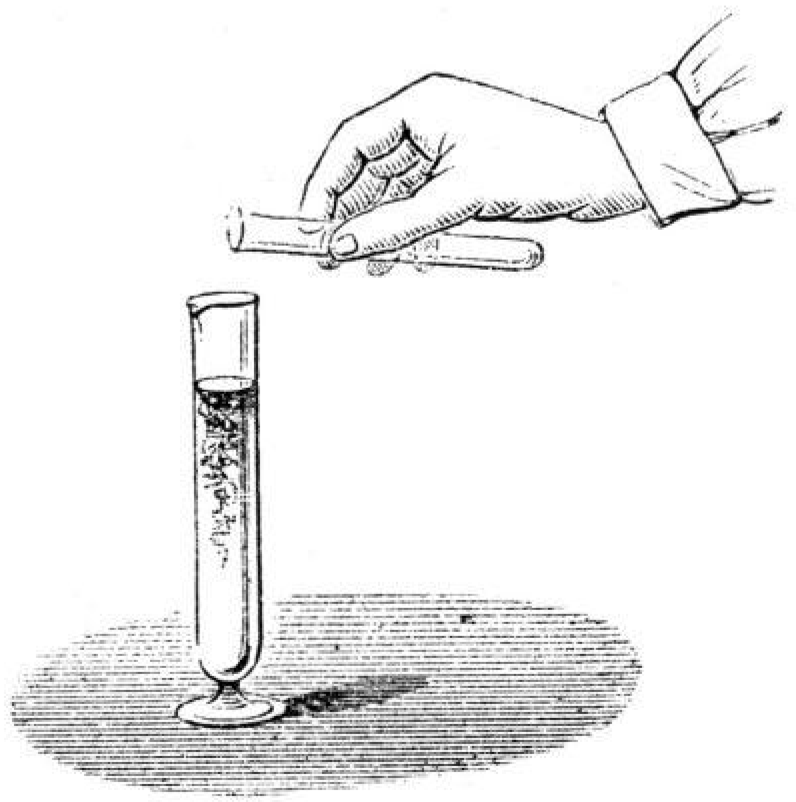
Man looks in vain elsewhere for such exhibitions of the power of chemical forces as are here displayed.

Lord Lytton[12] gives a most impressive description of an eruption of Mount Vesuvius, in “The Last Days of Pompeii:”

“In proportion as the blackness gathered, did the lightnings around Vesuvius increase in their vivid and scorching glare. Nor was their horrible beauty confined to the usual hues of fire; no rainbow ever rivaled their varying and prodigal dyes. Now brightly blue as the most azure depth of a southern sky, now of a livid and snake-like green, darting restlessly to and fro as the folds of an enormous serpent; now of a lurid and intolerable crimson, gushing forth through the columns of smoke, far and wide, and lighting up the whole city from arch to arch; then suddenly dying into sickly paleness, like the ghosts of their own life!

“In the pauses of the showers you heard the rumblings of the earth beneath, and the groaning waves of the tortured sea; or, lower still, and audible but to the watch of intensest fear, the grinding, hissing murmur of the escaping gases through the chasms of the distant mountain.

“Sometimes the cloud seemed to break from its solid mass, and, by the lightning to assume quaint and vast mimicries of human or of monster shapes, striding across the gloom, hurtling one upon the other, and vanishing swiftly into the turbulent abyss of shade; so that, to the eyes and fancies of the affrighted wanderers, the unsubstantial vapors were as the bodily forms of gigantic foes—the agents of terror and death.”



TESTING FOR GOLD WITH PURPLE OF CASSIUS.

Ex.—When gold is placed in a solution of Stannon's chloride and ferric chloride, a precipitate called Purple Cassius appears. Sometimes the color varies to brown or blue.

It is claimed that there are about three hundred extinct volcanoes, and many facts indicate that the convulsions in the earth's crust are much less frequent than formerly, yet one can easily conceive of its destruction by internal forces, when, as the poet has said,

"The cloud-capped towers, the gorgeous palaces,
The solemn temples, *the great globe itself*,
Yea, all which it inherit, shall dissolve,
And like the baseless fabric of a vision,
Leave not a wreck behind."

Revelation clearly announces the destruction of the earth: "In the which the heavens shall pass away with a great noise, and the elements shall melt with fervent heat; the earth also and the works that are therein shall be burned up."

THE CIRCLE OF THE SCIENCES.

MENTAL SCIENCE

Is the mind's knowledge of itself, of its faculties, and states. *Psychology* is now generally accepted as the most appropriate term to indicate that knowledge, and the studies that lead to its attainment. The *psyche*, as used by those ignorant of man's higher nature, means the vital principle supposed to animate all living bodies, whether of men or the lower animals. It is, with them, the same as *life*, and is regarded as a result of the organizations they see, and not their cause. Others more consistently hold that, even in the lowest sense, vital forces precede, secure, and determine the organisms they animate; and that in the case of man there is a nobler endowment, a superadded, distinct, self-conscious, personal intelligence. "There is a spirit in man, and the inspiration of the Almighty giveth him understanding." This *psyche*, or living soul, is a distinct, spiritual existence, however closely, for a time, allied with matter, and acting through bodily organs. It is capable of a separate existence, and while in the body, presents for our study phenomena peculiarly its own. Intellectual processes may be more subtle, and their analysis more difficult, than that of things external, because in the attempt the mind is, at once, subject and object, the observer and the observed. And, moreover, when greatly excited, it does not submit to immediate and direct investigation, as the effort at once arrests the excited feeling, and lowers the temperature, so that the state can be analyzed only as it is remembered. But, difficult of attainment as it is, the science that discusses the mind, proposing to show all that is known or may be learned respecting it, certainly challenges the interested attention of all who desire to know themselves. Whatever may be thought of the substance, or immediate origin of the active, thinking soul, consciousness affirms its presence, and its power to know and feel. When in a calm, thoughtful state, the phenomena are as real and as manifest as anything in physics or material things that are open to scientific investigation. By thorough introspection, the inquirer finds *himself* an invisible person, quite distinct from what is merely corporeal in his belongings, and of which he at once says: It is I; a person or being that he not only distinguishes from all others, but also from his own mental acts and states that are not himself. It needs no argument to prove that the physical frame, made of such material substances as gases, salts, earths, and metals, the particles of which are constantly changing, is not the man. It is not in the highest, truest sense, the body. Every particle of that frame may pass away while the body still remains. The real body is that which retains its organic sameness, amidst the incessant change of its materials. It is not the aggregation of gross substances, visible and tangible, but rather their connection and the life that unites them, that constitute a human body. We need not hesitate to say this life is the gift of God to man, made in his own image, and in his purpose an endowment far higher than mere animal life. When it is withdrawn, the organic structure built up as its earthly habitation is a ruin, and its material elements are scattered, the dust returning to dust again. Others may inquire for the "origin of souls," asking questions over which many have wearied themselves in vain, we here only confess our faith that the sovereign Lord, "God of the spirits of all flesh," has the relation of Fatherhood to his human children.

A perfect mental science would require first, the normal action of the intellectual faculties to give phenomena, and then the accurate observation, and orderly arrangement of the phenomena given. To have a starting place there must be the feeling that we are, and can distinguish between ourselves and the mental acts of which we are capable. This consciousness is the root of all our soul science, and without it there could be no fruitful study of the human intellect. It is more than mere feeling, as it implies that activity of mind by which a man distinguishes between his body and soul, the senses and their possessor. It is the self-conscious act of knowing what is within; and when the phenomena or state is presented, the knowledge is intuitive or immediate. No reasoning, or other mental process, is required. The soul confronts itself and its acts face to face, and knows them as they are. The endowment is natural and universal. Though a child at first may show no sign of the possession, it has the capacity, and if normally developed, soon claims the right to be itself and not another. Like other human powers, this also is capable of culture, and may be raised to a state of higher activity and clearer discernment. This improved reflective consciousness brings to view the more occult phenomena within, comparing and classifying things, that it may have a clearer, more discriminating knowledge of the facts considered.

Interrogating this witness, each finds in himself a power to think and reason. That is, an *intellectual faculty*, by the exercise of which there is intelligence, *memory* to retain or recall things once known, and *imagination*, that creates and represents things that are not, as though they were. These are distinct, though inseparable, faculties or powers. *Thinking* is necessary to exact or well defined knowledge, and until our ready impressions and conceptions are penetrated with thought, and we discern their nature, grounds, and connections, we have no science. Information may be received, facts committed to the memory, but if the treasures are jumbled together, and little thought given to either their analysis or orderly arrangement, they can be of but little value to their possessor. In its perceptions and sensations the mind is actively receptive; and by thought this normal activity is intensified. One who desires a correct knowledge of his own mind must connect his conceptions and impressions in some orderly manner, and think much. If there is an aversion to this, or hindrances arise from the almost incessant demands of business or society, and a tendency to mental dissipation is noticed, we may antidote the evil by mostly avoiding the popular light literature, and choosing, as the companions of our few leisure hours, standard works, in which are treasured the best thoughts of the world's great thinkers. The intelligent study of the outer world, of nature, having the divine impress on every feature, will also do much to cure the weakness that many are ready to confess, to themselves at least. Nature does not think—has not reason, as man has, but the phenomena presented are full of reasons, the embodiments of God's thoughts, that are above ours, high as the heavens are above the earth.

The *will* is the controlling motive power, and decides the question of character. A voluntary agent is responsible for his acts. Where there is conscious freedom, not only to *act* as he wills, but to *will* obedience to the dictates of conscience, character is possible. The freedom spoken of, and without which there can be no obligation or responsibility, is, of course, *human* freedom. The will power is man's, not that of the brute. The rational, voluntary agent, having conscience, moral ideas, sensibilities, and emotions, is, under law, blame- or praiseworthy, and personally responsible for what he is and does. His involuntary acts, if such are committed, are without moral character. There are some things that are not objects of his choice. When different ways of living are presented, he can freely choose which shall be his. But it is not given him to choose whether he himself shall have a moral character. That is inevitable; and his only option in the matter is as to whether it shall be good or bad.

LOGIC.

When the mind is employed in discriminating, arranging, judging, and reasoning, these several acts are all of a class, and are called rational or logical processes. Their importance can hardly be overestimated, as thus the reasoner gets assured possession of judgments or beliefs that are more or less general, and derives from them those that are particular and applicable in any exigency; or by the inductive method, from the particular facts within his knowledge, arrives at general propositions, and securely rests in them as true. In many, perhaps in most cases, both processes, the deductive and inductive, are used or implied. We understand phenomena or effects by their causes, and infer causes from their effects, explain the present by what has been, and anticipate the future by interpreting the past. We reason from what is seen to the unseen, from the facts of nature to nature's laws.

Systems of logic, if judiciously arranged, are of much value, and should be studied as guides and helps in our efforts to know the certainty of things. Method in reasoning is of much importance. But while comparatively few understand the rules, or adopt the exact technical terms used by scientific logicians, others, using methods and terms of their own, think vigorously, and reason well. The powers employed by the most thoroughly trained scholar and by the unlettered man may be equal, nor are their methods half so different as some suppose. Though the latter forms no expanded syllogisms,[1] says nothing of "subject," "predicate," or "copula," he as really has his premises, reasons from what he knows, and in many cases reaches his conclusions with about the same feeling of certainty. The knowledge he gains does not differ from that of those who are guided in their reasonings by the best rules that observation and experience suggest. Some of those, who in this matter of logic are a law unto themselves, not only reason well, but often very rapidly. Judgment is given so speedily on the presentation of the case that it seems intuitive. There is but a step from the premises of an argument, securely laid in what is conceded in the statement, or what they already know, to the conclusion that is legitimate, and they take it at once. Now, if this is true, and common sense reasonings often seem so easy, while those conducted by men of much science are often difficult and tedious, it may be asked what advantage, then, is there in the logic of the schools? A sufficient answer is found in the fact that the thoroughly trained logician can solve problems the other never attempts. In his processes the properties and relations observed are less obvious or more complicated than anything presented to the other. To apprehend them clearly, closer attention must be given than most men, without such training, ever give or can give. And then, the conclusions of the ready, rapid, though untrained, reasoner who investigates only common subjects, are really less reliable, because more likely to be founded on too superficial observations. The man of more science, and yet slower progress, is expected to handle the more difficult problems, and subject all their elements to a sharper scrutiny.

LANGUAGE

Is intimately connected with thought, not only as its expression, but as an auxiliary. Thoughts always become clearer and more firmly fixed in the mind by being expressed. Though words are not thoughts, and, carelessly uttered, may be quite meaningless, thoughts not only seek to embody, or clothe themselves in language, but our best thinking is done in the use of words, uttered or unexpressed. Though there may be no sound for the ear nor symbol for the eye, the word inly spoken serves to fix the otherwise transient thought so that it can be afterward recalled, and perhaps uttered, to stimulate the thinking of others. Hence the importance of the study of language, of words and their syntax, as employed to express mental processes. Grammar is important as an intellectual science.

ÆSTHETICS.

The science of the beautiful is an important and delightful branch of study; the knowledge gained being mostly through immediate perceptions and sensible impressions. Beauty, wherever discovered, appeals to the sensibilities, and raises pleasant emotions. As a means of culture it elevates and refines. Communings with nature in her lovelier moods subdue asperities, and inspire gentle, kindly dispositions, while the beautiful creations of architecture, statuary, and painting, of poetry and music, fill the souls of discerning, susceptible persons with indescribable pleasure. But though such emotions are frequently excited, and seem familiar, they are of all our mental phenomena least understood, and most difficult to analyze. Some of our most common experiences are, on examination, found the most inexplicable. All, in a general way, know that beauty of form, proportion and color, wherever seen, excites pleasurable emotions. But our knowledge of sensations and emotions is generally, though direct and immediate, imperfect, and can become thorough only when the first impression is retained, and the higher faculties employed in studying its character and its cause. Dr. Porter's[2] chapters on "Sense Perceptions" are, on the whole, satisfactory, and will help advance this branch of knowledge toward the dignity of a science. They give an analysis of beauty in objects that address the senses, and also of the emotions it awakens. Thoughtful students confess their need of more help. The science has its charms, but is still in its adolescence. Some things elementary are yet wanting, or known only by the names given them. Men talk of the "line of beauty" in architecture and sculpture, but do not yet know just what it is, or by what peculiarities it works on the sympathies of the beholder. We feel the exquisite pleasure but do not know just wherein the charms of the music that most delights us, consist, nor how it awakens the feeling it does. We can not tell just what it is in the poem we admire that gives its rhythm, figures of speech and imagery such enchanting power. The literature on the subject is extensive. We have, as all who read Ruskin's[3] works know, a rich treasure of astute observations, with keen, incisive criticisms, but yet no thorough analysis of all the materials necessary to complete the science of Æsthetics.

MORAL SCIENCE, OR ETHICS.

The science of duty, often called *moral* as relating to customs or habits of thought and action, discusses human obligations, or inquires what responsible voluntary agents ought to do, and why. Man has a moral nature; is so constituted, and placed in such relations that he feels certain things to be right for him, and others wrong; he says: I ought to do this, and that I ought not. These words, or their equivalents, expressive of obligation, can be traced in all languages of which we have any knowledge, and they voice the common sentiment of the race. Men differ widely in their intelligence, and consequently in their ability to discriminate with respect to acts or states that are purely intellectual. Their metaphysics may be cloudy and confused, so that their judgments on such matters will have neither agreement nor authority. But the moral sense discovers moral qualities more clearly. Its decisions are prompt, and their authority is acknowledged. Speculative questions on the subject are not all answered with the same agreement. If it is asked *why* a thing is right, different persons may answer differently. One says because it is useful; another because it is commanded by a higher authority; and another because it accords with the fitness of things. These are questions for the intellect and not for the moral sense. Its province is simply to decide whether the act or state is right or not, and there it stops. Whether the basis of the rectitude approved is in some quality of the act itself, in an antecedent, or a consequent, may be properly asked, and reasons assigned for the answers given. But such questions are speculative, and the answers do not have, even when the best are given, the force of a moral conviction. In saying a thing is right because it is right, we affirm our conviction of the fact, but tacitly confess we may not know all the reasons why. How the fact is known is sufficiently explained by a reference to consciousness. We are so constituted that when moral qualities, in ourselves or others, are fairly presented and understood, there arise feelings of approval or condemnation, corresponding to that which excites them. Of such convictions and emotions we are at once conscious, and can have no more certain knowledge of anything than of what is thus felt. Connecting them with their exciting cause, it, too, is known, not by any outward or sense perception, yet not less certainly by an inward moral sense, whose decisions are promptly given, and with authority. There are frequent occasions for men to distinguish between what is right and what is merely lawful. A villain, destitute of moral rectitude, who for his own pleasure, or gain, robs society of its brightest jewels, spreading ruin and desolation through the community, may violate no statute, and escape legal condemnation; but, though having no fear of the law or of the courts, he is not less certainly a guilty man.

Conscience, as a faculty of the soul, differs but little from consciousness. Both words are from the same root, and neither, in its primary, etymological meaning, implies anything as to moral character. Consciousness is self-knowledge, the mind's recognition of its own state as it is; and that a man has a conscience, or capacity for passing moral judgment on himself, is a condition that makes character of any kind possible. Each word, however, has now an additional meaning, sanctioned by general usage. The former generally implies emotions of approval or disapproval, and the latter that there is in the mind a standard of action, and a clear discrimination between right and wrong, with an immediate feeling of responsibility, or obligatory emotions.

Though thus richly endowed with intellect, sensibilities, and will, by nature capable of the highest mental activities, the structure of the soul would be strangely incomplete if the religious element were wanting. But it is not wanting. Man is a religious animal, and ever prone to worship. He has capacities that are not filled, longings unsatisfied, and must go out of himself for help and rest. Of all the sciences that concern him most, no other is half so important as the science of God, an infinite, all-wise, ever-present, personal God; our Creator, Redeemer, Benefactor. This science is transcendent, and confirmed by indubitable evidence. It satisfies and saves. "This is eternal life, to know the only true God, and Jesus Christ, whom he hath sent."

SOCIAL SCIENCE

Investigates, in an orderly, scientific manner, the principles of association, and whatever relates to the interests and improvement of men in communities. It has its basis in psychology, as that science of the soul reveals most clearly the elements of a *social* nature. By instinctive longings for sympathy and fellowship, men are drawn together, and readily consent to the restraints of society, whose earlier tacit agreements and maxims are at length formulated into rules and laws for their better government.

Civil governments, incomplete at first, and encountering many hindrances, often progress but slowly, and sometimes even recede from vantage ground that has been gained. Some known in history have made but little advancement during the nineteenth century, and still fail to adjust their political machinery to the wants and demands of the people. They will hardly survive much longer without some promise of better progress in the future. All really good governments are not equally good, and that is regarded best which secures the greatest liberty to the individual citizen, consistent with the rights of others and the public security. That end, when honestly proposed, may be, to some extent, secured under very different charters or constitutions, and very much depends on the wisdom of the administration. When the governing power is in the hands of one man, and he irresponsible for his manner of exercising it, it is called an *autocracy*, or *despotism*. When vested in one person, whose executive functions are exercised by ministers responsible to a legislative assembly or parliament, it is a *constitutional monarchy*. If the nobility, or a few principal men govern by a right, in some way claimed, and conceded to them, it is an *oligarchy*, or an *aristocracy*. If the power is in the hands of the people themselves, or their immediate representatives, as in the United States, it is a *democracy*, or *republic*.

Social science embraces a wide range of subjects of more than ordinary importance. It discusses both principles and facts, the principles that underlie all social institutions, and the practical, economic regulations that are wisely adopted in well ordered, prosperous communities. If the institutions are established, its province is to examine theories, collect, arrange, and generalize facts, that may have some bearing on any proposed corrective and reformatory measures. It scrutinizes public crimes, penal codes, judicial decisions, and prison discipline, with whatever else pertains to social life. It shows the relation of men to men, of the ruler to the governed, of the employer to his employes, of the rich to the poor, the fortunate to the unfortunate, and by its expositions instructs men how to act in their various relations. If the science were much better understood, the dangerous classes would be less dangerous; and the troublesome problems of pauperism, the liquor traffic, Mormonism, and the social evil, would be less appalling to average legislators and judges.

The experience of ages shows that the ameliorating, helpful agencies and influences that lift communities up to higher levels, often operate silently as the leaven, till the whole lump is leavened. In many tribes the advance from savagery and the usurpation by irresponsible leaders, of absolute power toward complete civil liberty and personal rights, has been slow. The change has come by means almost imperceptible, or by struggles that seemed at the time fruitless. The improved condition of society does not bring entire security, or freedom from assault. The yoke once taken from the neck, and the heavy burdens from the shoulders, new ideas of property, justice, and personal rights are developed. The spirit of enterprise is awakened, because each finds himself in the position of affluence and influence, to which his talents, industry and self-denial entitle him. Men become competitors, and inequalities of condition are inevitable. Incompetence, idleness, and extravagance bring want and misery. Wealth and poverty exist side by side, the rich growing richer, and the poor poorer. Class distinctions become odious. Capital and labor, that should be allies, are often in conflict, to the great injury of both. There may be occasion for complaint against those "who oppress the hireling in his wages," and "grind the faces of the poor." But many are envious without cause, and suffer only the penalty of their idleness and extravagance, become enemies of the community, and are deaf to remonstrance if they see, or think they see, any way of relieving themselves at the expense of those who have acted more wisely, and possess large estates. Here come in the functions of government, that is of society, with its better notions of right and justice, and power to enforce them. True "social science," founded on the experience of ages, recognizes the necessity of government, the obligations of the citizen, and the right of all to the property they have lawfully and honestly acquired. It demonstrates that real progress is in the way of a safe conservatism, while it admits the possibility of change and improvement, fully justifying the work of the reformer where reformation is needed. If existing institutions are inadequate because of some radical defect, have outlasted their usefulness, or become oppressive, revolution may be demanded. But any government, though unjust and despotic, is better than anarchy, and should be repudiated only when it is, under all the circumstances, possible to establish a better. When legitimate authority is resisted in the spirit of lawlessness or efforts at revolution prompted by an evil ambition, the actors are guilty. There have been many attempts, mostly abortive, to solve the problem of government, and reconstruct the social fabric. Some of them by good men, whose schemes were simply theoretical and impractical; others by malcontents and destructionists who mistake license for liberty. Plato, a man of probity and justice, but lacking the wisdom of the statesman, prepared a constitution for a model republic, which had too many defects for adoption; a republic with advantages for a select class, but slavery for the masses doomed to manual labor, which was made despicable. More wrote his “Utopia,”[4] regarded by some as a kind of program for a needed social reform. It had little influence with his countrymen, most of whom ranked it with works of the imagination, where it belonged, whether so intended or not. Campanella,[5] a radical communist of Stilo, in Calabria, wrote his Utopia, called “The City of the Sun,” a sensual paradise, in which there was to be a community of goods and of wives. For more than a century socialistic and communistic publications were numerous; many of them denouncing property as a sin, and advocating the greatest license in the intercourse of the sexes. Rousseau, in his discourse on the “Origin and Grounds of Inequality Amongst Men,” speaks with approval of “a state of nature,” something like that among our American Indians before they had any knowledge of civilization. He seems to have supposed there was no inequality, no vice, no misery, among untutored savages, and advised those who could, to return to a state of nature.

The skeptical Owen,[6] and the philosophical Fourier,[7] more practical than others, attempted to establish communities as models or examples of what could be accomplished on their theory, but failure attended their enterprises, or the communities were saved from utter disintegration by the tacit admission of principles that were once disavowed.

Modern socialism, akin to communism, and in all its tendencies subversive of good government, declaims over the poverty and misery of the unhappy masses, laments their insufficient shelter, food, and clothing, is sentimental on the subject of labor and wages, and seeks to rouse the abject sufferers to a sense of their sad condition. Its pictures of suffering are in many cases not overdrawn, and the greatest efforts can hardly exaggerate the facts. But socialism is not “social science.” It is utterly and perversely unscientific; it discusses effects, carefully leaving causes out of view; and would reform communities by corrupting and debasing individuals. With a vague notion that every man has a natural right to whatever he needs, it allows that the problem of equalization may be solved by violence, and thus all brought to a common level. The crimes against society, which have lately appalled us, are doubtless the result of the pernicious principles and teaching of “socialistic reformers.” But a reaction is sure to come, and the better instincts of the race will yet destroy the corrupt tree whose fruit is evil.

SUNDAY READINGS.

SELECTED BY CHANCELLOR J. H. VINCENT, D.D.

[*April 5.*]

Ecclesiastics 7:29: "Lo, this only have I found, that God hath made man upright; but they have sought out many inventions."

If we should stand amid the uncovered treasures which mark the site of ancient Nineveh or Babylon we would doubtless find in the objects, as they are this moment, very much to engage our most interested attention. We would regard with wonder and pleasure the specimens of beautiful architecture, the evidences of human skill and industry which modern exploration has disclosed to view. And yet, full of interest as such an occupation would be, we could not prevent our minds from engaging in another. Without any conscious exercise of will, our thought would revert to the day when these fallen structures stood in all their magnificence; when these halls, now filled with the sand of the desert, echoed to the strains of music and to the voices of the noblest and greatest of the land; when through these arches, now crumbling, armies marched gaily to battle, or returned in triumph, bearing the spoils of conquest. We would not be insensible to the value of the columns and capitals, the statuary and tablets before our eyes, and yet the very grandeur of these ruins would evoke the genius who would lead our thought by an irresistible constraint to look upon the prior vision of those cities in the day of their pristine perfection.

My friends, we do stand amid ruins. We walk day after day amid shattered greatness, in comparison with which the prized relics of Nineveh and Babylon, of Thebes, and Luxor,[1] and Troy, sink into insignificance. Far be it from me to underestimate the work of man, as we see him and know him to-day, or as we read of him in the records of the past. I am aware of what he is, and of what he has done. I am not insensible to his work, and yet I declare that man, great as he is—and he is great—is in certain respects not as great as he was. I mean that he is not what the progenitor of the race was. And viewed in comparison with that primitive condition—that condition at creation—man to-day, considered physically, intellectually, morally and spiritually, is a conspicuous instance of fallen grandeur—not worthless and valueless—far from that; but his perfection has departed; he is vastly inferior to what his great original was. Realizing this, we can not fail to revert in thought to that early day, and seek to see what the greatness was from which we have fallen.

Before, however, we attempt to look upon that picture, it will be necessary to establish and defend the theory of man's condition and history upon this earth, with which it is inseparably connected.

A view of human history, which has been strongly advocated of late, is that our race began physically, intellectually, and morally at the lowest possible point. Some even maintain that the first men and women were but the latest and highest developments of certain species of brutes. But whether this phase of the theory of evolution be included in it or not, the essential idea of the view to which we refer is that the progenitors of our race were the lowest kind of savages, in their powers and capacities, their tastes and pursuits scarcely distinguishable from the brutes around them, and that from this low beginning men have gradually come to the height of attainment and improvement which they occupy to-day. If this theory be true, the statement which we have made, and which we proposed to consider, is false. If such were the origin of our race, if the first men and women were in all the parts of their nature but a shade removed above the brutes of the forest and the field, of course we of to-day are in no respect their inferiors—of course ours is not, as has been declared, a fallen race. We maintain, however, that the theory which makes our race begin in a condition of barbarism and imbecility is untrue. I know that it is supported by eminent names; I know, too, that it has been pressed upon public attention with much noisy and confident assertion; I know all this, and yet I declare that the theory is unproven; more, I declare that it is untrue.

Let us look at a few considerations which may shed the light of truth upon this matter of the primitive condition of mankind, and by this I mean the condition of those who succeeded Adam himself on the stage of the world’s history:

1. We all know how long, in families which have lost position or power, the memory of former greatness is cherished. You will find in your charitable institutions, in the depths of poverty, and, perhaps, of wickedness, those who will tell you by the hour of the fortunes of their house in remote days, of the distinction which some ancestor, far removed, conferred upon the name. Such memories are preserved with care, and transmitted from generation to generation, and they become more and more precious as the descendants themselves have less and less honor of their own. The same principle operates with nations and with the great tribes of men, particularly when they have themselves sunk so low that they are conscious of no ground of glorying in themselves. Now it is an instructive fact that the traditions of all nations have more or less reference to a golden age, from which men have fallen. This is true in India, in China, in Egypt, among our own Indians, among the inhabitants of Central and South America—wherever traditional knowledge is preserved. It is a vague memory—nevertheless a memory cherished by the race, handed down through the ages, not of an era when humanized monkeys rejoiced in their promotion, but of a golden age, when men as gods dwelt upon the earth. The only explanation of such a wide-spread tradition is that there must have been a fact corresponding to it; there must have been a substance to cast this shadow over so many generations. Those who hold that mankind began at the lowest point, can not satisfactorily account for this tradition of the race.

2. Not only tradition, but history sustains our position. If the true explanation of man’s condition to-day, in the civilized countries, is that he has gradually raised himself from a state of absolute barbarism, we certainly ought to have in the records of authentic history the account of at least one nation, which, as matter of fact, before the eyes of the world, has done the same thing. But no such instance can be found, not one of a people, entirely barbarous, lifting themselves unaided, to the higher plane of even a comparative civilization. Archbishop Whately[2] says: “We have no reason to believe that any community ever did or ever can, emerge, unassisted by external helps, from a state of barbarism unto anything that can be called civilization.” And we may follow the course of civilization from our own land back to western Europe, from western Europe to Italy,

from Italy to Greece, from Greece to Egypt and the farther East, and still, as far as history takes us we see the barren portions of the earth continuing to be barren—continuing to bear only the wild fruits of barbarism, until the stream of knowledge, and culture, and civilization, is led to it from some other place. And that stream may be followed all the way back to the beginning of authentic secular history, and in no one instance does the dry ground yield fruit and flower of itself. We maintain that the vivifying stream began to flow because there was in the beginning, in the East, a fountain filled by God himself. Or, leaving the language of allegory, we assert that if our race was utterly barbarous at the beginning, it never would have risen from its barbarism, and authentic history can not adduce a single instance of a nation rising unaided from barbarism to overthrow this position. Mankind, therefore, did not begin at the lowest point, or, judged by all the analogy of history, it would be there still.

3. Again, the records, outside of the Bible, which have come down to us from early times, are few and imperfect, but the oldest of those which do remain indicate the existence of nations in a high state of civilization in the earliest periods of human history. In Egypt, China, Chaldea, these earliest records, whether monumental or written, bear evidence, not of universal barbarism in the previous ages, but of powerful and enlightened nationalities. Such a state of things is inconsistent with the theory which makes the history of our race a gradual development from a brutal and degraded beginning.

4. Still further, the science of paleontology comes forward to prove the same thing. There have been found in Belgium and in France, some human skulls and skeletons, unquestionably of very great antiquity. Concerning them one of the most competent of human judges, Principal Dawson, of McGill University, Canada, says: “These skulls are probably the oldest known in the world.” But what sort of men do they indicate as living at that remote day? “They all represent,” he says, “a race of grand, physical development, and of cranial capacity equal to that of the average modern European.” Further he says: “They indicate also that man’s earlier state was the best, that he had been a high and noble creature before he became a savage. It is not conceivable that their great development of brain and mind could have spontaneously engrafted itself on a mere brutal and savage life. These gifts must be remnants of a noble organization, degraded by moral

evil. They thus justify the tradition of a golden and Edenic age, and mutely protest against the philosophy of progressive development, as applied to man." Again, he concludes from a careful study of these remains: "The condition, habits and structure of Palæocosmic[3] men correspond with the idea that they may be rude and barbarous offshoots of more cultivated tribes, and therefore realize, as much as such remains can, the Bible history of the fall and dispersion of antediluvian men. We need not suppose that Adam of the Bible was precisely like the old man of Cromagnon.[4] Rather may this man represent that fallen yet magnificent race which filled the antediluvian earth with violence, and probably the more scattered and wandering tribes of that race, rather than its greater and more cultivated nations."—*Nature and the Bible, pp. 174-179.*

5. In addition to these arguments from tradition and history, and monumental and written records, and an actual study of human remains, which experts pronounce to be older, probably, than the flood, we have evidence within ourselves. We are not unfamiliar with stories of children of noble, perhaps of royal, descent, who have been abducted and brought up among people of low tastes and habits. But ever and anon, in gesture or inclination or bent of life, the blood shows itself, and to an attentive eye tells of the gentler and loftier sphere from which it came. So with us. Our consciousness reveals within us remnants of a former greatness; aspirations which this world has never taught us, longings for peace and purity which we feel we ought to have, but which we know this world never imparts. These things are the impress of the joys of that golden age which all these centuries have been powerless to erase from the souls of Adam's sons. Morally, we know we are not what we ought to be; we are conscious of our degradation. As regards intellect, we retain powers which have, indeed, accomplished marvelous results; and yet, let some abnormal stimulus affect the brain—whether it be that produced by sudden excitement or passion, or that caused by powerful artificial agencies, and we see flashes of power yet in reserve which intimate what this wondrous human mind may once have been. And physically, our frail bodies, quickly tired and quickly crumbling to dust, tell us daily that here, at least, the theory of development from imperfection to perfection has signally failed.

[*April 12.*]

From these considerations we deem it evident that the theory of man’s development from a primitive condition of barbarism is untrue. The various glimpses which we have been able to obtain of the early ages reveal man as in an advanced condition. To all this the representations of the Bible correspond. It is not the design of the inspired volume to give a minute description of the customs, habits, knowledge, employments of the nations of the world. All that it says upon these subjects is incidental, yet no one in reading its accounts of those early times, and of the people then living, could possibly imagine that the men and women of whom it speaks were such as the rude Hottentot or the savage Caffre of to-day. The picture of man in the primitive times drawn from the Bible, is the same as that which is drawn from all these other sources; a being of physical and intellectual power; not a savage, not a barbarian, but an enlightened, capable, efficient man. How much he knew, how much he could accomplish, what acquaintance he had with the forces of nature, which we are now beginning to understand, must be matters of conjecture. Sin had commenced its blighting work, but we must remember that man in those early days had inherited from the first man splendid powers, and probably varied and extensive knowledge. His physical strength and his length of days were still great, and doubtless in all respects, save morally and spiritually, he was even yet a magnificent creature, and a power upon the earth.

Still, this was not the point which, in this discussion, we wished to reach. All this was the greatness of man after he had begun to deteriorate, after the poison of sin had begun to do its certain and terrible work. This vision of primitive man in his physical and intellectual strength is the splendor which abides a little while in the sky after the sun has set. Nevertheless, the sun had set, and there is a world-wide difference between this picture and that unto which we would lead you—the picture of that sun in its glory—the picture of unfallen man—the first man—the perfect man. Now let us look upon it. The long ages of preparation have rolled away and the earth is a garden of loveliness. Upon the splendors of its landscapes, the beauty of its lakes, the grandeur of its mountains and oceans, the sun looked down from his pavilion in the sky by day, and the moon and the stars by night. The magnificent domain waited in harmony and beauty for its inhabitants, “And

God said, let us make man in our image, after our likeness." "So God created man in his own image: in the image of God created he him." "And God saw everything that he had made, and behold it was very good."

In the place of honor and dominion in that waiting world of beauty, enthrone a being who shall be the counterpart of those words of infallible description—a *man*, made in the image of God, and receiving the unqualified commendation of his divine Creator. We may, without danger of mistake, consider him to have been physically a being of magnificent stature, and of matchless perfection of feature and form, with a body ignorant of weakness or disease, free from the seeds of sickness and of death. That a change would afterward have been necessary to fit his body of flesh and blood for its immortality is possible, but such change would not have been what we understand by death. Age would not have brought infirmity to him. Nature would have had no debt to pay to the grave.

Enshrined in this perfect body as in a glorious temple was a mind corresponding, doubtless, in excellence to its habitation and to the terms which describe its creation. Intellectually as well as morally, he was created in the image of God. He was possessed of reason and of actual knowledge. When the various classes of animals were passed in review before him, he had such an apprehension of their distinctive characteristics as to be able to give to them all appropriate names. And as he knew thus much of nature, how shall we limit his familiarity with her mysteries? And what shall we say of the powers of discernment, of intuition, of memory, of judgment, of the facility in working, of the unwearied and the unending delights and achievements of a mind made in the image of God, and not yet marred or weakened by sin!

But the crowning glory of that first man, that which marks his distance from us more than any physical or intellectual superiority, is that in his moral and spiritual nature he bore a likeness to his divine Creator. This being, whose body knew no imperfection, whose mind was rich in its possessions, and mighty in its power to acquire and enjoy every kind of truth, was holy. No thought arose in that heart which could not be published in heaven—none which could mantle his cheek with the blush of shame, or cause his manly eye to drop in consciousness of wrong, or make a ripple of disquiet in the sea of perfect peace which filled his soul. His thoughts were

God’s thoughts. His loves, his wishes, his purposes, in harmony with God’s, ascended and descended like the angels Jacob saw, in perfect and blessed communion between heaven and earth, and earth was heaven begun. Of this world, with its abounding life, he was the acknowledged king. Within him was the consciousness of peace, and joy, and immortality. All about him was beauty, and amid the glories of his Eden home, God himself condescended to walk with him and be his friend.

Such is a faint outline of the picture on which I would have you look—the picture of man as he was in the beginning. Does not the sight justify the assertion that we are a fallen race? Does it not confirm the teaching of our text, that “God hath made man upright, but they have sought out many inventions?”

I need not delay to prove that this picture is not a representation of our present condition. Think of these frail physical existences, begun with a cry, continued in pain and weakness, and extended with difficulty to their three score years and ten. Think of the ages through which the intellect of the most favored portion of the world has been struggling to its present attainments. Think particularly of the moral and spiritual condition of the race, in comparison with that heavenly vision of Godlikeness which stands at the beginning of our history. I need not delay then to prove that mankind has fallen. I will, however, ask you to remember when you reflect upon the sad disorders of the present state, upon the sorrows and weaknesses and wickednesses of men to-day, that God did not thus create the progenitors of the race. If we see ruins about us and within us, let us remember that the temple as it was fashioned by the supreme architect was perfect. Let us remember also the real and only cause of this terrible catastrophe. It was sin—sin that always has ruined and always will destroy the beautiful, the pure, the true. Men did indeed go from that height of exaltation into the depths of barbarism; it is true enough that the pages of remote history show us men living in caves, and almost reduced to the level of the brutes—and sin led them there! Men did lose the moral beauty of our first parent; they did lose much of the intellectual and physical strength which lingered for a season in his immediate descendants—and sin was the despoiler that remorselessly stripped from them those glorious robes! You and I might have been as Adam—ignorant of sorrow and of suffering, and the world still an Eden about us, but sin has cast us down.

But let us remember also, with infinite gratitude and hope, as we look upon that picture of primeval perfection which we have sought to restore, that that condition may be regained. The crumbled arches, the fallen walls, the shattered foundations of Nineveh's or Babylon's palaces can never by any human skill be made to reproduce the glory that has departed, and yet the temple of man's Eden estate, though cast down with a more fearful destruction, can be restored! Yea, made more glorious than it was before, and established upon a foundation, so that through the eternal ages it can never again be moved! Thanks be unto God, this is possible to us. Jesus Christ has come from heaven and has undertaken the accomplishment of the stupendous task. Jesus Christ has promised to effect it for every one who will yield to his influences. And he can do it, for he is the Savior, he is the Redeemer, the buyer-back of that which was lost, and of nations and of regions as well as of individual souls. … His spirit is the inspiration of the life which here is lived. That is enough to lift up any place or any people. And just as certainly will it lift up any human soul. Just as certainly will it redeem it from the consequences of the race's fall. Not in this life, indeed, may we expect to have again the perfection of power and the freedom from sorrow which our first parent had; but the work of bringing men back to all the blessedness which Adam enjoyed, with new elements of blessing added, will be done—yea, it is even now going on, through the power of an indwelling Christ in souls that are here to-day. Let us all take the loving and mighty hand which is extended for our uplifting; let us seek our birthright, and though, through the first Adam our Paradise was lost, let us yield ourselves to the second Adam, by whom a better Paradise shall be regained. —*The Rev. Dr. E. D. Ledyard.*

[*April 19.*]

Monarchs reign, but their dominion is merely external. They do not and can not enter into the realm of the soul; but "there is another king, one Jesus," whose right it is to sit enthroned in every heart, to direct every conscience, and to have dominion over every thought and action. Have you given him the sovereignty of yourself?

Sin reigns, and that king, alas! holds sway in many—I ought to say in the vast majority of human souls. But he is an usurper; for "there is another king, one Jesus," who is the rightful lord of the heart. Under which king are you? He who repudiates the royalty of Jesus over him is guilty of treason against the majesty of heaven, and is but courting his destruction.

Death reigns, and day by day he is sweeping in new multitudes into his silent realm. The mightiest and the meanest alike must yield to him who is the terror of kings, no less than he is the king of terrors. At one time he rides on the hurricane, and dashes the laboring vessel and the freighted souls within her on the roaring reef; at another he drives through the city streets riding on his pestilential car, and spreads desolation round him. Now he careers upon the boiling flood, and sweeps whole villages before him into swift destruction; and again he leaps in the lightning-flash upon some devoted building, and kindles a conflagration that burns many in its flames. He laughs at men's efforts to elude his grasp; and as we look upon the settled countenance of the loved one whom we are preparing to lay in the grave, we are almost compelled to own him conqueror. But no! "there is another king, one Jesus," who is "the Resurrection and the Life," and "who hath abolished death and brought life and immortality to light by the Gospel." Let us, then, be undismayed by this last enemy. He is a vanquished foe. Our Lord Jesus has gone into his domain, and having conquered him there, has brought him back with him to his palace, to be there the page who opens the door for his friends into the chamber of his presence. Yes! as we stand by the remains of our Christian dead, and under the influence of sight are moved to speak of Death as king, we recall in another sense than they were meant, but in a sense which faith recognizes as true, the words "There is another king, one Jesus."

Paradoxical as it may seem, these two things always go together. Where Christ is owned as the sole sovereign, there his service is perfect freedom; but where his supremacy is either ignored or given to another, there comes the slavery of superstition, or the tyranny of priestcraft, or the cold domination of philosophy, and it is hard to say which of these is the most degrading.—*W. M. Taylor.*

[*April 26.*]

He who would preach the Gospel with power must be himself a believer in the Lord. The secret of true, heart-stirring eloquence in the pulpit is, next after the power of the Holy Ghost, that which the French Abbé has very happily called “the accent of conviction” in the speaker. Behind every appeal that Paul made to sinners, there was the memory of that wonderful experience through which he passed on the way to Damascus; and therefore we are not surprised that he *so* preached as either to secure men’s faith or to rouse their antagonism. But his conversion alone, without his Arabian revelations, would not have made him the apostle he became. In the desert he met his Lord, and received from him many important spiritual communications. There, too, he meditated on the truths revealed to him, and poured out his heart in prayer for a thorough understanding of their meaning and a full realization of their power. Thus he came back to Damascus, if not with a face glowing like that of Moses when he descended from Sinai, at least with a heart filled and fired with love to Him who had there unfolded to him the mysteries of the Gospel. Now, what Paul thus received from the Lord has been given to us by evangelists and apostles in the New Testament Scriptures. Our Arabia, therefore, will be the study and the closet in which we pore over these precious pages, and seek to comprehend their many sided significance, as well as to imbibe the spirit by which they are pervaded. He who would preach to others must be much alone with his Bible and his Lord; else when he appears before his people, he will send them to sleep with his pointless platitudes, or starve them with his empty conceits. Get you to Arabia, then, ye who would become the instructors of your fellowmen! Get you to the closet and the study! Give your days and nights to the investigation of this book; and let everything you produce from it be made to glow with white heat in the forge of your own heart, and be hammered on the anvil of your own experience!—*W. M. Taylor.*

EASY LESSONS IN ANIMAL BIOLOGY.

CHAPTER I.

Biology is the science of life, the true doctrine concerning all living things. Animal biology is that branch of the science which relates to animals, as distinguished from plants. It tells of these animals what we know about them, where and how they live, what food they eat, how it is received, and how they grow and multiply. Of all the sciences, this seems most extensive, having for its field a world of numberless forms, alike in that they all live, and have some characteristics in common, yet showing great diversity in their structure, appearance, and mode of life. In this summary of facts we shall simply classify, or methodically arrange in groups, according to their distinguishing peculiarities, the members of this vast family.

The animal kingdom is divided into the following sub-kingdoms, each of which is subdivided into classes. The following table shows these divisions in their proper order, beginning with the lowest:

Sub-kingdom I—*Protozoa*.	{ Class I—Monera. { Class II—Gregarinida. { Class III—Rhizopoda. { Class IV—Infusoria.
Sub-kingdom II—*Spongida*.	
Sub-kingdom III—*Cœlenterata*.	{ Class I—Hydrozoa. { Class II—Anthozoa. { Class III—Ctenophora.
Sub-kingdom IV—*Echinodermata*.	{ Class I—Crinoidea. { Class II—Asteroidea. { Class III—Echinoidea. { Class IV—Holothuroidea.

SUB-KINGDOM V—*Vermes*.	{ Class I—Flat Worms. { Class II—Round or Thread Worms. { Class III—Rotifera. { Class IV—Polyzoa. { Class V—Brachiopoda. { Class VI—Annelidæ.
SUB-KINGDOM VI—*Mollusca*.	{ Class I—Lamellibranchiata. { Class II—Gasteropoda. { Class III—Cephalopoda.
SUB-KINGDOM VII—*Articulata*.	{ Class I—Crustacea. { Class II—Arachnida. { Class III—Myriapoda. { Class IV—Insecta.
SUB-KINGDOM VIII—*Tunicata*.	
SUB-KINGDOM IX—*Vertebrata*.	{ Class I—Pisces. { Class II—Reptilia. { Class III—Aves. { Class IV—Mammalia.

SUB-KINGDOM I.

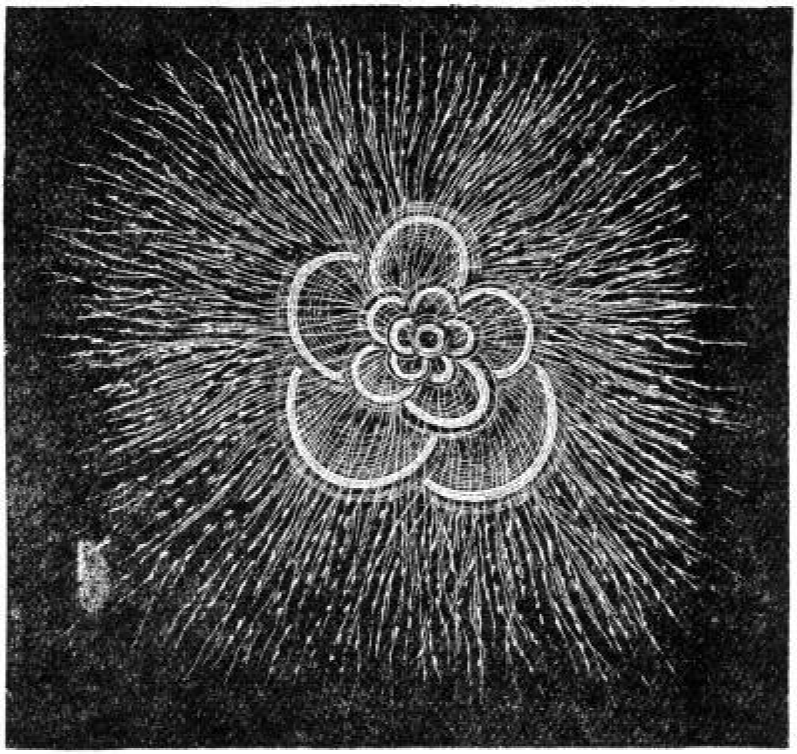
Protozoa (first animals). These earliest formed animals are distinguished for the simplicity of their structure. In some cases their animal nature was long ago in doubt, and they were, for a time, put down as probably belonging to the vegetable kingdom. The border line between the two has never been very definitely located. Biologists may fail to tell just what special quality distinguishes the minute animal from the microscopic plant. This is not wonderful, when it is remembered that myriads of animals, known to be such, are so small that it requires a lens of strong magnifying power to discover them. Three thousand of them, placed side by side, would make a line but little over an inch long.

CLASS I.—*Monera* (single). These are the simplest forms of microscopic aquatic animals. They are entirely homogeneous, and without any developed organs; mere particles, of a jelly-like, but living, or life-supporting substance, called protoplasm, or more properly, bioplasm. This, all admit, is the physical basis of life, and the medium of its manifestation, just as the conductor is a medium of manifestation of electricity. But is it not a stupid blunder to confound the mere medium of its manifestation with the life itself? In neither case is the recognized physical basis of the manifestation necessary to the existence of that manifested by its means. Electricity exists without the conductor, and life may exist without the bioplasm.

Class II.—*Gregarinida* (living in herds). Minute animals which are found in the intestines of the lobster, clam, and cockroach. They are worm-like in form, and of a very simple, cell-like structure, the only organ being a nucleus.

Class III.—*Rhizopoda* (root footed). The representative forms of this class are the *Amœba*[1] and *Foraminifera*. The amœba is an indefinite little bit of bioplasm, as structureless as the monera, only that it is made up of two layers of the substance, has an apparent nucleus, and a contractile cavity within. These first animals vindicate their right to be recognized as such, by moving, receiving food, growing, and reproducing their kind. They move by a contractile force, throwing out at will processes of their soft bodies, to serve them as feet and arms. True, these are blunt, and without digits, but they answer the purpose. They eat either by simple absorption, or by wrapping their soft bodies around the food, and holding it in the extemporized stomach till it is, in some way, assimilated.

The work of reproduction is carried on principally by self-division, and budding. The animal rends itself into two or more parts, each having all the elements of the whole, or it throws out buds that mature and drop from the parent mass, having the vital element, and a portion of the bioplasm, or medium necessary to its development.



FORAMINIFERA, SHOWING ROOT-LIKE FEET.

The *Foraminifera* (perforated animals), of this class, have several peculiarities. The body is even more simple, being apparently without layers or cavity. The processes thrown out as arms and legs are not blunt or massive, but long and slender. And, moreover, small as it is, it has the wonderful property of secreting about itself an envelope, whose thin walls are built of the carbonate of lime. The delicate little structures are often singularly beautiful. Some are monads, having but a single shell; others, by a process of budding add new cells or chambers, often in a spiral coil. These are marine shells, and their numbers in many parts of the ocean are astounding. The bottom of the sea, for many degrees on both sides of the equator, wherever examined by dredging, is found covered with them, either still in their organic state, or ground by attrition to a fine lime dust. It is estimated that a single pound of the sea bottom in some localities contains millions of them, and they are the principal material of the chalk hills.

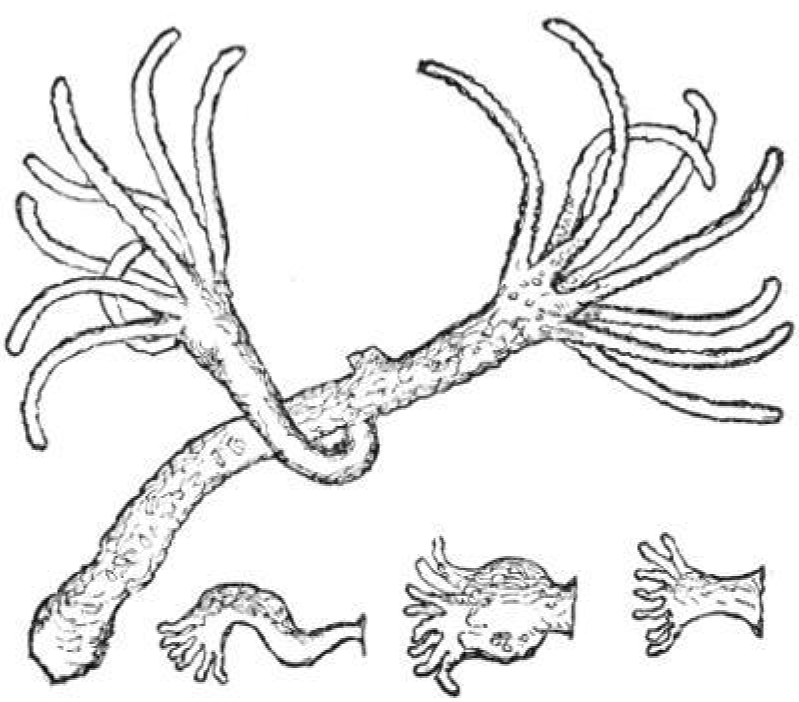
Class IV.—*Infusoria*. This class includes *Vorticella* (wheel animals), *Flagellata* (whip-shaped animals), *Tentaculata* (having tentacles), and others. Their general characteristics do not differ widely from those already mentioned. As the name imports, they are mostly found in vegetable infusions that have been exposed for some time, and are directly the product of invisible cells or animal germs that were floating in the atmosphere till a lodgment was found favorable to their development. Those called *Vorticella*, to the eye seem simply mould on the plant to which they are attached, but under the glass their animal qualities appear; and they multiply with amazing rapidity. Every drop of water from a stagnant pool is full of these animalculæ, of various shapes and dimensions, some of them constantly in motion, propelled by numerous cilia, or slender, hair-like appendages that fringe the circumference, and are used as oars. Their whole organism, though delicate, and having a thin membranous covering, is imperfect. There are two contractile openings, with a slight depression at the mouth, leading to a funnel-shaped throat, into which the nutritive substances descend.

SUB-KINGDOM II.

Spongida, or sponges, are especially worthy of mention. When much less was known of their nature and habits, they were classed with vegetables, but since their mode of reproduction has been discovered, they are known to be animals. The sponge is formed of an aggregation of ciliated cells, built together on a skeleton or framework of calcareous or silicious substance, that extends by slow external secretions as the animal body grows. There is a central cavity toward which there are numerous channels, from openings on the surface, through which water is continually received, and one through which it is discharged. The animal part is a sensitive, gelatinous film, extending through the growing mass, and spreads out over the surface. It is perforated, in places, and adapts itself to all the sinuosities and intersections of the canals that run in every direction. The little animals are provided with exceedingly fine cilia that they keep in almost constant motion; no one knows how, or for what, but it is supposed they thus sweep in the water that circulates through all the channels and chambers formed for it. After death, the soft matter, like all animal tissues, decays, or is dried up; and by beating and washing, it and any calcareous substances are removed. The horny, elastic fibers are found so exquisitely connected about the internal canals and cavities that water is freely admitted, or by pressure expelled. Sponges are found in every latitude, but are more numerous and grow larger in warm climates. Those in our markets are mostly from the Bahamas and the islands of the Mediterranean. They are obtained by diving, often to great depths.

SUB-KINGDOM III.

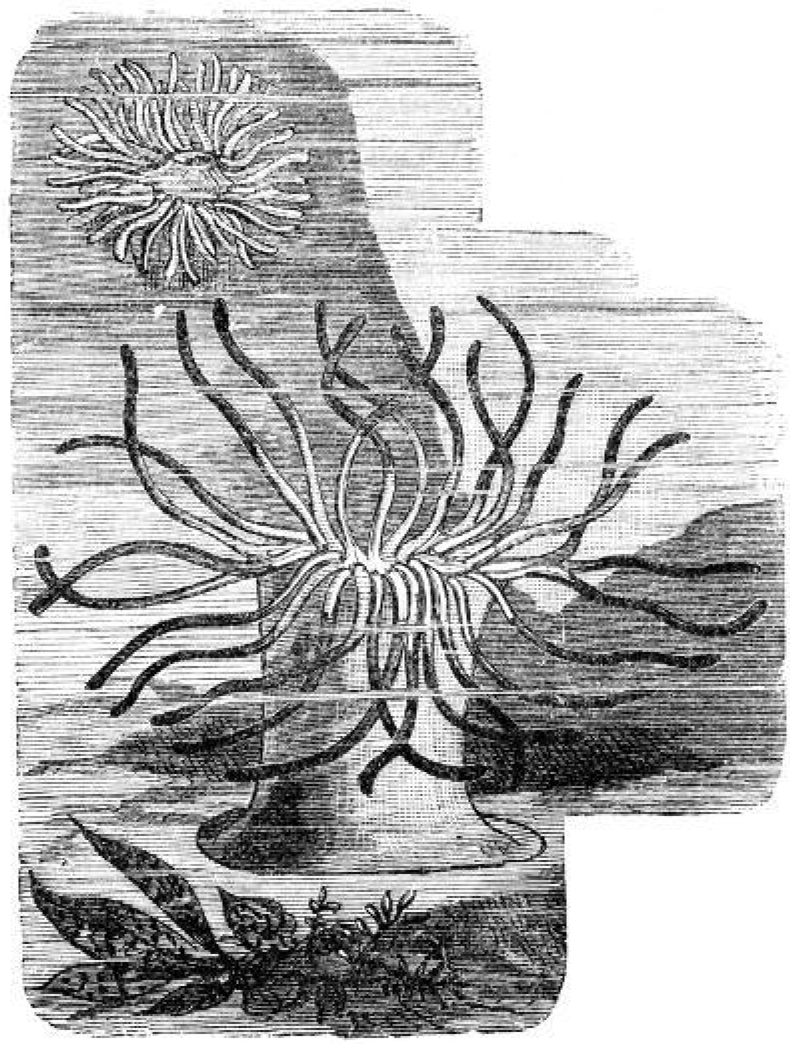
Cœlenterata (hollow intestines). These are radiate animals, have a distinct digestive cavity, and always two layers of tissue in their walls. They have minute sacs containing a fluid, and barbed filaments capable of being thrown out as stings. The classes of the Cœlenterata are the *Hydrozoa*, *Anthozoa*, and *Ctenophora*. The best known representative of the former is the fresh water *hydra* (water animal). It gives but slight evidence of a nervous system, has no stomach, or digestive sac, but is a simple tube into which the mouth opens. The sensitive little body is in color and texture, to the casual observer more like a plant than an animal. It is attached at one end to a submerged aquatic plant, while the mouth at the free end is provided with tentacles,[2] by which it feeds and moves. It buds, and also produces eggs. The young hydra, when hatched or thrown off, attaches itself to plants, as did its parents. Some hydroids attach themselves to shells, and are supported by horny, branching skeletons, specimens of which are numerous, and may be seen in almost any museum.



HYDRA, SHOWING BUDS.

A second representative of this class is the jelly fish. It has a soft, gelatinous, circular body, that floats or swims on the surface in calm weather, with the mouth downward. Tubes radiate from the center to the circumference that is fringed all around with pendant tentacles, sometimes of great length and of considerable contractile power. They are of various sizes, some quite small, others as much as eight feet in diameter. They move about by flapping their sides, after the manner of opening and shutting an umbrella. When dried, the thin, filmy covering is very light. One variety, called *Lucernaria*, is found attached to grasses along the Atlantic shore. But the ordinary jelly fish is free, and borne on the surface of the sea.

The *Anthozoa* (flower animals) are small, but not microscopic animals, having a double cavity, the inner of which is the digestive sac. The best known of the class is the *Actinia* (rayed), or sea anemone, so called from its resemblance to a plant or flower of that name. The body is somewhat like a flower in shape. The disc has a central orifice, very contractile, and surrounded with tubular tentacles of various forms, which it elongates, contracts, and moves in different directions. They are so many arms by which the animal seizes its prey, and when expanded for the purpose, being tinted with brilliant colors, present an elegant appearance, and make vast fields of the ocean look like beautiful flower gardens. They feed voraciously on little crabs and mollusks, that often seem superior to themselves in strength and bulk, but they have power in their little tentacles to seize and hold their prey, and when they engulf large bodies the stomach is distended to receive them; and their digestion is good. The purple sea anemone is very common on the southern shores of England, and one species, found on the shores of the Mediterranean, is said to be esteemed a great delicacy by the Italians. At night, or when alarmed, the animal draws in its arms, shuts the door, and seems but a rounded lump of flesh, a huge oyster without a shell. The coral polyps (many footed), belonging to this class, are little folk, but of importance from their well-earned reputation as reef builders. They are very diminutive creatures, mere drops of animal jelly, often not larger than the head of a pin, but their works show unmistakable evidence of life, and an organic structure. They live in communities, closely united, but each, having a distinct individuality, by the sure process of secreting a portion of the calcareous matter within reach, prepares for himself a house, as all his ancestors have done, and his neighbors are now doing. They build together, their foundations having strong connections, and thoroughly cemented. There in his own little palace the polyp lives, his food being brought to his lips; and, having sent out a numerous progeny to do likewise for themselves, there he dies. Life and death, as in all mundane communities, being in close proximity, the old dying, a new generation builds houses over their sepulchres.



SEA ANEMONE.

The great variety of corals seen in any extensive collection, some very beautiful, others rough and unsightly, are from different branches of a very extensive family. *Astrea* (star shaped), from the Fiji islands, is a kind of coral hemisphere, covered with large and beautiful cells.

Mushroom coral is disc shaped, not fixed or attached, and is the secretion, not of many, but of a single huge polyp.

Brain coral is globular, and the surface irregularly furrowed or corrugated.

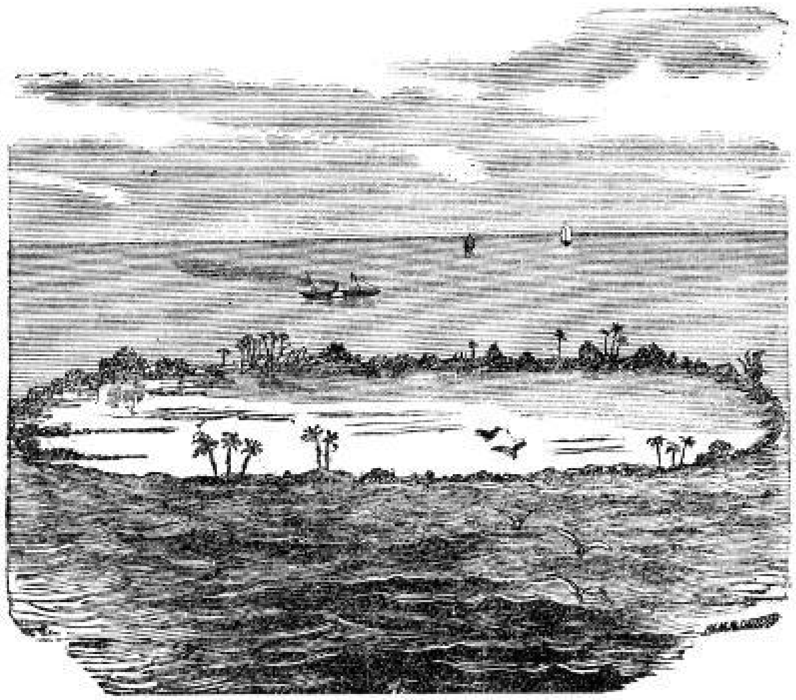
Madrepora (spotted pores) *coral* is neatly branched, the branches having pointed extremities ending in single minute cells.

Porites, or sponge coral, is also branched, but the branches are not pointed, and the surface smoother.

Tubipora, or organ pipe coral, shows some very striking peculiarities. A section of the vast structures built by them resembles a collection of regular, smooth, red colored pipes, firmly bound together by cross sections.

Coral rocks are of slow formation, but have attained prodigious dimensions; whole islands are of coral origin, and in some seas the concealed rocks make navigation dangerous. The reefs are often 2,000 feet thick, though it is estimated that not more than five feet are added in a thousand years. The little architects were at work early.

Corallium rubrum, or red coral, much sought after and precious, is shrub-like, of fine texture, and a beautiful crimson color. In a living state its branches are said to be covered over with bright polyps, and the skeleton receives a very fine polish. It is used for ornaments. Professor Dana says: “Some species grow in large leaves rolled round each other, like an open cabbage, and another foliated kind consists of leaves more crisped, and of more delicate structure. ‘Lettuce coral’ would be a significant name; each leaf has its surface covered with polyp flowers. The clustered leaves of acanthus and oak are at once called to mind by this species.”



CORAL ISLAND.

The Ctenophora (comb-bearing) are considered the highest of the Cœlenterata, having a more complex digestive apparatus, and better developed nervous system. Their long tentacles, and comb-like cilia are used for swimming.

SUB-KINGDOM IV.

Echinodermata (spiny skinned) have all their parts symmetrically arranged about a central axis, a contractile heart, good digestive organs, and a peculiar system of radiating canals extending through the organism. They are a numerous family of exclusively marine animals, and their characteristics furnish an interesting study. Most naturalists mention four classes.

Class I.—*Crinoidea*, or sea lilies, now nearly extinct, are fixed to rocks, or the sea bottom, by what seems simply a stem, but is the body of the animal. At the top is the mouth, resembling an expanding bud or flower that opens upward, surrounded by long tentacles, or arms, not unlike the sea anemone. It is supported by a calcareous skeleton consisting of many members, stiff-jointed, crossing and interlacing with one another. When living, the gelatinous animal partly envelops this framework.

CLASS II.—*Asteroidea*, or star fish, have a flat disc, five or more radiating arms extending some distance from the body at the center, and containing a part of the viscera. The mouth is where the arms meet, and opens downward. The upper surface is studded over with rough knobs, between which are the openings of many little tubes, for the passage of water into and out of the body. The round mouth is very dilatable, enabling it to receive large and solid objects for its food, which there is no attempt to masticate. After the digestive organ is somehow wrapped around the shell fish on which it feeds, it is held in its firm embrace till the nutritive portion is disposed of, and then thrown out. They are voracious eaters, devouring all kinds of garbage that would otherwise accumulate along the shores, valuable as sea scavengers, though destructive of living crustaceans and shell fish.

CLASS III.—*Echinoidea* (hedgehog-like) are covered with spines which they move either by the enveloping membrane or by small muscles properly situated for the purpose. The thin, horny, and, when dry, very light skin is peculiar, in that it is composed of regularly shaped, pentagonal plates, arranged in radiating zones, every alternate plate perforated with small holes; and among the spines, but capable of extending beyond them, are little arms, provided at the end with forceps, probably for seizing their prey, or for ridding themselves of troublesome parasites. These are also used for locomotion. They are less active than some others of the family, live near the shore among rocks, or under the seaweed, feed on crabs, and are oviparous.[3]

CLASS IV.—*Holothuroidea* (whole mouthed). They are elongated, like a cucumber, and the head end terminates abruptly, the mouth being a circular opening surrounded with feathery tentacles. They have remarkable muscular power, by which they can disgorge the contents of the stomach, throw off their tentacles, and even eject most of their internal organs, and survive the loss, afterward producing others, perhaps more satisfactory or in a healthier condition. In tropical climates, they have been likened to the sea urchin, without a shell, but are proportionally longer, and their axis horizontal.

SUB-KINGDOM V.

Vermes (worms). Animals having head and tail composed of segments. The digestive organ is tubular, and the nervous system a double chain of ganglia[4] on the ventral[5] surface. There are six classes of vermes. The animals differ greatly in appearance.

Class I.—*Flat worms* are best known as the parasites that infest animals, such as the liver fluke of the sheep, and the tapeworm. The flat worms pass through a very peculiar metamorphosis, some varieties taking as many as seven different forms.

Class II.—*Round or Thread Worms* are represented by the pin worm and *Trichina*. The latter is the dangerous worm which finds its way into the human system from pork flesh, in which it is imbedded.

Class III.—*Wheel Animalculæ*, or *Rotifera*. A most interesting microscopic worm, abounding in fresh water and in the ocean. They will remain dried up for years, and then recover life. Their shapes are very peculiar.

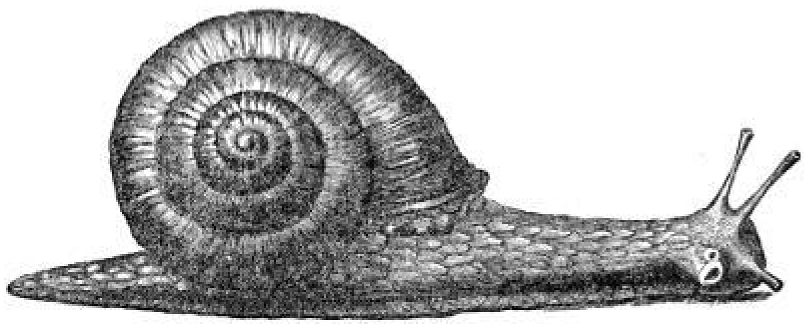
Class IV.—*Moss Animals*, or *Polyzoa*, are the animals which form a coral-like shell. They are abundant on the seashore, and are called sea mosses.

Class V.—*Lamp Shells* (*Brachiopoda*). These worms are marine, and form a bivalve shell, the valves being on either side of the body. The body has long arms on one side of the mouth, which bear fringes; the motion of the fringes draws food into the mouth. They are also used in respiration. But few species of the Brachiopods are now living, though they were once very plenty.

Class VI.—*Annelidæ*. This last class includes the leeches, a flat worm, whose body is divided into segments; the earth or angleworm, a familiar worm of many segments, and the marine worms. Each segment of the latter bears clusters of bristles, used in swimming.

SUB-KINGDOM VI.

Mollusca (soft). General characteristics—Mollusks, a numerous branch of the animal kingdom, are so called from the softness of their bodies, which usually have no internal skeleton or framework to support them. They are covered with a tough, muscular skin, and generally protected by a shell. They have a gangliated nervous system, in some cases well developed, the medullary[6] mass not enclosed in a cranium or spinal column—of which they are mostly destitute—but distributed more or less irregularly through the body. They have hearts, and an imperfect circulative system, the blood being pale or blueish. Some breathe in air, some in fresh, more in sea water. Some—both of those on land and those in water—are naked, others have a calcareous covering or shell. The larger marine mollusks are often guarded by very strong, heavy shells. Some are viviparous, others oviparous, and multiply rapidly.



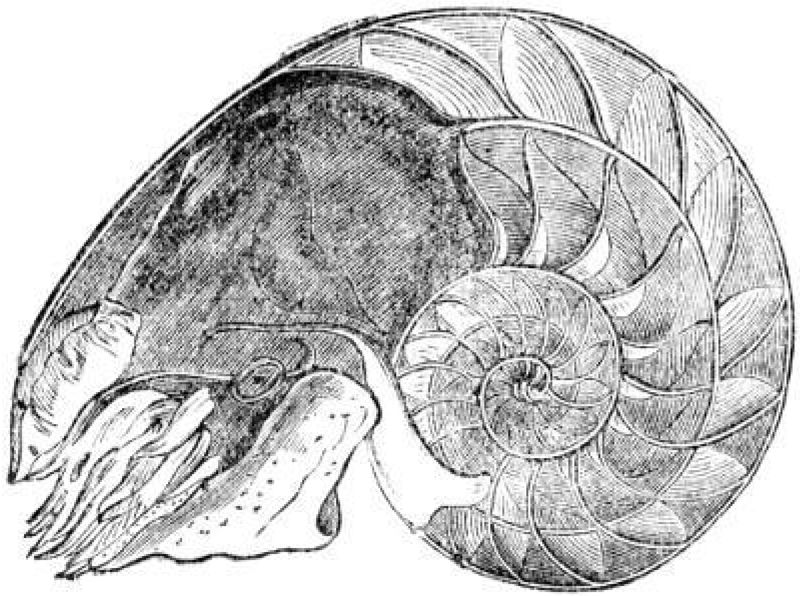
SNAIL.

Three general, and many subdivisions are recognized. The total number of living species is said to exceed twenty thousand, of which only a few can be mentioned. The classes under this division are *Lamellibranchiata*, *Gasteropoda*, and *Cephalopoda*. The chief representatives of the first class are all ordinary bivalves.

Ostrea (oysters) are well known bivalve mollusks. The shells are so irregular in surface and shape that it would be impossible, as it is unnecessary, to describe them. The animal itself is very simple in structure, proverbially stupid, low in the scale of animal life, but highly esteemed as a delicious article of food. They are found in almost all seas, and in water of from two to six fathoms, but never very far from some shore. They multiply so rapidly that, though the consumption increases with the increase of population and the facilities for distributing them, the supply is equal to the demand. Boston is mostly supplied from artificial beds, and the flats or shallow bays in the vicinity of our maritime cities abound in such “plants.” Baltimore and New York have each an immense local trade, and the oysters exported from the Chesapeake Bay fisheries amount to about $25,000,000 annually.

CLASS II.—*Gasteropoda* (stomach-footed). This class, including the great snail family, is a very large division of terrestrial, air-breathing mollusks. Their light shells vary much in form; when spiral and fully developed they have as many as five or six whorls, symmetrically arranged. The shell can contain the whole body, but the animal often partially crawls out and carries the shell on its back. The locomotion is slow and accomplished by the contractile muscle of the ventral foot.

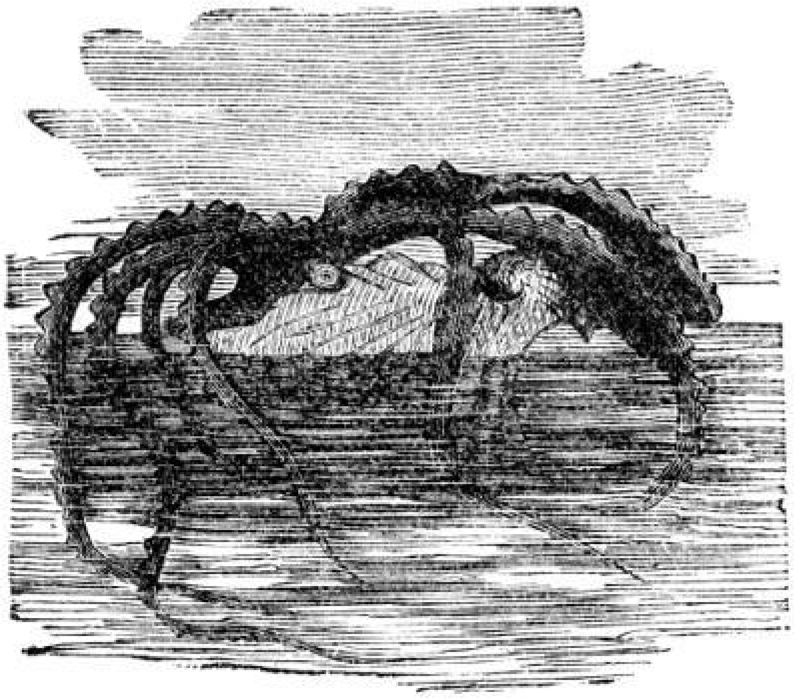
CLASS III.—*The Cephalopoda* (head-footed) have distinctly formed heads, large, staring eyes, mouths surrounded with tentacles or feelers, symmetrically formed bodies wrapped in a muscular covering; they have all the five senses, and are carnivorous. This class is entirely marine, and breathes through gills on the side of the body. The naked *Cephalopoda* are numerous, and furnish food for many other animals. Those living in chambered shells, once numerous, are now known principally by their fossils, the pearly *Nautilus* (sailor) being their only living representative. This has a smooth, pearly shell, and is much prized as an ornament. It is a native of the Indian Ocean, dwelling in the deep places of the sea, and at the bottom. The shell is too well known to need description, and but few specimens of the living animal have been obtained.



NAUTILUS.

The following facts as to the physical organization and habits of this interesting species of univalves are gleaned from the American Cyclopædia, and given for those who have not access to any extensive work on the subject; they were mostly copied for the Cyclopædia from Professor Owen's celebrated memoir, and the "Proceedings of the Linnæan Society of London:"

The posterior portion of the body containing the viscera is soft, smooth and adapted to the anterior chamber of the shell; the anterior portion is muscular, including the organs of sense and locomotion, and can be drawn within the shell. The mantle is very thin behind, and prolonged through the calcareous tube of the occupied chamber as a membranous siphon, and through all the divisions of the shell to the central nucleus; on the upper part of the head is a broad, triangular, muscular hood, protecting the head when retracted and used as a foot for creeping at the bottom of the sea, with the shell uppermost. … The mouth has two horny mandibles, like the beak of a parrot reversed, the lower overlapping the upper, moving vertically, and implanted in thick, muscular walls. … There are ninety tentacles about the labial processes and head. The internal cartilaginous skeleton is confined to the lower surface of the head, a part of the cephalic nervous system being protected in a groove on its upper surface, and the two great muscles which fasten the body to the shell are attached to it. The funnel is very muscular and is the principal organ of free locomotion, the animal being propelled backward by the reaction of the ejected respiratory current against the water before it. … The nautilus, though the lowest of the Cephalopods, offers a nearer approach to the vertebrate animals than does any other invertebrate, in the perfect symmetry of the organs, the larger proportion of muscle, the increased bulk and concentration of the nervous centers in and near the head, and in the cartilaginous cephalic skeleton. The mature nautilus occupies but a small part of the shell, the parts progressively vacated during its growth are one after another partitioned off by their smooth plates into air tight chambers, the plates growing from the circumference toward the center, and pierced by the membranous siphon. The young animal before the formation of these chambers can not rise from the bottom of the sea, but the older ones come to the surface by the expansion and protrusion of their bodies producing a slight vacuum in the posterior part of the chamber unoccupied, and, some say, by the exhalation of some light gas into the other deserted chambers. They rise in the water as a balloon does in the air, because lighter than the element surrounding them. They float on the surface with the shell upward, and sink quickly by reversing it. By a nice adjustment, in the completed structure, between the air chambers and the dwelling chamber, the house and its inhabitant are nearly of the same specific gravity as water. … In parts of the Southern Pacific, at certain seasons of the year, fleets of these little ships are carried by the winds and currents to the island shores, where they are captured and used for food.



ARGONAUT.

The Paper Nautilus, or argonaut, secretes a thin, unchambered shell in which its eggs are carried; has tentacles or arms with which it crawls on the bottom, and swims backward, usually with the back down, squirting water through its breathing funnel. The argonaut differs from the true nautilus in having larger arms of more complicated structure, with sucker discs, and partially connected by a membrane at the base. It has an ink gland and sac, for its secretion. It has a great number of little cells, containing pigment matter of different colors, whose contractions and expansions give it a remarkable power of rapidly changing its tints. There is no internal shell, and it is ascertained that the external shell is peculiar to the female, and is only an incubating and protective nest for the eggs. The eggs are attached to the involuted spire of the shell, behind and beneath the body of the female. From the fact that the animal has no muscular or other attachment to its shell, and has been known, after quitting it, to survive sometime without attempting to return, the argonaut has been supposed to be a parasitic occupant of the cast off shell of another, but is now pretty clearly proved to be the architect of its own shell, which it also repairs when broken by the agency of its palmated arms. It is said the argonaut rises to the surface, with the shell upward, turning it downward when it floats on the water; by drawing the six arms within the shell and placing the palmated ones on the outside, it can quickly sink. This explains why the animal is so seldom taken with the shell. The shell is flexible when in water, but very fragile when dry. The largest known specimen is in the collection of the Boston Society of Natural History; it is 10 by 6½ inches. For a full account of this animal see “Proceedings of the Boston Society of Natural History,” vol. v, pp. 369-381.

End of Required Reading for April.

JERRY McAULEY AND HIS WORK.

BY COLEMAN E. BISHOP.

Extremes of life meet in a great city. “Man, the pendulum betwixt a smile and tear,” here swings between the utmost extremes of squalor and splendor, misery and enjoyment, sin and virtue. That conflict between good and evil—old as the human soul, its arena—is waged nowhere as fiercely as here, where life is intense and assailable souls are legion. You have to go but a little below the surface of city life to find a worse than Dante’s “Inferno;” and if that were the whole story, if there were no compensating charities, one would feel it a mercy to call down on the city the desolation, and the peace of Sodom.

But, thank God! there *are* those redeeming, reforming influences to give one new hope for civilization, new faith in humanity, or new faith in divine grace. Its missions and charities are the sunny side of New York. There are over one hundred and thirty established missions in the city, with a million and a half of dollars permanently invested, beside the other millions required to support them; and the eleemosynary and relief institutions of New York outshine the charities of all other cities, proportioned to numbers. Some liberal, devout souls seem to be looking after every conceivable phase of suffering and sin, and if the devil seems generally to be getting the advantage, let us believe that it is because his antagonists are not more numerous, rather than because he is any smarter or attends more strictly to business. Indeed, the ingenuity of some of the foes of sin, might put to confusion the proverbial originality of the great adversary. Of these exceptional efforts, perhaps there is none more unique than the work of the late Jerry McAuley, nor one that has wrought so great results with so little human aid; nor one to which the Christian believer can point as a testimony of divine agency with greater confidence.

Yet, a thoughtful study of the man and his work will reveal the rationale of it, and help us to understand why it took hold of a certain class in the way it did; that is, why he proved so exact a means to that exact end. The characteristics and training that made Jerry McAuley a successful criminal made him, when his nature and purpose had been transformed, a successful missionary. The son of a counterfeiter, he was educated in the worst streets and dens, graduated a tippler and petty thief in his boyhood, and took his degrees of gambler, drunkard, burglar and wharf rat; and at the age of nineteen was convicted and sentenced to fifteen years' imprisonment. Short and inglorious career of sin! To be followed by a long and glorious one of righteousness, and crowned at last with a triumphant death in faith! Here for seven years he hardened under perfunctory gospel ministrations and prison discipline, until at the right time Orville Gardner, known as "Awful Gardner, the Reformed Prize-fighter," found Jerry and led him to that change which he always called his "transformation." He was pardoned out, only to meet the killing, chilling reception that society gives to one who has passed the bars. Now followed seven years of struggles for an honest life. Only his soul and its Maker know what these were. At one time he relapsed into his old ways, but he was sought and reclaimed by agents of the Howard ("Five Points") Mission. Strange, is it not, that good people are so much more alert to recall the fallen than to aid the struggling and keep the rescued secure? Why is it that interest in the unfortunate is deferred till interest seems useless?

Jerry now conceived the plan of a mission to people of his own old life, men and women tempted in all points like unto himself. He applied for advice and help to one or two clergymen and some wealthy church members, only to meet with mortifying coldness or refusal. We can readily understand this caution, considering McAuley's antecedents and qualifications. He could hardly read and write his own name. Churches are so fiercely criticised that they have to be very chary about espousing unpromising enterprises. It was a *natural* caution if not a *Christian* charity; worldly if not spiritual wisdom. Besides, are there not still things that He hides from the wise and prudent, and reveals unto babes? At length, McAuley found men able and willing to help, and with their aid he opened his Water Street Mission, in 1872. This situation is one of the worst in the lower part of the city, in the "Bloody Eighth Ward," the haunt of river thieves, sailors and abandoned women of the lowest degree. It was a "rum start," indeed, to the denizens of Water Street when one of their leaders was graduated from prison to prayer meeting; and by scores they “came to scoff, and remained to pray.” This mission was a success from the start, and to-day remains in full tide of beneficent operation.

Two years ago Jerry was able to carry out his long-entertained desire to “do something for up-town sinners” by the establishment of the “Cremorne Mission.” It was a more daring undertaking because it had to do with more respectable sinners. He said it was a mistake to speak of those in the gutter as “hard cases;” their pride and reliance are gone, and they are not likely to be affronted or resistant when told they need a Savior; the prosperous and successful are the hard hearted; as you ascend the scale in means, intelligence and pretension, the harder the sinner becomes.

On West Thirty-Second Street, just off Third Avenue, was the infamous Cremorne Gardens, one of the most dangerous because *not* one of the lowest resorts of abandoned men and women. In this vicinity are many houses of ill repute of the higher rank; gamblers and sporting men have their “runways” in that part of the city. Jerry “carried the war into Africa” by leasing a building next the “Cremorne Garden,” so that in all respects the sad satire of DeFoe was and is reversed,

“Wherever God erects a house of prayer
The Devil always builds a chapel there;
And ’twill be found upon examination
The latter has the larger congregation.”

The passenger by the elevated railroad, or one of the several street car lines that converge at the intersection of Sixth Avenue and Broadway, may see any evening a brilliant prismatic sign of “Cremorne Garden,” and seemingly just above it this more brilliant legend:

JERRY MCAULEY’S CREMORNE MISSION.

He will reach the Mission first, as he approaches this strange conjunction of stars; Jerry said he wanted the Lord to have the first chance at a sinner when he could.

The doors are open night and day, and some one is always there to welcome the visitor. A “protracted meeting” is held here all the year, and all the years, around. Going in, you stand in a long, narrow hall, with high ceilings modestly decorated; an aisle down the middle flanked by rows of oaken settees terminates at a low platform on which are chairs, a cheap desk, and a grand piano. The place seats five or six hundred. The hall is brilliant with electric lights, and the walls are illustrious with such Scripture quotations as, in the words of one of the converts, have been found most apt to “fetch ’em”—*i. e.*, sinners. By the platform are conspicuous notices that speeches are limited to one minute each, a rule that is easily enforced in the case of the converts, because they have only facts to tell, and do not seem to be in love with that sweetest music on earth, the sound of their own voices. “One minute each,” Jerry would say: “Don’t be too long; cut off both ends and give us the middle; you need not get mad, as some people have done, if I ring this bell.” “All right,” replied one easy speaker; “If I get long-winded pull me down by my coat tails;” whereat all laughed heartily, as they frequently do.

Jerry McAuley’s method, first and last, was unique, but simple and very effective. Testimonies are the great reliance. They teach salvation by object lessons, prove the truth of conversion by concrete examples. There is no argument, no exhortation, no didactics, no theological disquisition. What need of these in the presence of these living examples? A man stands up and says: "For twenty years I was a common drunkard and thief. Five years ago I found Christ here. I have not touched a drop nor stolen a thing since." McAuley was won by *proof*. He said: "It was a testimony that brought me. I was 'sent up' for fifteen years and six months; I listened to preaching there for over seven years, but I was still unmoved. Then a man came along who gave his experience. He had been a wicked man. That man told just my history; but he was saved, and since there was hope for a desperate man like him, I knew there was hope for me. And there was! Now you have heard the biggest debtor to grace that is in the room, let the next heaviest debtor follow me." Others were won by the undeniable transformation of Jerry. These things were irresistible. Men describe the effect on themselves of these living witnesses:

> "The Bible reading and the prayer did not have any effect upon me, but the testimonies of some who had been, as I could not help admitting to myself, just as I then was, did affect me. I felt that I was in the same boat as they had been in. The conviction of my state forced itself upon me, whether I wanted to think of it or not."
>
> "I came to this meeting three weeks ago. I was drunk when I came in here, but drunk as I was, those testimonies, such as you have heard to-night, reached me, and I went forward to those chairs. There I gave my heart to Christ after serving the devil forty-seven years."
>
> "When I first heard the testimonies here I thought those who spoke had great impudence to tell all about their past lives, but by and by I felt that they were describing my case. Then, as they told how Jesus saved them, I felt I needed to be saved."
>
> "As testimony after testimony was given I would say to myself: that's me, that's me."
>
> Another said that the prayer and Bible reading did not affect him, but the testimonies were "like shot after shot fired at him."

These effects are driven home and clinched by direct personal efforts with penitents, by attentions that follow them to their homes or shops, or into evil haunts, by relief and creature comforts—in a word, by an interest vigilant, ceaseless, and tender as divine love, because inspired by it.

As the method is peculiar, the atmosphere of the meeting is. One familiar with ordinary devotional meetings, and, more, with revival efforts, can not fail to notice here the contrast. Speaking is uniformly in an ordinary tone, and in a conversational, matter-of-fact manner—an effect that is heightened by the use of phrases common in the resorts where some of the converts learned their vernacular. And prayer is specially subdued and low toned—the more impressive and reverential on account of it.

Then, one feels the momentum of the exercises and the tone of cheerfulness and joy that prevail. There is none of that exhortation to “improve the precious time;” none of that dismal bewailing of spiritual barrenness and besetting doubts, fears and temptations, which sometimes make devotional exercises mechanical and dreary, and furnish stumbling-blocks to young believers. These converts do not dwell much on their *enjoyment* of religion; albeit, they do one and all give thanks without ceasing for their deliverance. One notes, too, the absence of cant, of quotations and set phrases; everything is original. There is little exhortation of others. In short, like Bartimeas, they know “Whereas I *was* blind, *now* I see;” and unlike the blind man, they know who worked the miracle.

It was the founder and leader of the two missions who gave all this tone to their services. He was of a medium height and slender, with a heavy, wiry moustache, keen eyes, a nervous temperament, energetic, quick-witted, sympathetic; one readily caught good feeling and confraternity from his presence. He would flash out at a hymn, a text, or a testimony, with a bit of experience. Before two sentences had passed his lips he probably would leap down from his place on the platform, saying, humorously: “I can’t talk up on that stage,” and go down the aisle as if to hold pleasant converse with his audience. It was a strange *melange* of earnestness, experience, humility and wit, with not the least attempt at eloquence, and apparently no study of effects. He describes one case of conversion:

> “This man had come into the meeting with his head about twice the usual size, and his eyes as red as a red-hot poker. You could have squeezed the rum out of him. He asked me to pray with him, but I hadn’t much faith in the man, but that’s just where God disappointed me. I was deceived. The man meant business.”

The missions are supported entirely by voluntary contributions left in the boxes by the door. Sometimes these run low. One night Jerry asked all who were glad to be there to hold up their hands. All hands up. “Now,” said he, “when you take your hands down put them way down—down into your pockets, and fetch something out to put in the boxes by the door.” He said, that to run a mission successfully required “grace, grit and greenbacks.” I fancy considerable of his influence was due to his knowledge of the secret hearts, the personal experiences of his auditors; he was like a priest at confessional, when out of meeting. There are no *verbatim* reports of his talks extant, and if there were they would give the reader no proper idea of the man, because the grotesqueness of his language would probably be the most striking feature of them. He must have been heard to be understood, and even then I think he could not have been understood save by those whose experience and modes of life gave them the touchstone.

Sister Maggie is a typical convert. At one of the Cremorne missions a Sunday-school worker from the East Side told of his class of fifteen street boys and girls. “I asked them,” he said, “how many of them drank beer, and all promptly put up their hands. I asked how many thought it a bad practice, and only four or five responded. I then told them of the evils that drink led to, and cited them numerous examples within their knowledge, and finally asked how many of them would resolve never to drink any more beer. About half of them kept their hands down, and in a way that showed they meant business.”

This discouraging incident brought to her feet, on the platform, a tall, thick-set, strong-featured woman. She spoke with a strident, energetic voice, a Bowery accent, and a manner to which all thought of shrinking or embarrassment was a stranger. It was Sister Maggie. She said, as nearly as I recall the words: “This teaching children beer-drinking is the beginning of all the deviltry. I was passing a dive yesterday and I saw a little kid come out with a pail of beer that she had been sent for, and no sooner was she outside the door than the pail went to her head. That’s the way I used to do, and learned to drink and steal at the same time. But God can help us to reclaim even beings so badly educated. He helped me, and there can’t be a worse case on the East Side. There is not a dive over there that I haven’t been drunk in. Brother S. knows how often I have drank with him at old C.’s dead-house (rum-shop). Sixteen years ago I was a leper, a confirmed sot. If you had seen me you would not have believed that I could have been saved. Christians said I was too far gone; they said there was not enough woman left in me to be saved. I had had the jim-jams twicet. [Laughter.] The first time I had ’em I thought my back hair was full of mice—oh, that was awful! [More laughter.] I was just getting over the tremens when I first come in here. I was a walking rag-shop, and if you’d a seen me you’d give me plenty of room on the walk. I staggered in and sat down on the very backest seat. Now they let me sit up here on the platform. What d’ye think of that? God helped me, and he has helped me now for sixteen years, and I am going through. I am happy. I have friends and good clothes, and more than all, I have *a good home*, and that is what I never knew before.”

At the mention of the word “home,” all the woman’s instinct asserted itself, and for the first time her voice softened, and her manner melted; she sobbed, and sat down.

I can give no complete idea of the effect of this, because the reader can know nothing of the surroundings, the antecedents of the speaker and of many of her listeners, and the keen rapport that ties them together. These worshipers are a class and an organization by themselves; they have no church affiliations, and their worship is *sui generis*; many of them were outcasts of society in former years, they stand alone since their reformation, and they are drawn close together in their isolation. True, there are many among them who were always respectable members of society; many who since their conversion here have joined churches; there are richly dressed and cultured-looking people scattered in these audiences; but the fact remains that the genius and distinctive *personnel* of the meetings are of the order of which Jerry McAuley and “Sister Maggie” and their ministrations are representatives.

I know of no religious exercises better calculated to inspire the true religious feelings of faith, charity, humility, gratitude, and rejoicing—the distinguishing marks of the original Christian following. But they differ from the noisy demonstrations which sometimes are taken for “primitive Christianity,” as they do from the cold and conventional worship which advertises itself in brownstone structures and double-barred mahogany pews. If one wants to get a breath of vigorous faith and wholesome humility, he should attend the Sunday evening services at one of the McAuley Missions.

Jerry McAuley died suddenly, but not unexpectedly, last September. His funeral, held at the great Broadway Tabernacle, was one of the largest ever seen in this city, and was attended by hundreds of abandoned characters who had been reclaimed through his instrumentality, and who were probably never inside of a church before, and may never be again. Women with painted faces, but with tears in their eyes and bits of crape fastened at their throats or arms, stood with downcast heads beside other women who, under other circumstances, would have shunned them. Thus did all classes testify to the power of simple faith and devotion in a poor, uncultured outcast. Over the platform in his chapels are his last, characteristic words:

“IT’S ALL RIGHT.”

“The workman dies, but the work goes on.” There is no calculating the power and extent of the influences this humble worker has set in motion. Beside the hundreds of living examples of his labor here, the seed has scattered to the four winds of heaven, and sprung up in various forms to bless the world—in other cities and other lands. The Missions publish *Jerry McAuley’s Newspaper*, which, extensively circulated, especially in prisons and “the slums” of cities, carries the glad tidings of the testimonies to do a silent and unknown work. An affecting feature of the private work of these converts is their efforts to hunt out and reclaim missing boys and girls. Letters are received from agonized parents, from distant points as well as the city, imploring the help of missionaries to find these estrays; and their efforts are often successful.

I close with one example of this radiating, ramifying influence: Michael Dunn was an English thief by inheritance, for his parents were thieves, and as he expresses it, “he had thieving on the brain.” He had spent the greater part of thirty-five years in different prisons, and continued the same life after coming to this country. One evening an unconverted man sent him into the McAuley Mission as a joke, but it proved to Dunn a blessing, for he found a Savior, and the desire for stealing was all cast out. He was moved to undertake that most important work, the provision of a home for refuge and work for his brother ex-convicts. After many trials and difficulties he finally established the Home of Industry, No. 40 Houston Street, New York, where many lost men who were a terror to society, have been made honest Christian citizens, and are working to save others. Nor did the work stop here. In the autumn of 1883, Dunn was called to San Francisco, California, to open a similar Home in that city, where his labors are as successful as they were in New York.

EIN FESTE BURG IST UNSER GOTT.

TRANSLATION OF LUTHER’S FAMOUS HYMN.

Our God's a fastness sure indeed,
A trusty shield and weapon;
He helps us free in every need
That unto us may happen.
The old wicked foe
Now in earnest doth go,
Deep wiles and great might
In his fell store unite,—
The earth holds not his fellow.

By strength of ours is nothing done,
Full soon are we dejected!
But on our side's a champion
By God himself elected.
And who may that be?
Christ Jesus is he,
The Lord God of Hosts!
All gods else are vain boasts,
Our camp is in his keeping.

Though demons rage both far and near
And gape our souls to swallow;
Not all too great shall be our fear;
Success our steps shall follow.
The prince of this world,
Though threats he hath hurled,
To us can do nought,
For if to judgment brought
One word declares his sentence.

To let the word stand they are fain,
And small thereby their merit;
He dwells among us on the plain
With gifts and with his spirit.
What though they take life,
Goods, name, child, and wife,
We need not rebel—

No profit those to hell,
While ours must be the kingdom.

THE WEATHER BUREAU.

BY OLIVER W. LONGAN,
Of the War Department.

In an article on the “War Department” in The Chautauquan for December, mention was made of the weather observations by the Signal Corps of the army. This novel service—novel both in its character and in its assignment to a military department—was commenced in 1870, under a resolution of Congress, approved February 9th of that year, which required the Secretary of War “To provide for taking meteorological observations at the military stations in the interior of the continent, and at other points in the states and territories of the United States, and for giving notice on the northern lakes and on the sea coast, by magnetic telegraph and marine signals, of the approach and force of storms;” and in June, 1872, the provisions of the service were extended to include “the agricultural and commercial interests” of the country. The plan of organization and the superintendence of the service were imposed upon the chief signal officer of the army, then General Albert J. Myer,[A] to whose memory the signal and the weather services are living monuments.

If failures are of value in guiding succeeding attempts in the same line, General Myer had the advantage of a number, both in this country and Europe, but attributing those failures to a want of proper agents rather than to mistakes of method, and being thoroughly imbued with the idea that efficient service from a body of men employed in the same enterprise can be obtained only by enrolling them under the oath of enlistment as subjects of military discipline, he could adopt all the methods of operation which had already been tried by others, and, uniting with them his own, could undertake the work with a confidence in men as much as in measures, and make sure progress over the same road that had been too difficult for others to travel. The signal service, which had been organized by him as a special and distinct department of the army, was well prepared to operate the most important agent in the work, the magnetic telegraph, and it has constructed its lines over a great extent of country not yet reached by civilization, connecting frontier military posts with each other and with the lines owned and operated by private companies.

The office division first established under the law of Congress, which has been mentioned, was called by the appropriate but too extensive name of “Division of Telegrams and Reports for the Benefit of Commerce,” but great works which have been proved and not found wanting may, like great men, afford to adopt simpler titles without loss of favor, or if they will not do so voluntarily, such titles will be fashioned for them and fastened to them by the people, who have not time to regard the official proprieties that would hold them off at a respectful distance by an appearance of gravity of demeanor or by an impressive name, so the office has come to be familiarly known as the “Weather Bureau,” and the officials in charge have accepted the designation without objection. It detracts nothing from the appreciation of the work they accomplish in giving information—premonitory—of wind and rain, heat and cold, frost and snow, river flood and ocean tide, and much more of interest and value day by day—yes, and night by night as well—for the inhabitants of this continent, and in adding to the knowledge of meteorology which is eagerly sought by the scientists of the world.

The service contemplated by Congress was at first intended—if we may judge by the language of the law—only to benefit persons interested in the commerce upon the great lakes and the ocean. Then the agriculturists were permitted to take share in the advantage afforded by a prevision of the weather. But we are all too greatly interested personally in the kind of weather expected to-morrow or the next day not to seek to profit by the work done for those engaged in special business when there may be great gain for us as individuals without robbery of their peculiar rights. Our interest moves us to speech almost unconsciously, as we meet our friends by the way and tell them, what they already know as well as we, about the kind of weather then prevailing, or express our hopes or fears for what soon will be. Work and play are sources of profit and pleasure, according to the influences of the weather, and the signs of olden time are numberless, to which we give our confidence, whether they come from the beasts of the field, or the fowls of the air, or the fishes of the sea, or are indications based upon a correct philosophy, well known but not well understood. The masses of the people will not give up their attachment to these signs, nor for the old-fashioned almanac which their fathers consulted for their weather predictions, but now that they have the aid of a great government institution conducted by men who study the weather as a science, and who have patiently held out to them the benefit of their investigations until the good natured skepticism and the raillery of the multitude who dubbed the genius of the weather service "Old Probabilities," has all been banished and has given place to full faith and credit, they ought to acknowledge, and no doubt do, that their personal wants in this respect have been recognized, and they will take interest in the methods by which they are met and satisfied.

Every feature of the signal service has been brought into requisition and use for the work of weather observations and storm warnings, and the original bureau seems to have been so wholly absorbed in the new one that the corps will be known, except perhaps in official circles, only by its operations in this special field, until war shall call for the more frequent use of the flags and rockets and lights, by the aid of which a great part of the rapid communication pertaining to the business of warfare is carried on. The *personnel* of the corps comprises a chief signal officer (brigadier-general), twelve second lieutenants, one hundred and fifty sergeants, thirty corporals, and three hundred and twenty privates. Prior to 1878 there were no signal officers in the regular establishment except the chief, but in that year authority was given to appoint two second lieutenants annually from the sergeants, and the selection for these appointments is made by competitive examination. Officers are assigned to duty with the corps by detail from the regiments of the army, and after a course of instruction return to their proper stations, and are succeeded by others. A few civilians are employed in the office in Washington city, in various capacities, including that of professional scientist and instructor, but the great work of observation is done by the army force, so called because every member of it is ready at any moment to lay aside his special duties and take up arms for any emergency. The pay of the officers is that of their grade in the army: $5,500 per annum for a brigadier-general; $1,400 for a second lieutenant, with an increase of ten per centum after each five years of service, until the increase reaches forty per centum (after twenty years' service), when no more is added. (This increase, called longevity pay, is not given to any officer above the rank of colonel.) As the government is supposed to furnish a habitation of a certain number of rooms for each commissioned officer, there is quite an augmentation of the pay when the duty requires a station where there are no public quarters, as is the case in most of the service for the weather bureau, and commutation is paid. The pay of the enlisted men varies according to their rank and station of duty, but is based upon the army pay table. The pay of a sergeant, including all allowances, averages monthly within a few cents of the following amounts: At a military post where quarters and rations are provided in kind, $40; at an observing station, $80; at the office in Washington city, $100; that of a corporal $25, $65, and $85; that of a first class private $22, $62 and $82, and a decrease of four dollars for a second class private, the respective stations being those mentioned for the sergeant. The great difference in the pay of the same man at different places is made by the "extra duty pay," and commutation of allowances (rations, quarters, fuel and clothing) when they can not be furnished in kind. The entrance into the service is by an enlistment of five years. Every man must pass a rigid examination into his physical and educational qualifications. The service presents advantages not found in any other branch of the army. The inducements attract a well educated class of men, many of them graduates of colleges, who find in the scientific part of the work at least, a congenial pursuit.

The school of instruction is at Fort Myer (formerly Fort Whipple), a military station in Virginia, on the bank of the Potomac river, nearly opposite Washington city. The course embraces the drill and discipline of the soldier, the code of signals, the construction and operation of telegraph lines, the use of meteorological instruments, and the method of taking observations. The central office in Washington occupies an ordinary looking brick building of uncertain age, half a square west of the War Department building. It was originally two two-story dwelling houses, but has received an additional story and "Mansard" roof, and has been fitted for its present use. The plain and dingy exterior escapes critical notice by the display of the mysterious looking machinery and fixtures upon the roof. An immense arrow (anemoscope), with broad feather-tip end, turns its point ever "in the wind's eye," and is imitated by two smaller ones at a lower elevation. Several sets of spinning instruments of various sizes stand in different places. These are the anemometers for measuring the velocity of the wind. The part of each visible from the ground is simply a vertical rod with four branches on the top, each branch having a hemispherical shaped cup upon its outer end, so placed as to catch the wind in its convex side. The force of the wind gives the cups a rotary motion in a horizontal plane, which causes the vertical rod to revolve and record by connecting mechanism and dials the velocity of the wind in miles per hour. Near the center of the roof may be seen a vessel with a funnel shaped top into which the rain falls and is measured to ascertain the depth in inches, of which accurate record is kept. Close by is a framework supporting a small cage-like structure with lattice sides and tight roof, within which are hung the thermometers, barometers, and other instruments consulted regularly at intervals, to ascertain the temperature, pressure, and humidity of the atmosphere. These fixtures and instruments are used for the purpose of obtaining indications of the weather, and for the instruction of "Observers," and serve as well for standards by which instruments to be sent to other places may be tested. Only a portion, however, of the business of the office is done in the building mentioned, a number of others in the vicinity being occupied for the several departments of the work.

The central office is connected by telegraph with stations in the principal cities of the country, at the sea and lake ports, and at points along the Atlantic and gulf coasts, uniting with stations of the United States Life Saving Service, for it is there that the work of greatest value, the saving of human life, is done. The number of stations is limited only by the amount of money provided by Congress for their maintenance. The top of Pike's Peak and the summit of Mount Washington both contribute their share of upper air phenomena, and the lowest valleys in the interior give up their secrets of the atmosphere at its greatest depth. What an ocean it is, with its surface which we are told is a certain number of miles above us, and its bottom, which we know is under our feet, its shoreless currents, some as gentle as the breath of an infant, others more fierce and swift than the whirlpool rapids of Niagara. To learn of these currents their force and direction, whence they come and whither they go, the laws to which they are subject, has been the fascinating study of amateur meteorologists and the pursuit of men of professional attainments in science for many years. But their discoveries were of little practical value before Professor Morse, on a May day forty years ago, sent over a telegraph wire between Washington and Baltimore, his first message in the form of a question which since has been answered in wonders by the same agent then employed, and the same now used to send warning of the coming storm, whose swift wings have no other rival.

The weather stations are distinguished under a classification made by a special service performed at each, and are known as telegraph, printing, display, special river, cotton region, and sunset stations. A number of them may have all the special features indicated by the different names, while others may have but the one feature which gives it its place in the class. The last mentioned are so called, not, as the name may possibly suggest, because they are located away off toward sundown, but because of the special observation taken at the time of sunset which affords so good an indication of the probable weather for to-morrow. It recalls the fact that the Jews were reminded more than eighteen hundred years ago of their habit of observation in this particular, which was a rebuke for their failure to read "the signs of the times." A great number of reports are received at the Weather Bureau weekly and monthly, by mail from volunteer observers, from medical officers at military posts, from agricultural and scientific societies, and at regular times from meteorological societies in the old world, but as none but the telegraphic reports have more than a relative connection with the work of weather indications for which we are looking day after day, it is sufficient for the purposes of this article to simply mention the fact that there are several hundred of these mail reports from which record of permanent and daily increasing value is made for study and information of the climate in various sections of the country.

The station service necessarily commenced under the disadvantage of having no observers of experience, but there was only one way to get them, and the beginning of the work was the commencement of the education of the men who were to perform it. At this time there is not only the course of instruction at Fort Myer, but men as assistants at stations are perfecting themselves to occupy the places of those who have become masters in the profession, or to take charge of new places. The telegraph stations number about one hundred and fifty, and at each one is an "Observer Sergeant," with one or more assistants. Their equipment in instruments is similar to that which has been mentioned in connection with the central office, viz.: For ascertaining the temperature of the air; the weight or pressure and relative humidity of the atmosphere; the direction, force, and velocity of the wind; and the depth of the rainfall; also at river stations for taking the temperature of the water, and for measuring its rise or fall; and at display stations signal flags and lanterns are included. The observers take the record from their instruments at regular intervals every day and night, Sundays and holidays included. There are three of these observations—taken at 7:00 a. m., 3:00 p. m., and 11:00 p. m., Washington time—telegraphed to the central office. Those taken at other hours are not telegraphed unless called for, but are recorded and enter into the weekly and monthly mail reports. The dispatch is in cipher, which permits the sending of a long message in from five to twelve words, giving pressure, temperature, direction, and velocity of wind, depth of rain or snow fall, appearance and movement of clouds and any special meteorological phenomena present, and adding from river and coast stations the stage of water in the rivers and the ocean swell. By preconcerted arrangements with telegraph companies, the reports pass over the wires without delay or interruption, and all reach the central office within about forty minutes after the observations are taken. They are at once translated and entered upon graphic charts—outline maps of the United States—on which each station is marked by its geographical location. By the use of symbols and figures all the meteorological conditions of each locality are exhibited, and so perfect is the system of arrangement for reporting and drafting that in less than two hours from the time the record was taken by the "Observers," the officer who is to make the weather predictions has all the reports before him in the central office.

The ever-changing conditions of weather are in a manner photographed, and serve as guides for the work which immediately follows the making up of the charts. First the "synopsis" of conditions is made up, then the predictions or "indications" of the kind of weather expected, and the places where storm warnings are to be shown are determined, and in the form of a bulletin they are telegraphed to all parts of the country as the "Press Report," to observing stations to be reproduced and furnished to local papers, posted in public offices and mailed to postmasters for exhibition in their offices, several hundred postoffices in some instances being supplied from one station. They are also placed in railroad stations and distributed from trains at points along their lines. Thus the people are advised of the kind of weather prevailing over the district of country in which they live, and informed of the changes that may be expected within twenty-four or forty-eight hours.

"Observers" at their several stations publish in connection with the "indications" telegraphed from the central office, the conditions prevailing in their own localities, and in large cities where weather maps are hung in the rooms of boards of trade, merchants' exchange, or other important offices, they place or change the symbols used to indicate the conditions at all the stations, as they receive them from reports passing them to the central office or repeated from the latter.

The development and progress of all storms are as clearly delineated upon the charts prepared in the central office as it is possible for sensitive instruments to reveal them, and special attention is given to indications of high winds approaching the coasts. Orders are sent by telegraph to the maritime stations within a region likely to be visited by dangerous winds, and a "cautionary signal"—a square white flag with a square red center by day, or a red-center light by night—is displayed, remains out until notice is received from the central office that the danger has passed. This signal is used as the storm approaches. As the general direction of storms upon the Atlantic coast, or approaching it, is easterly or northeasterly, and the direction of the wind is circular and opposite to the motion of the hands upon a watch, with an inward tendency toward the "storm center," as the storm departs from a station the wind will probably blow from the north or west, and a "cautionary off-shore signal," a square white flag with square black center above the red flag, by day, or a white light above the red by night, may be ordered. Special record of the velocity of the wind is made at the stations when storm signals are ordered, and if it reaches twenty-five miles an hour the display of the signal is regarded as "justified." These signals have become a necessity, and the occasions are exceedingly rare in which a vessel will leave port with one in sight until the captain or master has first made inquiry at the station for "particulars." Coasting vessels are dependent in a great measure upon them; if they pass a station with the storm-signal displayed, they frequently escape encountering destructive gales by putting into the nearest port until the danger has passed.

The coast telegraph lines connecting the signal and the life saving stations have been constructed by the government, and are operated by the signal corps. The weather stations are equipped for making connections with the main line at any point, and many instances may be found recorded in the official reports, of shipwrecks on the coast to which relief has been brought in a very few hours by the prompt action of “Observers” opening telegraphic communication from a point abreast the wreck, direct to the central office in Washington, and sending information to be repeated, with the weight of official authority, to the nearest port from which steamers could be sent to the rescue. A vessel in distress, or in need of any information, if in possession of the international code of marine signals—a number of nations have adopted the American code—may communicate with the shore stations. By this code a number of small flags of various shapes and colors, used singly or together, answer to certain words and sentences, and these being translated into other languages, convey from the American or Englishman to the Frenchman, German, or Spaniard the question or answer desired, as plainly as an affirmative nod of the head from one to the other would mean yes.

The river reports are an important feature of the service. The temperature of the waters, surface and deep, taken regularly, makes a record for the benefit of those engaged in the propagation of food fishes, which is becoming an important government work. The stage of water taken in connection with the reports of rainfall and temperature of the atmosphere in their influence upon deep beds of snow and ice-locked streams affords ground for warnings, when needed, to persons engaged in any river traffic, or exposed to floods upon the banks.

The “waves” of temperature have become as real to us as those upon the ocean, and the prosecution of very many kinds of business, or the transportation of perishable produce is guided by the reports of the Weather Bureau, as it foretells the coming of heat or cold. The interior of the country will no doubt soon have the benefit of signals, as well as “bulletins.” A large, white flag, with a square black center, is now displayed at stations in advance of an expected “cold wave,” and before a great while we may expect the “limited express” upon the different railroads will be made the bearer of signals to forewarn the inhabitants of the country through which it passes of the change of weather rapidly following its track. The possibilities of the service seem to be unlimited, but the most careful watchfulness of the “Observers,” and the keen vigilance of the officers who direct them have not yet brought the elements to reveal all their movements. Sometimes, as a wary foe, a storm will steal in between the sentinels, or descend from the upper air, and gathering all its strength into one narrow channel, will drive destruction through town and country, and leave behind it evidences of power which we can hardly credit, except by sight.

One of the many specimens exhibited in the National Museum at Washington is a section of a young oak tree four inches in diameter, with a pine board, one inch thick, four inches wide at one end, and twelve inches wide at the other, which has been driven through the tree more than half its length (eight feet, the label states), and is now held as in a vise, the tree above and below the board being unbroken. This has been deposited in the Museum as an evidence of the force of the wind in a tornado that visited the vicinity of Wesson, Mississippi, April 22, 1883.

The progress of work in the Weather Bureau has been first toward encompassing great interests in the fields indicated by law, then to take up the smaller needs pertaining to individual benefit and pleasure, and as time passes and the service widens there will be personal contact that will give an intimate knowledge and impression of its value which narrative can not.

FOOTNOTES

[A] Died at Buffalo, New York, August 24, 1880.

HOW TO WIN.

BY FRANCES E. WILLARD,
President National W. C. T. U.

CHAPTER II.

Last month, taking the Past for a background, I tried to picture the opportunity which the Present holds up before the daughters of America. Let me now, for a brief space, coming freshly from the field of active service, where banners wave and squadrons wheel, try to talk about the conditions of success, in this wonderful battle of life. First, then, I would give this not at all startling bit of advice: *Keep to your specialty*; to the doing of the thing that you accomplish with most of satisfaction to yourself, and most of benefit to those about you. Keep to this, whether it is raising turnips or tunes; painting screens or battle pieces; studying political economy or domestic receipts; for, as we read in a great author who has a genius for common sense: “There is not one thing that men ought to do, there is not one thing that ought to be done, which a woman ought not to be encouraged to do, if she has the capacity for doing it. For wherever there is a gift, there is a prophecy pointing to its use, and a silent command of God to use it.” Such utterances as these are assertions of the “natural and inalienable rights” of the individual as such. They are deductions of the Christian philosophy which regards you and me, first and chiefly, as human beings, and makes the greatest possible account of personal identity. In all ages there have been minds that saw this truth. The intellects which towered like Alpine peaks above the mass of men, were the first to reflect its blessed light. Two thousand years ago, Juvenal made the heroine of a famous “Satire” say to the hero: “I like our Latin word for *man*, which equally includes your sex and mine. For you should not forget that, in all things highest, best, and most enduring in our natures, I am as much a man as you are.” The sun of truth looms high above the far horizon in our day, and even the plains of human thought and purpose are glowing with the light of this new inspiration. “Personal value,” “personal development,” these will be the noontide watchwords, “when the race out of childhood has grown.” Only yesterday I heard a fashionable butterfly, in the surroundings of a luxurious home, saying with sudden enthusiasm: “Of one thing I am sure; every woman that lives is bound to find out what is the very best thing she can do with her powers, and then she’s bound to do it.” In creating each of us with some peculiar talent, God has given us each “a call” to some peculiar work. Indeed, the time is almost here when the only call that will be recognized as valid, in any field, must involve in him who thinks he hears it, both adaptation and success. Each one of us is a marvelous bundle of aptitudes and of capacities. But, just as I prefer the active to the passive voice, I prefer to put the aptitudes first in my present inventory. Besides, the world has harangued us women on our capacities, from the beginning, and it is really refreshing to take the dilemma of our destiny by the other horn, at last! Civilization (by which I mean Christianity’s effect on the brains and hands of humanity), wonderfully develops and differentiates our powers.

Among the Modocs there are but four specialties—assigned with remarkable fairness, in the proportion of two for the squaws and two for the braves. The last hunt and fight; the first do the drudgery and bring up the pappooses. Among the Parisians, on the contrary, the division of labor is almost infinite, so that the hand perfectly skilled in the most minute industry (as, for instance, in moulding the shoestrings of a porcelain statuette), needs no other resource to gain a comfortable livelihood. Among the Modocs, skins are about the only article of commerce. Among the Parisians, evolution has gone so far in the direction of separating employments formerly blended, that you can not buy cream and milk in the same shop.

By some unaccountable perversion of good sense, the specialties of human beings who are women, have been strangely circumscribed. But they were *there*, all the same, and now, under the genial sun of a more enlightened era, they are coming airily forth, like singing birds after a thunder storm. And wonderfully do they help some of us to solve the toughest of all problems: *What is life for?*

Let us see. Lift the cover of your sewing basket; there are thimble, scissors, spools of thread, and all the neat outfit needful to a seamstress, but minus the needle they have no explanation and no efficiency. Unlock your writing desk: what are paper, ink, and sealing-wax, without the pen? They are nothing but waste material and toys. So it is with you and me. We have no explanation that is adequate; we have no place in the work-box and portfolio of to-day; no place in the great humming hive of the land we live in, save as some predominating aptitude in each of us explains why we are here, and in what way we are to swell the inspiring song of voluntary toil and beneficent success. Suppose that here, and now, you proceed to take an "inventory of stock," if you have not been thoughtful enough to do that already. Made up as you are, what is your *forte*, your "specialty," your "best hold," as men phrase it? Be sure of one thing, at the outset: The great Artificer, in putting together your individual nature, did not forget this crowning gift, any more than he forgets to add its own peculiar fragrance to the arbutus, or its own song to the lark. It may not lie upon the surface, this choicest of your treasures; diamonds seldom do. Miners lift a great deal of mere dust, before the sparkling jewel they are seeking gladdens the eye. Genius has been often and variously defined. I would call it *an intuition* of one's own best gift. Rosa Bonheur knew hers; Charlotte Cushman recognized hers; George Eliot was not greatly at a loss concerning hers. As for us, of less emphatic individuality, sometimes we wait until a friend's hand leads us up before the mirror of our potential self; sometimes we see it reflected in another's success (as the eaglet, among the flock of geese, first learned that he could fly, when he recognized a mate in the heaven-soaring eagle, whose shadow frightened all the geese away); sometimes we come upon our heritage unwittingly, as Diana found Endymion, but always it is there, be sure of that, and "let no man take thy crown." As iron filings fall into line around a magnet, so make your opportunities cluster close about your magic gift. In a land so generous as ours, this can be done, by every woman who reads these lines. A sharpened perception of their own possibilities is far more needed by "our girls" than better means for education. But how was it in the past? If there is one thought which, for humanity's sake, grieves me as no other can, it is this thought of God's endowment bestowed upon us each, so that we might in some especial manner gladden and bless the world, by bestowing upon it our best; the thought of his patience all through the years, as he has gone on hewing out the myriad souls of a wayward race, that they might be lively stones in the temple of use and of achievement, and side by side with this, the thought of our individual blindness, our failure to discern the riches of brain, heart and hand, with which we were endowed. But most of all, I think about the gentle women who have lived, and died, and made no sign of their best gifts, but whose achievements of voice and pen, of brush and chisel, of noble statesmanship and great-hearted philanthropy, might have blessed and soothed our race through these six thousand years.

There is a stern old gentleman of my acquaintance, who, if he had heard what I have felt called upon to say, would have entered his demurrer, in this fashion: “That’s all fol-de-rol, my friend; a mere rhetorical flourish. If women could have done all this, why didn’t they, pray tell? If it’s in it’s in, and will come out, but what’s wanting can’t be numbered. You can’t pull the wool over my eyes with your vague generalities. I went to the Centennial; I saw Machinery Hall, and what’s more for my argument (and less for yours), I saw the ‘Woman’s Pavilion,’ too.”

He would then proceed to ask me, with some asperity, if I thought that any of my “gentle myriads” could have invented a steam engine? Whereupon I would say to him, what I now say to you, “most assuredly I think so; why not?” And I would ask, in turn, if my old friend had studied history with reference to the principle that, as a rule, human beings do not rise above the standard implied in society’s general estimate of the class to which they belong. Take the nations of Eastern Europe and Western Asia; “civilized” nations, too, be it remembered; study the mechanic of Jerusalem, the merchant of Damascus and Ispahan; in what particular are the tools of the one or the facilities of commerce familiar to the others, superior to those of a thousand years ago? Surely, so far as oriental inventions are concerned, they have changed as little as the methods of the bee or the wing-stroke of the swallow. We hear no more of man’s inventiveness in those countries than of woman’s. Why should we, indeed, when we remember that both are alike untaught in the arts and sciences which form the basis of mechanical invention? They are inspired by no intellectual movement; no demand; no “modern spirit.” It is not “in the air” that *men* shall be fertile of brain and skilled of hand as inventors there, any more than it is here that women shall be, and where both knowledge and incentive are not present, achievement is evermore a minus quantity. None but a heaven-sent genius, stimulated by a love of science, prepared by special education and inspired by the *prestige* of belonging to the dominant sex, ever yet carved types, tamed lightning or imprisoned steam. Besides, in ages past, if some brave soul, man or woman, conscious of splendid powers, strove to bless the world by their free exercise, what dangers were involved! Was it Joan of Arc? the fagot soon became her portion; or Galileo? on came the rack; or Christopher Columbus? the long disdain of courtiers and jealousy of ambitious coadjutors followed him; or Stephenson? his fetter was the menace of the law; or Robert Fulton? he faced the sarcasm of the learned and the merriment of boors. Even for the most adventurous inventors of to-day (as the aeronaut experimenters), what have we but bad puns and insipid conundrums, until he wins, and then ready caps tossed high in air and fame's loud trumpet at his ear—when death's cold finger has closed it up forever.

Times are changing, though. The world grows slowly better and more brotherly. The day is near when women will lack no high incentive to the best results in every branch of intellectual endeavor and skilled workmanship. Not a week passes but from the Patent Office comes some favorable verdict as to woman's inventive power. Wisdom's goddess deems herself no longer compromised because places are assigned us in her banquet hall. "The world is all before us where to choose," and I, for one, appeal from the "Woman's Pavilion" of the first, to that which shall illustrate the second hundred years of this republic.

FORTRESS, PALACE AND PRISON.

BY EDITH SESSIONS TUPPER.

It is believed by many that the Tower of London is, as its name would seem to indicate, but a single lofty pile, while in reality it is a vast collection of grim towers and frowning bastions; a great walled town in the heart of busy London.

The Tower, or the White Tower, built in the time of the Conqueror, is surrounded by twelve smaller towers—Bloody, Bell, Beauchamp, Devereux, Flint, Bowyer, Brick, Martin, Constable, Broadarrow, Salt and Record. In turn, these are environed by the ballium walls and bastions, and the moat guarded by Middle, Byward, St. Thomas, Cradle, Well and Devlin towers.

As one descends Tower Hill, the eye takes in the whole immense and hoary mass, fit emblem of the stormy and tempestuous times in which each separate tower arose. Like black shadows of the past casting their gloom over the present, rise the lofty turrets above the roofs of modern buildings. Sternly they look down upon throbbing London, each with its own history, each with its own awful secrets locked in its stony breast.

Amid the terrific conflict of the days when the Norman was trampling the Saxon under foot, the Tower, the Great Tower, or the White Tower, as it was variously called, arose.

William the Conqueror caused it to be built as a fortress for himself in case his Saxon subjects might rebel against his hard and iron rule. Among the ecclesiastics who possessed the richest sees of those days was Gundulph, bishop of Rochester, who was also a fine military architect. To him the Conqueror gave the commission to build his New Fortress in 1079-80. Gundulph selected the site just without the city then, and to the east, on the northern side of the Thames. The tower is quadrangular in shape, one hundred and sixteen feet from north to south, ninety-six from east to west, ninety-two feet in height, and its external walls are fifteen feet in thickness—an imposing and superb specimen of Norman architecture. It is three stories high, not counting the vaults. There are some slight traces remaining of the grand entrance on the north side, but visitors enter by modern doors on both the north and south sides.

In the northeast corner of the White Tower is a massive staircase, connecting the three stories. The column around which the stairs wind is a remarkable and well preserved specimen of ancient masonry. A wall seven feet in thickness runs north and south, which divides the tower from base to summit. Another wall extending east and west subdivides the southern portion into unequal parts, forming in each story one large and two small rooms. The smallest division on the ground floor is called Queen Elizabeth's Armory, being filled with armor and trappings of her day. On the north side of this room, one is shown a cell formed in the thickness of the wall, ten feet long and eight wide, receiving no light save from the entrance. In this dark and dismal room was imprisoned for twelve long years the gay and brilliant courtier, Sir Walter Raleigh, on suspicion merely of being implicated in a plot to place the Lady Arabella Stuart, the niece of the unfortunate Queen of Scots, on the throne of England. This ill-fated lady also perished in the tower, her reason having been dethroned by her long and cruel captivity. In the year she died, Sir Walter was released and sent to South America to search for gold mines; returning unsuccessful he was remanded to the Tower, and beheaded in 1618 to please the Spaniards. James First wished to gain their favor, as his son Prince Charles was to be married to the Infanta. Raleigh's bravery and valor had been too often directed against the Spaniards for them not to exult over his cruel fate. In this wretched and gloomy cell, it is said, he wrote his "History of the World." In the center of this armory are various instruments of torture; about the room are stands of weapons, halberds, battle axes, maces and bills, and military instruments for cutting the bridles of horses; at the end of the room is a figure on horseback, representing Queen Bess. Her dress is copied from an ancient portrait, and she is attended by officers and pages. Just back of these figures hangs a very old picture of St. Paul's cathedral. But the most terribly fascinating objects in this room are the block, the headsman's hideous, grinning mask and the original axe. With horror the visitor looks upon the block, dented here and there where the executioner's nervous blows struck wide of the mark, and upon the ponderous axe, whose blade has cleft the necks of so many royal and noble victims.

One is glad to leave this chamber of horrors and go above into St. John's Chapel, which is considered one of the finest specimens of Norman architecture left in the kingdom. It terminates in a semi-circle, and the twelve enormous pillars are arranged in similar fashion. These pillars are united by arches which admit the light into the nave from the windows. In the reign of Henry III. three immense windows of stained glass were added to the chapel. It is not known at what time or from what cause it ceased to be used for religious purposes. A large room directly above, on the third floor, was used as a council chamber by the kings, when they held their court in the Tower. It was in this room that the infamous Richard, Duke of Gloucester, ordered Lord Hastings to instant execution in front of St. Peter's Chapel.

This room is now used as a depository for small arms, and the arrangement of weapons in the form of various flowers is wonderful and artistic, the entire ceiling being covered by curious and intricate combinations of these arms.

Encircling the great White Tower, as has been said, is a row of twelve smaller towers. Perhaps the one first in interest is that directly opposite the Traitor's Gate, and rightly named, known as the Bloody Tower. It is rectangular in form, being the only one of that shape in the inner ward. It closely joins Record or Wakefield Tower on the west. Its grand gateway was built in the reign of Edward III., and is the entrance proper to the inner ward. The massive portcullis gives signs of immense age. It was in this tower in 1483, that the most infamous order of the hateful Gloucester, the murder of the innocent princes, the children of Edward IV., was consummated.

"The tyrannous and bloody act is done—
The most arch deed of piteous massacre
That ever yet this land was guilty of."

The little victims are supposed to have been buried at the foot of the staircase in the White Tower, but a strange mystery surrounds their fate.

Joining Bloody Tower is the tower next in size to the Great or White Tower, known as the Record Tower, from its having been for many centuries the depository for the records of the nation, and Wakefield Tower, from the imprisonment there of the Yorkists, after the victory of Margaret of Anjou, the Amazonian queen of the good but weak Henry VI., at Wakefield, in 1460. This victory gave the House of Lancaster the ascendency for a short time. The next year the Yorkists were successful, Henry was remanded to the tower, and was soon after found dead, murdered by Gloucester’s command, it is supposed.

“Within the hollow crown
That rounds the mortal temples of a king
Keeps death his court; and there the antic sits,
Scoffing his state and grinning at his pomp;
Allowing him a breath, a little scene,
To monarchise, be fear’d and kill with looks;
Infusing him with self and vain conceit,
As if this flesh which walls about our life
Were brass impregnable; and humor’d thus,
Comes at the last, and with a little pin
Bores through his castle wall—and farewell, king!”

Wakefield Tower is very ancient, having been built in the time of William Rufus, in 1087.

On the opposite side of the inner ward looms up the gloomy and famous Bowyer Tower, so named from its having been the residence of the Master Provider of the King’s Bows. In a dungeon-like room of this tower, “false, fleeting, perjured Clarence,” younger brother of Edward IV., was drowned in a butt of Malmsey. Together with the detestable Gloucester, he stabbed the young son of Henry VI. in the field at Tewkesbury, but retribution was swift. He soon incurred the displeasure and jealousy of his royal brother, and perished in this wretched manner.

“O Brackenbury, I have done these things
  That now give evidence against my soul—
  For Edward’s sake, and see, how he requites me.”

But a short distance from the Bowyer is the Brick Tower, which acquires a mournful interest from the fact that tradition has assigned this as the prison of the martyr of ambition, the lovely Lady Jane Grey. Fuller says of her that at eighteen she possessed the innocence of childhood, the sedateness of age, the learning of a clerk, and the life of a saint. Gentle, modest and retiring, fond of her studies and books, little dreamed she of her short-lived honor and cruel fate. Forced upon the throne by the insatiable ambition of Northumberland, she ruled for ten days. It is asserted that Mary wished to spare her cousin’s life, but that Wyatt’s rebellion so alarmed her that she determined to make an example of Lady Jane and her boy husband, Guildford Dudley.

Not only her piety, grace and beauty excite our admiration, but also her sublime heroism, which caused her to refuse to bid her young lover and husband farewell, lest the parting should unman him. Dudley was executed on Tower Hill, and the same day the lofty spirit of his wife joined his.

“Now boast thee, death; in thy possession lies
  A lass, unparallel’d.”

Next to the Brick is the Jewel or Martin Tower, where the crown jewels were formerly kept. They are now preserved in the Record Tower. On the wall of the Martin Tower we saw inscribed the name of Anne Boleyn. It is said that one of the unfortunate gentlemen who lost their heads on her account traced it there. Diagonally across the inner ward rises the Bell Tower, thus named from the alarm bell which crowned it. This was the prison of Princess Elizabeth during her enforced stay in the tower. Some little children used to bring her flowers here, until it came to the ears of Mary, who forbade this innocent service.

Only a short walk from Bell loom up the frowning walls of Beauchamp Tower, than which there is no more interesting place in the entire enclosure. Its architecture is of the reign of John and Henry III. Its name is derived from Thomas De Beauchamp, Earl of Warwick, imprisoned here during the reign of Richard II. This tower is in the center of the western side of the inner ward, and in a half circle projects from the strong ballium walls, and is two stories in height. Its walls are covered with inscriptions made by different prisoners; some indicative of their mental agony during their captivity; many, indeed the most, expressing Christian fortitude and pious resignation to their hard lot.

The first name noticed is that of Marmaduke Neville, near the entrance. He was one of the Nevilles, Earls of Westmoreland, and was implicated in a plot to place Mary Stuart upon the throne during Elizabeth's reign. In the southern recess is shown an inscription in old Italian: "Dispoi che voie la fortuna che la mea speranza va al vento pianger, hovolio el tempo per dudo; e semper stel me tristo, e disconteto. Wilim: Tyrrel, 1541." The mournful burden of which comes like a sigh of despair from out the past, "Since fortune hath chosen that my hope should go to the wind to complain, I wish the time were destroyed; my planet being ever sad and unfavorable." Alas, unhappy one! Your words no doubt were but the echo of many sad hearts that found the times were indeed out of joint.

Over the fireplace is a beautiful and touching inscription: "Quanto plus afflictionis pro Christo in hoc sæculo, tanto plus gloriæ cum Christo in futuro. Arundell, June 22, 1587," which being interpreted is, "The more affliction for Christ in this world, the more glory with Christ in the next."

This was written by Philip Howard, Earl of Arundel, whose devotion to the Romish church during Elizabeth's reign, brought so much odium upon him, and made for him so many enemies that he at last resolved to leave his country, friends, and his wife, to whom he was devotedly attached, and go into voluntary exile for his better safety. He informed the Queen in a most pathetic letter of his intention, designing she should not receive it until he was well out of way, but by some freak of fortune, the letter fell into the hands of his foes, and he was seized as he was setting sail from the coast of Sussex. He was sent to the Tower and kept a close prisoner for forty years, when worn out with his long and cruel confinement and sorrow he died, realizing at the end, we hope, the truth of the touching words he traced upon his prison walls.

There are several interesting inscriptions made by Arthur and Edmund Poole, great-grandsons of the Duke of Clarence. These young gentlemen were also accused for conspiring for Mary Stuart, adjudged traitors, and pined away their lives in hopeless captivity. Well might the White Rose of Scotland have said with Helen of Troy:

> “Many drew swords and died; where’er I came,
> I brought calamity.”

One inscription reads:

> “IHS. A passage perillus maketh a port pleasant. Ao. 1568. Arthur Poole, Æ sue 37, A. P.”

A passage perilous indeed, hadst thou, poor soul; God grant thou madst a fair haven at last.

Another contains these words:

> “IHS. Dio semine in lachrimis in exultatione meter. Æ. 21, E. Poole, 1562.” “That which is sown by God in tears is reaped in joy.”

In all the inscriptions left by these ill-starred gentlemen, there breathes the same spirit of noble and pious submission.

The greatest interest clusters about one little word, supposed to have been traced by the hand of one to whom that name was sacred. Directly under one of the Poole autographs is the word “IANE,” supposed to have been the royal title of Lady Jane Grey, written there by her husband, Lord Dudley, who was confined in this tower. Scarcely can the eyes be restrained at this touching reminder of the fate of those two unhappy children, the victims of circumstance and greedy ambition.

In the corner next the Beauchamp is the Devereux Tower, named from the brilliant Robert Devereux, Earl of Essex, the chivalric soldier and courtier, first a petted favorite, then a victim of Queen Elizabeth. His story is one of thrilling and fascinating interest. Meteor-like he flashed through his court and army life, and after gaining the zenith of his power, sank as suddenly as he had risen. It is said that he was one of the many with whom the royal and fickle spinster coquetted, and that he really touched her haughty heart. The government of Ireland was in his hands, but enemies at court plotted his overthrow. He in turn plotted against these foes and rashly attempted to cause their removal. He was arrested and arraigned in Westminster Hall for high treason, pronounced guilty, and doomed to the block.

Elizabeth had a terrific struggle between revenge and affection, but the baser passion got the victory, and the accomplished general, statesman, and courtier trod the same hard road to death that so many knew full well.

> "I have reached the highest point of all my greatness!
> And from that full meridian of my glory,
> I haste now to my setting: I shall fall
> Like a bright exhalation in the evening,
> And no man see me more."

The towers of the outer ward are comparatively of but little interest, with the exception of St. Thomas's Tower or the Traitor's Gate. This is a large, square building over the moat, the outside of which is guarded by two circular towers, which exhibit specimens of the architecture of the time of Henry III. The gate through which state prisoners entered the Tower is underneath this building.

The rain was falling drearily on the day we visited the Tower. Somber and heavy skies looked sullenly down on the gloomy scene. Thoughts as somber and heavy weighed down our minds as we stood before the Traitor's Gate; thoughts of countless numbers that had gone in at that gate never to come forth again. In the clang of those iron portals behind them they heard their death knell. The royal, the noble, the illustrious, the pious passed under these frowning battlements, leaving behind grandeur, brilliancy of courts, dreams of glory, home, friends, all that makes life sweet, to receive in exchange, the dungeon, the scaffold, the block, the axe.

They who entered there left hope indeed behind.

Through this gate went Elizabeth, expecting naught but death; dreaming little of the hour when all England should lie within the hollow of her white hand. Under these portals three short years after she issued from the Tower in all the full flush of her pride and triumph, received by lords and dukes, amid the blare of trumpets, and peal of bells and roar of guns. Elizabeth's hapless mother, Anne Boleyn, returned to the Tower. No nobles in her train now; no burst of music; no chime of bells nor roar of artillery; alone, save with her jailers; her fair fame blackened; her triumphs, glories—all shrunk to this little measure. The husband she had stolen from another, in turn lured from her, wearied of her, longing to be rid of her, hurrying her to her fearful doom with brutal haste.

> “A dream of what thou wast; a garish flag,
> To be the aim of ever dangerous shot;
> A sign of dignity, a breath, a bubble;
> A queen in jest only to fill the scene.
> …
> Where is thy husband now?
> …
> Who sues and kneels and says God save the queen?
> Where be thy bending peers that flattered thee?
> Where be the thronging troops that followed thee?
> …
> For one being sued to, one that humbly sues;
> For queen, a very caitiff crowned with care;
> Thus hath the course of justice wheeled about
> And left thee but a very prey to time;
> Having no more but thought of what thou wert,
> To torture thee the more, being what thou art.”

What thoughts must have chased each other in lightning rapidity through the mind of beautiful, brilliant, witty Anne Boleyn, during those short seventeen days she passed in the Tower before she was led out to execution. What experience of life had she not compressed into those three little years of usurpation, during which she spurned “heads like foot balls,” laughed, danced, and jested away her poor, butterfly life? What remorse must have been hers when the sad, pale face of Katharine arose before her! What unspeakable anguish when the coquettish features of Jane Seymour swam before her weeping eyes!

On the 19th of May, 1536, when hedge and field were bursting into bloom, when birds sang and soft breezes played, when all nature must have breathed of beauty, hope and life—Anne Boleyn went forth the second time from the Tower to receive her crown; not this time an earthly diadem, glittering with jewels, but the thorny crown of martyrdom. Not in cloth of gold and blazing with gems, but in sable robes went she to this coronation, and her only salute was the dull boom of the cannon which announced to the royal ruffian waiting at Richmond that he was free.

The Tower of London has been used not alone for a fortress and prison, but also for a palace. All of the kings from William to Charles II. held occasional court in the Tower. A palace occupied a space in the inner ward, between the southwest corner of the White Tower and the Record, Salt and Broadarrow Towers. The queens had a suite extending from the Lanthorn to the southeast of the White Tower. And near the Record Tower was a great hall which demanded forty fir trees to repair it at the time of the marriage festivities of Henry III. and Eleanore of Provence.

When Edward the Black Prince took prisoner King John of France, he lodged his royal captive in this palace, and King John gave an entertainment for his captor in this great hall. The beautiful Elizabeth of York, daughter of Edward IV., and queen of Henry VII., resided for a time in this palace, and passed from thence to her coronation. Sixteen years later she lay in state twelve days in the royal chapel in the White Tower, where her knights and ladies kept solemn vigil beside her bier. What an impressive scene it must have been! The windows all ablaze with lights, and an illuminated hearse holding the royal dead.

Queen Mary held court in the Tower directly after she had defeated Northumberland and the Dudleys. The ancient custom of a state procession from the Tower to Westminster was observed for the last time at the coronation of Charles II.

Very near the Devereux Tower is a plain, unpretentious building—the chapel of St. Peter Ad Vincula. It is small, having but one nave and one side aisle, and is quite without ornamentation. But marvelous interest invests it. Here Lady Jane was buried; here the body of poor Anne Boleyn was thrust into an old chest and hastily interred in the vaults; here lies the dust of Northumberland, Thomas Cromwell, Somerset, Surrey, and Essex, teaching the terribly solemn lesson that ambition, talents, fame, form no sure bulwark against death.

> “To-morrow, and to-morrow, and to-morrow
> Creeps in this petty pace from day to day,
> To the last syllable of recorded time;
> And all our yesterdays have lighted fools
> The way to dusky death. Out, out, brief candle.”

Just in front of this chapel is the spot on which the scaffold was built; the spot where the best blood of England flowed like water; the spot which mars the escutcheon of the Tudors with an ineffaceable stain; the spot where Englishmen first looked upon the spectacle of the blood of their countrywoman flowing beneath the blows of a foreign headsman. Here fell the heads of two of the wives of Henry VIII.; here the hapless Lady Jane was despatched, and the gallant Essex met his death by orders of Henry’s daughters, fit representatives of their father; here was enacted that revolting scene, the butchery of the venerable mother of Cardinal Pole, the Countess of Salisbury. She was sister of the Earl of Warwick, and daughter of the murdered Clarence. Her only crime seems to have been her royal blood. When brought out to execution, she refused to lay her head on the block, saying haughtily, “So do traitors use to do, and I am no traitor.” The sequel is almost too sickening to be rehearsed. The executioner pursued his victim around the scaffold, striking at her with his axe, and finally dragged her by her white hairs to the block. Thus miserably perished the last of the Plantagenets.

Heavier fell the rain and wilder blew the wind as we slowly took our way toward the outer entrance to the Tower. In the patter of the rain upon the stone flagging beneath us, we seemed to hear the footsteps of a countless, headless throng; in the slow drip, drip of the raindrops from the gloomy walls, the drip, drip of warm life blood trickling down and ebbing away; borne on the wail of the wind there seemed to come sighs of anguish, moaning voices long since silenced, voices from out a dreadful past, voices that cried aloud for vengeance. And as the great gates of the Tower clanged behind us, in a tremendous peal of thunder, there seemed to come an answering voice from heaven, “Vengeance is mine, I will repay.”

GEOGRAPHY OF THE HEAVENS FOR APRIL.

BY PROF. M. B. GOFF,
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THE SUN,

With its immediate attendants, Mercury, Venus, Earth, Mars, etc., has been the “theme of our discourse” for the last eighteen months. Except an occasional reference to one of the planets as being located near some fixed star, or in some constellation, little has been said about the 3,391 “fixed” stars, visible to the naked eye, many of which are located on maps of the heavens, just as villages, cities, mountains, rivers and plains are located on maps of the earth; nor of the somewhere between 30,000,000 and 50,000,000 which are visible only through powerful telescopes, and whose distances from the sun are so great as to make that of Neptune appear like a little walk “across lots” to visit a neighbor. Nor is it proposed now to enter upon such an extensive subject, except so far as may be necessary to present a single thought. As we know, our sun is a bright body, whose light and heat (so great is their power) we can hardly estimate. Both these qualities render it visible to us and make us realize its presence. The other bodies, as Mars, Jupiter and the Moon are seen only by reflected light, and were they as distant as the fixed stars, would not be at all visible. These 30,000,000 to 50,000,000 stars must be suns. How many satellites has each? We do not know, for they can not be seen. Suppose each had as many as our sun. Then instead of 30,000,000 to 50,000,000 of heavenly bodies, we have within reach of the telescope from 240,000,000 to 400,000,000. How many are outside of these? No man can number them. We shall have to wait till our minds can grasp the infinite. Are these millions of bodies standing still, or are they in motion? Does our sun stand still and permit us to go around him once every year, or is he, and are we along with him, making our way through other vast multitudes and moving around some other central orb? Observation proves that the sun is only a sergeant in a great army of generals, and marches his squad in an appointed way to their assigned duties. How do we know? The records of patient watchers for centuries reveal the fact. “If we suppose the sun, attended by planets, to be moving through space, we ought to be able to detect this motion by an apparent motion of the stars in a contrary direction, as when an observer moves through a forest of trees, his own motion imparts an apparent motion to the trees in a contrary direction. All the stars would not be equally affected by such a motion of the solar system. The nearest stars would appear to have the greatest motion, but all the changes of position would appear to take place in the same direction. The stars would appear to recede from that point of the heavens *toward* which the sun is moving, while in the opposite quarter the stars would seem to crowd more closely together.” Proceeding upon this principle, Sir William Herschel was in 1783 enabled to announce that the observed proper motion of a large portion of the stars could be accounted for on the supposition that the sun was moving toward the constellation *Hercules*. Later investigations not only established the fact that the sun moved, but that it was moving nearly toward the star *Rho*, in *Hercules*, and Struve estimated its motion at about five miles per second; though Professor Airy places it at about twenty-seven miles per second. It is also highly probable that its motion is not in a straight line, but in obedience to the same laws that govern the motions of its own satellites, it with other suns revolves about a center located nearly in the plane of the Milky Way, and with an orbit so great “that ages may elapse before it will be possible to detect any change in the direction of its motion.” Meantime, finite beings are interested in knowing how its light and heat affect their interests, and how these qualities may be made most profitable to mankind. For ourselves, we must at present be content to know that on the 1st our sun has reached a point 4° 48′ north of the equator, and that by the 30th he will be 14° 58′ north, an increase in northern declination of 10° 10′, and, as a consequence, our daylight will be increased about one hour and thirteen minutes, and the time “from early dawn to dewy” twilight will be seventeen hours and thirty-five minutes. On the 1st sunrise occurs at 5:43 a. m., sunset, 6:24 p. m.; on the 16th, sunrise, 5:19, sunset, 6:40; on the 30th, sunrise, 4:59, sunset, 6:54.

THE MOON.

The phases for the month are as follows: Last quarter, 7th, at 9:34 a. m.; new moon, 15th, at 12:43 a. m.; first quarter, 21st, at 6:12 p. m.; full moon, 29th, at 1:06 a. m. Rises on the 1st, at 8:38 p. m.; sets on the 16th, at 8:28 p. m.; rises on the 30th, at 8:21 p. m. In latitude 41° 30′ north, least elevation on the 6th, and equals 30° 20′; greatest elevation on the 19th, equals 66° 44′ 29″.

MERCURY

Will be an evening star during the month; it will have a direct motion of 12° 25′ 59″ up to the 17th, after which, to the end of the month, a retrograde motion of 5° 22′ 11″. On the 8th, at 2:00 a. m., will be at its greatest eastern elongation (19° 26′); on the 16th, at 11:55 a. m., will be 6° 21′ south of the moon; on the 17th, at 5:00 a. m., will be stationary; on 27th, at 10:00 p. m., will be in inferior conjunction with the sun—that is, will be between the earth and sun; and next day, at 1:00 p. m., will be 1° 42′ north of Venus. A few days before and after the 8th may be seen as a pale, light star, near the western horizon. Its times of rising and setting are as follows: On the 1st, rises at 6:21 a. m., sets at 7:51 p. m.; on the 16th, rises at 5:50 a. m., sets at 8:00 p. m.; on 30th, rises at 4:53 a. m., sets at 6:29 p. m. Diameter increases from 6.4″ to 11.8″.

VENUS,

Like Mercury, will be evening star throughout the month, and near the 28th the two will keep “close company,” but will so completely hide themselves in the light of “Old Sol” as to be entirely indifferent to the gaze of the “vulgar crowd.” On the 1st Venus rises at 5:34 a. m., and sets at 5:34 p. m., being just twelve hours above the horizon; on the 16th, rises at 5:19 a. m., sets at 6:09 p. m.; on 30th, rises at 5:06 a. m., sets at 6:42 p. m. Diameter diminishes during the month two tenths of a second; motion, 34° 38′ 45″ eastwardly; on the 14th, at 3:00 p. m., six minutes north of the moon.

MARS

Has a direct motion of 21° 13′ 16″, and his diameter increases two tenths of one second. On the 14th, at 12:27 a. m., 12′ south of the moon. On the

1st, rises at 5:29 a. m., sets at 5:25 p. m.; on the 16th, rises at 4:56 a. m., sets at 5:24 p. m.; on the 30th, rises at 4:27 a. m., sets at 5:23 p. m.; on the 14th, at 12:27 a. m., twelve minutes south of moon.

JUPITER

May well be called this month the “Ruler of the Night.” From twilight till near the dawn his broad face looks condescendingly upon our little world, and by his example cheerily bids us “pursue the even tenor of our way.” Jupiter rises on the 1st at 2:26 p. m., sets next morning at 4:02 a. m.; on the 16th, rises at 1:24 p. m., sets on 17th at 3:02 a. m.; rises on 30th, at 12:29 p. m., and sets next morning at 2:07 a. m. Before the 21st, retrograde motion amounts to 36′ 26″; after that date to end of month, direct motion equals 8′ 42″; diameter diminishes three seconds, from 40.4″ to 37.4″. On 21st, at 3:00 p. m., stationary; on 23d, at 2:05 p. m., 4° 37′ north of the moon. It might be observed in passing that as a mean result of five years’ observations at the Dearborn Observatory, Chicago, the time of Jupiter’s rotation has been discovered to be greater by three seconds than was supposed in 1879.

SATURN

Sets at the following times: On the 1st, at 11:49 p. m.; on 16th, at 10:57 p. m.; on the 30th, at 10:09 p. m.; is, therefore, an evening star, and will remain so till the 18th of June. On the 18th, at 8:20 p. m., 4° 1′ north of the moon. Diameter diminishes from 16.6″ to 16″. Makes a forward (direct) motion of 3° 2′ 30″. For observation, this month is preferable to May. Can be found a little northwest of *Zeta*, in the constellation *Taurus*.

URANUS,

Unlike Saturn, retrogrades nearly one degree of arc during the present month, and shines from early eve to break of day, rising on the 1st at 5:18 p. m., and setting on the 2d at 5:22 a. m.; on the 16th, rising at 4:17 p. m., and setting next morning at 4:21; and on the 30th, rising at 3:19 p. m., and setting May 1st at 3:23 a. m., and can be seen all night by those who know where to find him (a little southwest of *Eta*, in the constellation *Virgo*). On 26th, at 12:16 a. m., 1° 17′ north of moon.

NEPTUNE,

Not only the father of waters, but water himself, scarcely visible at best, “hangs out” all day, rising soon after the sun, and setting as follows: On the 1st, at 9:36 p. m.; on the 16th, at 8:40 p. m.; on the 30th, at 7:46 p. m. Has a retrograde motion of 58′ 16″; and on the 16th, at 8:42 p. m., is 2° 13′ north of the moon.

We have now passed the boundary of the first century of our existence as an independent nation. We are as a people engaged in a confused struggle with the problem of our own national self-consciousness. We want to know what is the spirit that is in us as a nation. We must know this in order to be properly master of ourselves and of our destiny. We must know this in order to know our place in universal history.—*George S. Morris.*

ENGLAND AND ISLAM.

BY PRESIDENT D. H. WHEELER, D.D., LL.D.

Within two years there have been three prophets in Egypt. Arabi Pasha is in exile; Chinese Gordon is dead; El Mahdi, the mysterious voice in the Soudan wilderness, mutters his prayers in the mosque of Khartoum. England bombarded Alexandria; Arab loss in dead perhaps 5,000. Then England fought and conquered Arabi in the open field, captured him, and sent him into exile; Arab loss in dead perhaps 7,000. Next there is trouble on the Red Sea, and another English army killed perhaps 9,000 Arabs. And last a battle or series of battles in the heart of the Soudan; Arab loss in dead perhaps 12,000. Probably not less than 30,000 have been slaughtered by Englishmen in less than two years. English loss, a few hundreds. The butchers have been liberally rewarded; one soldier has become a "lord;" promotions and extra pay and pensions have fallen in a silvery shower on "our brave fellows" in Egypt. Only one Englishman got nothing. He disappeared one day in the desert, and his dromedary was said to carry the destiny of England; and perhaps it did. He was a soldier seeking peace at the meeting place of the Niles. Chinese Gordon entered Khartoum in triumph, and almost at once there rose a cry: "We must rescue Gordon." Then came the long delayed march of an army in search of the English prophet at Khartoum; then the butcheries, called battles of Metemneh, and what not. And then in the last days of January there was a slaughter, not this time by Englishmen in person, and perhaps 5,000 more Moslems perish by Moslem steel in the sack of Khartoum. Then a wail rises on every breeze in Christendom; "*Alas! alas! Gordon is dead!*" The story of his death is a parable: "Stabbed in the back while leaving his house." Make the "house" stand for England, and the knives that pierced him the indecisions, tergiversations, and infidelities of an English ministry with a great Christian statesman at its head. The world has supped full of the horrors of that kind of Christian statesmanship. We have believed in it; we have hoped that it meant something, even in those bloody Egyptian campaigns.

We are nearly at the end of our confidence. It is not merely the shade of Disraeli which calls mockingly for explanations; the world that believed in Gladstone when Disraeli was playing at fantastic military statesmanship, wants to know why Christian statesmanship in Egypt has, in a short time, spilt almost as much blood as was shed by one army in that American conflict which Mr. Gladstone thought so cruel and so useless. We can not even condone Mr. Gladstone's offense against civilization by saying that it has been a less bloody assault on humanitarian ideas and plans than Disraeli's was; for Gladstone has butchered twenty men to Disraeli's one. There has, in fact, been nothing so bloody in this century—I mean no such large butchery by a small army. Ten years of such statesmanship would fill the Nile valley with human bones. It is high time to call for a full explanation. What does Mr. Gladstone mean? What does he expect to accomplish? If he has intended something exalted and noble, which we should wish to believe, it is time to say so with the breadth of statement and accuracy of detail by which he obtained renown. The personal question stands at the front, because England is governed by one man. It is a happiness of Englishmen that they are able to know whom to blame when things go wrong. Mr. Gladstone is the head of a government for whose acts and failures to act he is perfectly responsible. What England does in Egypt Gladstone does. It is the one governmental luxury of the English people—they know exactly who governs them. Mr. Gladstone has not been compelled to do this or that by parties or circumstances. If he turns butcher in the Delta, on the Nile, or on the Red Sea, he alone does it, and he does it because he chooses to do it. For, at any moment, he can shift disagreeable duties to another; three lines in the form of a resignation will relieve him of the burden of responsible government. So long as he remains at the head of the English ministry he is the man who shoots down Arabs by the thousand. In this country politicians have divided, dispersed and destroyed responsibility to such an extent that the people know not whom to blame for evil events. It is a devil's art, from whose manipulations England has by some special favor of heaven escaped. There the ghosts of murdered men and things can "shake" their "gory locks" at the Prime Minister; and he may not reply:

"Thou canst not say I did it."

Many of us have expected Mr. Gladstone to retire when each of these bloody episodes in Egypt has begun. His retention of power may be explained as an old man's insane appetite for office, or as the surrender of a statesman to the logic of a situation. The first explanation we respect Mr. Gladstone too much to accept; the second is embarrassed by the absence of a clearly defined policy. We should understand Disraeli; but he would help us to understand him by making distinct proclamation of his purpose to govern and bless the Moslem world. He would have butchered less, but he would have planted an imperial stake on every battle-field. We should have known that he meant conquest and dominion. There would have been no meaningless carnage. A humanitarian war is a difficult conception; but it is not impossible to conceive of wars that produce beneficent results. We could conceive of the subjugation of Islam to British sway, and rejoice to see the Soudan like India, slowly but surely rising into civilization under English rule. But an army thrusting down no imperial stakes, going home after each slaughter to be paid, promoted and fêted, is not doing work which opens any vistas of smiling peace and advancing light. It is only a bloody carnival. No petty cabinet differences, no outcry of public opinion, no Jingoism in the army or the royal family, no temporary exigencies of party, no domestic dangers nor foreign rivalries can explain and justify the responsible man's conduct. Mr. Gladstone's garments are dripping with Moslem blood, and the world can not find an explanation which explains.

It seems to the spectators that England is doing the one thing she should most carefully avoid doing. She is uniting Islam, and teaching Islam how to make war. In each new campaign the Soudanese are better armed, fight with better method, and kill more Englishmen. England is training them into sturdy and disciplined soldiers. A Moslem victory is proclaimed in every Arab tent, and in every Indian village. Such a victory is not merely a victory for El Mahdi; it is a hope for the whole Moslem world. Moslem defeats travel less swiftly, and mean only a delayed victory. What fierce resolutions are begotten in Moslem bosoms by Mr. Gladstone's campaigns of butchery, we can easily imagine. Meanwhile, Christendom can only say: "Premier of England, your garments are soaked with blood; and, may God forgive you, the blood is not your own. We can not understand you, but we are painfully certain that you are arousing all Islam against us." Meanwhile, the ancient spears are giving place in the Prophet's armies to repeating rifles, and

Krupp guns may soon guard every height along the Nile. Islam is strong in numbers. There are 75,000,000 of Soudanese, with a very large proportion of men just civilized enough to make terrible soldiers. It may happen some day that a military leader will arise in the front of this vast army, and that an effeminate Europe may find that its military science has gone over to the Moslems. Probably no one man's policy could effect its transfer more rapidly than Mr. Gladstone's. When that dark wave of the Moslem millions is gathered into conquering masses by a capable leader, it will have mighty winds of religious enthusiasm behind it, and plenty of room before it. The southern and eastern shores of the Mediterranean would be swept clean of the petty European military establishments in a month. Morocco, larger than France, holds at least half of the western gate of the Mediterranean, while the Turk holds the eastern gate; and a month's campaign might convert that sea into a Moslem lake, and leave its Italian, French, and Spanish and Greek shores to be ravaged again as in the crusading centuries, by Moslem piracy and brigandage. The one thing the Arab can learn thoroughly is the art of war. He was once great on the sea. That man may exult too soon, who, remembering the leviathans of the deep which destroyed Alexandria, trusts Europe's safety to the great navies of Europe. Islam has some great ships in the Bosphorus, and is rapidly learning where the great ships grow. It is true that if splendid leadership does not arise, Islam may continue to bleed and die in vain; but wars produce great soldiers as regularly as oaks bear acorns. There *is* danger. Ten years of Gladstoneism in the Nile valley would make the danger a terrible reality. Christendom should rise up and condemn the bloody education which England is imparting to Islam.

Meanwhile, Germans and Frenchmen are in the armies of the Prophet, teaching the rude but vigorous men of the desert how to use arms of precision with deadly effect. In the process of creating a terrible peril for Europe, greed, personal ambition, and national jealousies are contributing to perfect the lessons in modern warfare which England is giving to Islam. No doubt it is true that a great man is needed to weld together the forces of Islam. But why should no strong man be expected to arise in a race so rich in warlike memories? When the Prophet is once crowned with the diadem of military success, there is an army of Mohammedans in India wearing the queen's uniform, there are vast resources at Constantinople ready to fall from the helpless hands of the Sultan; there are millions of soldiers who require no pay, and have no scruples about the rights of private property. If one gives rein to his imagination, he is soon in a world of awful possibilities. There are two hundred millions of Mohammedans waiting for a leader to restore the glories of Islam.

The relations of England to Islam are logically and historically friendly. England has a Moslem army in India, and has long protected the head of Islam at Constantinople, from the consequences of his vices, extravagances and follies. The Indian mutiny had a religious source, but this was denied, and the spring covered up so successfully that, until Mr. Gladstone attacked Disraeli's policy in the name of Christianity (such as it is) in Bulgaria, England had successfully encountered the difficulties of her position as a Christian power ruling directly and indirectly half the Moslem world.

Does Mr. Gladstone foresee an "irrepressible conflict" between England and Islam? Is he instinctively bringing on a conflict which will be the less perilous the sooner it comes? Will history add to his rare good fortune by making him glorious as the beginner of a defense of Christendom which he has never dreamed of organizing? Disraeli's conception followed logic and history. He made a Christian queen empress of India, and he contemplated with composure a time when the descendant of Victoria should be born in India, and be reared in the faith of Mohammed, the center of the British empire having gone to its proper place. Against such ideas Christian England revolted. Is Mr. Gladstone reversing centuries of history and setting the Moslem and Christian worlds by the ears again? If he is moving on that line, his armies should conquer and hold Egypt and the Soudan, the Nile and the Red Sea, the ancient Delta and its modern canal, with the grip England once laid on North America. The audacity of the conquest would provoke diplomatic criminations; but it were easier far to face them than to answer the hard questions which are provoked by fruitless slaughter in Moslem lands. England is only the heart of the British empire. A quiet and gentle England is a possible dream; but the empire is war, conquest, dominion, at the expense of weak peoples. The empire can not survive the definitive abandonment of an imperial policy. The empire must dominate by force or fall to pieces.

It is not worth while to seek in the history of the Egyptian debt, and the "grasping disposition of the English bondholders," for the key of the present situation. Those who make a religion, or at least a philanthropy, of heaping abuse on bondholders anywhere and everywhere, are the least reasonable of Christians. It is not a crime to deny ourselves, save money and lend it to others. To refuse to pay debts freely contracted is not the first of virtues or the best of policies. The bondholders are commonly poor people who have saved a little by pinching themselves, and have bought bonds for the holy purposes of family forethought. Repudiated debts are baptised in the self-renouncing spirit which is at the heart of our religion, and repudiators make war on the foundations of character and society. In so far as England protects her money lenders, she protects her noble middle class, whose honest thrift lies at the foundation of her wealth. It is among the strangest perversions of feeling that prodigals and prodigal governments should get the sympathy of mankind. Let England foreclose her mortgage on Egypt, and the honest world will thank her for abolishing one nest of spendthrifts. Much is said of the miserable Egyptian peasants, from whom the taxes to pay interest are wrung with every form of despotic oppression. Let us not be deceived by such false-toned appeals for sympathy. The fellaheen of the Nile are oppressed irrespective of the bondholders. Arabi or El Mahdi would maintain the oppressive systems if they were in power. If there were no bondholders, the backs of the miserable fellaheen would smart under the lash of the oppressor. The despotism is Egyptian, not English. English rule would gradually emancipate the oppressed classes. Nowhere, not even in Ireland, has England conquered a people without improving the condition of the poor. The interest on debts which she surrenders in the valley of the Nile does not go to the relief of the peasants; it is squandered in the harems of Cairo and Alexandria. The issue is strictly between the splendid, many-concubined lords in Egypt and the honest and self-denying people who have lent honest money on the faith of England.

Which way, then, will events march? Toward a war between Islam and Christendom, or back to the old imperial policy of England? It would seem that the world's hope lies in a restoration of the ancient policy of the British empire. The events of 1885 in Moslem land will be full of interest, perhaps pregnant with destiny. A larger English army, perhaps 25,000 men, will soon be in Egypt. It will probably face a better trained foe. There will be more English graves in Egypt. To what end? The London *Times* says: "Gordon must be avenged." England repeats the cry. But what end will the vengeance serve? And what if Arabi Pasha and the Emirs killed in the late battles, and the 30,000 to 40,000 Moslems slain, should be avenged? Soon or late—if she does not attempt it too late—England must return to her historical policy and stand among Christian powers the foremost ally of the sons of Mohammed. It is the inexorable logic of her greatness. Let us shut our eyes upon the horrible vision of the new crusades, as useless as the old and far bloodier. Christendom can hope for no more fortunate disposition of Mohammedanism than that it should be locked fast in the iron arms of the British empire; and on the other hand the failure of the British empire would involve the greatest possible disasters for Christendom. Many foolish things have been promised in the name of “manifest destiny.” Perhaps destiny is never so manifest that it may be read off by uninspired prophecy; but there is no other Power which seems fitted to play England’s imperial rôle; and it does not appear how the progress and happiness of mankind can go forward without such an imperial force as England has been for two centuries. While we deprecate the effects of the Jingo spirit which Disraeli fostered, and repudiate the indifference to the progress of Christianity which the Hebrew statesman scarcely concealed, we can not look with complacency upon changes of British policy which would disintegrate the empire. Let England’s drum-beat go round the earth with the sun; for the sunrise of progress and civilization will awaken wherever that martial music falls upon the ear of mankind.

THE ART OF FISH CULTURE.

BY PROF. G. BROWN GOODE.

PART I.

When any portion of the earth is colonized by civilized man, an era of change and readjustment at once begins. The untilled plain, the primeval forest, the bridgeless river, the malaria-breathing swamp, and the jungle—lurking place for beasts of prey—are all obstructions which must be removed from the highway of social and industrial progress. Until a new environment had been created, the colonists of Virginia and New England were like helpless children, compared with the Indians whom they had come to disinherit. The hills were soon cleared, and the water-courses dried up, swamps were drained, and lakes were made in the valleys, the plains were plowed and planted with exotic vegetation, and great regions of land were entirely changed in character by irrigation and the use of manure. The New World has in two centuries become in very truth a new world, for its physical features have been entirely reconstructed. The aboriginal man retreated before the advancing strides of civilization, and has now been practically exterminated, at least east of the Mississippi River.

The manner in which the man of European descent has eliminated and replaced the son of the soil is fairly typical of changes which have occurred in the animal and vegetable life of the continent. Bear, moose, caribou, deer, wolf, beaver, and all other large animals have been entirely destroyed in many parts of the country, and the time is not far remote when they will exist among us only in a state of partial or entire domestication. The prairie chicken once reared its brood in Massachusetts, but is now never seen east of the Alleghenies. The alligator is fast being exterminated in Florida and Mississippi, and the buffalo is now rarely to be seen except in captivity. The sea cow of the north Pacific, the great auk of New England and Newfoundland stand with the dodo, the moa, and the zebra in the list of animals which have become extinct within the memory of man, and the list will continue to increase. A similar story might be told for birds, reptiles, and plants. The rattlesnake is retreating to the mountain tops, the turkey, the pigeon, the woodpecker and hosts of others are disappearing, the medicinal plant ginseng, once so important in the Alleghenies, is almost a rarity to botanists.

The aboriginal animals and plants go. They are replaced by others, which in that struggle for existence which plays so important a part in determining careers for plants and animals, have become particularly well fitted to be man's companions. The clover, the ox-eye daisy, the buttercup, the thistle, the mullein, the dandelion, followed the European to America, and with them the broad-leaved plantain, which, as every one knows, the Indians called "the white man's foot," because it sprung up at once in every meadow where the soles of his shoes had touched. With these came the European mouse, the rat, the cat, the dog. The browsing herds of deer and buffalo were replaced by oxen, horses and sheep, and the greedy, quarrelsome, impertinent sparrow was permitted to drive out the native birds which many of us would have been glad to keep as relics of the old dispensation.

Not less important in many regions have been the changes in the life in the waters. In many of our streams and lakes the fish, formerly abundant, have been entirely exterminated. Sometimes, perhaps we may charitably say usually, this has been the result of ignorance, but often, I fear, it may be ascribed to recklessness or cupidity.

Fishes may be grouped, according to their habits, into two classes—resident and migratory. Representatives of each of these classes may be found both in fresh water and in the sea. Among resident fresh water fishes may be mentioned the perch, the catfish, suckers and dace, the pike and pickerel, the black bass. Resident sea fishes are typified by the flounders, cod, sheepshead, blackfish and sea bass, which are found near the shore in winter as well as summer. In cold climates, resident fishes always retreat in winter into deeper water to avoid the cold, and if they can not get beyond its reach they subside into a state of torpidity or hibernation, in which all the vital functions are more or less inert. The carp, and many other kinds of fish, at this time, burrow into "kettles" or holes in the mud in the bottom of the pond, where they remain for months. A hybernating fish may be frozen solid in the middle of a cake of ice, and emerge when thawed out, unharmed.

Migratory fishes, on the other hand, are those which wander extensively from season to season. There are migrating fish in the sea, which, like the mackerel, the bluefish, the menhaden and the porgy come near our northern coasts only in the summer, and in winter retreat to regions either in the south or far out at sea unknown; others, like the smelt and the sea herring, which retreat northward in summer and only appear in quantity on the Atlantic coast of the United States in the colder months of the year.

Then there are migratory fishes which live part of the year in the rivers. Such are the shad and the river-herrings or alewives, which leave the sea in the spring and ascend to the river heads to spawn, and the salmon, which does likewise, to spawn in the brooklets in November and December. Still more remarkable is the eel, which breeds in the sea, where the male eels always remain, while the young females, when as large as darning needles, ascend in the spring to inland lakes and streams, there to remain, until, after three or four years, they are grown to maturity, when they descend to salt water, to reproduce their kind and die.

There are also migratory fish in fresh water, like the white fish, the salmon, trout, and the siscowet, which live in the abyssal depths of the great lakes and swim up into the shallows and creeks in winter to spawn their eggs, and the brook trout and dace, which for a similar purpose ascend from the pools and quiet meadow stretches to the pebble-paved ripplets near the spring sources of the brooks in which they live.

Having, in a general way, classified fish according to their habits, we are in a position to consider the manner in which man has succeeded in exterminating them. As a general rule, fish deposit their eggs in shallow water, and the time of egg-laying is very closely dependent upon the temperature of the water. The eggs of a fish are, as every one knows, enclosed in two sacs, or ovaries, which are situated close to the walls of the abdominal cavity, and separated from the water by thin walls of skin and flesh, rarely, even in the largest fish, more than a sixteenth of an inch in thickness. Experiment has shown that the temperature of the blood in the abdomen of a fish deviates very little from that of the water in which it is floating. Experiment has also shown that as soon as the water has reached a certain degree of warmth, variable with each kind of fish, the eggs are sure to be laid within a very few hours. This being the case, it usually happens that great schools of fish always congregate together at one time upon the spawning grounds. Since the spawning grounds of many kinds of fish are in shallow water, and the fish are at that time most easily caught, it happens that many of the most extensive fisheries are carried on in the spawning season. The delicious little smelts which our neighbors in Maine and New Brunswick send us by the hundred car-loads each winter, packed in little boxes of snow, are always full of eggs—so are the lake white fish, when they are caught, and the shad, and the Potomac herrings, and the cod, and the mullet, and the herring, and in early spring the mackerel.

Now consider how easy it is, taking these fish so much at a disadvantage, to diminish their numbers, simply by catching them. The man who catches a spawning cod destroys anywhere from 5,000,000 to 7,000,000 of eggs, a spawning halibut at least 2,000,000, a shad from 50,000 to 2,000,000. Is not the American breakfasting on broiled shad roe a modern representative of him who killed the goose which laid the golden egg? When we consider that the yearly catch of mother-fish along the New England coast does not fall short of ten to fifteen millions of individuals, we may gain an adequate idea of the destruction of fish life by the fisheries.

Still it is not necessary to be alarmed at these figures. They are presented simply in illustration of the immense possibilities of destruction when the fisheries are carried on at the spawning season. As a matter of fact, cod are just as abundant along our coasts as they ever were, and it has not yet been demonstrated that any kind of sea fish has ever been diminished in numbers by hook and line fishing or by netting them at a distance from the shore.

Some kinds of fishes, however, enter narrow bays and estuaries to spawn, and if they are there recklessly destroyed, the local supply at least may be permanently interfered with. This has apparently been the case with certain species in Narragansett Bay, Rhode Island. For instance, the scuppaug or porgy has been seriously diminished in numbers in certain seasons, years ago; the supply will probably be replenished from adjoining waters by the reparative tendencies of nature, if this indeed has not already been done. So, too, the halibut has been exterminated in Massachusetts, where it was once so abundant as to be regarded as a nuisance by the fishermen. It is in the inland or freshwater fisheries, however, that the work of extermination has been thorough, and here, from the nature of the case, the work once accomplished, it is beyond the power of nature to remedy the damage. If I could take the reader with me next May to one of the many little streamlets of Cape Cod, flowing southward into Nantucket Sound, I could show him a scene which he would never forget. The little rill has been encased at bottom and sides with planks, so that it flows for a mile or two, down to its junction with the sea, in a straight trough not over fifteen inches wide, and a foot in depth. At a convenient level place a shed has been built over the trough, and in the floor is a kind of cistern, through which the waters of the brook flow as it goes on its course. In the shed stand two men, each with a great scoop of netting, with which they labor, dipping the fish out of the cistern as they fill it, swimming up the trough from the sea. Several barrels are taken out every day, and in some of these streams one or two thousand barrels always reward a season's work, the brook being the property of the township, and the privilege of fishing being sold at auction for the benefit of the public. Dip! dip! dip it is all day long, and as the little alewives are tumbled into barrels and carts, the eye of the practiced observer notes the plump sides and the brilliant iridescent coloring of the silvery scales, which indicate that the fish are loaded with a precious burden of eggs, to deposit which in the pond at the head of the stream is the motive which leads them to press forward so blindly into the trap men have set for them.

In these enlightened days the town laws generally require that the brooks shall be unobstructed for one or two days each week, and so a few fish get by the barriers and are allowed to perpetuate their kind. In the past, however, many excellent "herring brooks" have been completely deprived of their fish.

This illustrates how completely man has the destinies of river fish under his control. Suppose that instead of a fish house with movable barriers, an impassable dam had been built. Of course the fish would have been locked out, and their kind exterminated in that immediate region. This is precisely what has happened in almost every river and stream on the Atlantic coast of the United States. Shad and salmon were formerly abundant in every river of New England—and shad and alewives in every considerable stream south to Florida. Now, they are excluded, either entirely or in great part from the waters in which they once swarmed in great schools. Take, for instance, the Connecticut River. In colonial days, salmon were there in immense numbers. All summer long they were swimming up from the sea to the headwaters of the river, to Massachusetts, New Hampshire, and Vermont, where they deposited their eggs in the cool, clear rapids of the main river and its tributaries. They were so abundant that the shad fishermen used to require their customers to take one salmon with every shad, and, as the hackneyed old story goes, the apprentices were accustomed to stipulate in their papers that they should not be required to eat salmon above three times a week. In 1798 a dam was built across the river at Miller's Falls. Next year many salmon were seen at the base of the dam, the following year a smaller number, and in less than ten years salmon had entirely disappeared from the Connecticut. Not a salmon was seen in those waters until seventy years later, when, in 1871, a single artificially bred fish was caught at Saybrook. I could show you a map prepared by an associate of mine, in which the present and former limits of the shad are shown, and you would see how they once ranged clear up into the mountains, far up the Susquehanna into New York State, up the Connecticut into New Hampshire and Vermont, and how now, in many rivers, they are confined to very restrictive stretches at the river mouths.

The dams operate in still another way. We have considered hitherto only their influence upon the sea fish which ascend the rivers to spawn. Their effect upon the resident fish is quite as baneful. As the suckers, and the bass, and the cat fish, and dace, and trout, grow large, they naturally go down stream in search of deeper water and wider pools, where they get more room and better food. If they luckily escape the baskets and traps set for them in every dam, they never can get back. The streams are gradually sifted out and left tenantless.

Little need be said of the manner in which ponds are drained dry in order to get all the fish in them, in which immense seines are hauled in little lakes, clearing out everything, great and small, of the use of explosives, lime, or *cocculus indicus*, in the work of wholesale destruction. The fact stands undisputed and undisputable, that in many parts of the United States the native fish are actually exterminated, and the mud turtles, muskrats and fresh water clams left as sole occupants. Even the harmonious bull-frog has been devoured by man, and only his diminutive cousins, the cricket frogs and hylas left—the aquatic choir can henceforth perform only soprano and contralto songs, unless the fish culturist finds some way of bringing back the basso whose obligatos we once admired.

Oysters, scallops, and lobsters are going the same way. Although they live in free waters, they are stationary in their habits, and wholesale gathering will soon complete the work of extermination so recklessly begun. The forthcoming census reports on the fisheries will show conclusively the need of immediate protection.

What is the remedy for these great evils? One hundred thousand men are actively engaged in the fisheries of the United States, and at least one fiftieth of the entire population of the country are, to a large extent, dependent on the fishery industry. Fish is the poor man's food, for unlike any other food product it may be had for the taking. A fish swimming in the water has cost no man labor. There floats four pounds of savory shad, fifty pounds of nutritious sturgeon, a hundred barrels of whale oil; there lies a bushel of oysters, or a barrel of sponges. They are God's gift, and man has only to gather them in, and possibly submit them to a very simple process of preparation, to be the possessor of a valuable piece of property. If the matter can be properly regulated, good fish ought to be sold in every town and village for two thirds or half the price of beef and pork. As it is, poor fish often cost more than beef and pork, and in many localities good fish can not be had at any price. It is a great problem in political economy, and one which we are, as yet, far from thoroughly understanding.

We are confronted with the question, What can be done to neutralize these destructive tendencies?

There are evidently three things to do.

1. To preserve fish waters, especially those inland, as nearly as possible in their normal condition.

2. To prohibit wasteful or immoderate fishing.

3. To employ the art of fish breeding.

a. To aid in maintaining a natural supply;

b. To repair the effects of past improvidence, and

c. To increase the supply beyond its natural limits, rapidly enough to meet the necessities of a constantly increasing population.

The preservation of normal conditions in inland waters is comparatively simple. A reasonable system of forestry and water purification is all that is required, and this is needed not only by the fish in the streams, but by the people living on the banks. It has been shown that a river which is too foul for fish to live in is not fit to flow near the habitations of man. Obstructions, such as dams, may, in most instances, be overcome by fish ladders. The salmon has profited much by these devices in Europe, and the immense dams in American rivers will doubtless be passable, even for shad and alewives, if the new system of fish-way construction devised by Col. McDonald, and now being applied on the Savannah, James, and Potomac, and other large rivers, fulfills its present promises of success.

Up to the present time, however, although much ingenuity and expense have been lavished upon fish-ways by the various state fish commissions of this country, there has been little practical outcome from their use. Our dams are too high, and the shad and alewives, which we are especially desirous to carry over these obstructions, do not seem to take kindly to the narrow, tortuous defiles of the fish ladders.

The protection of fish by law is what legislators have been trying to effect for many centuries, and we are bound to admit that the success of their efforts has been very slight indeed. Protective legislation rarely succeeds. The statute books of each state are crowded with laws which no one understands, least of all the men who made them, and which the state governments are powerless to enforce. Every one remembers Whittier’s grand old hero, Abraham Davenport, the Connecticut statesman, who, “on a May day of that far old year 1780,” when the earth was shrouded in darkness, and he and all his colleagues in the State Assembly felt that the judgment day had come, stood up, “albeit with trembling hands and shaking voice, and read an act to amend an act to regulate the shad and alewife fisheries”—and then went on to rebuke those around for their fearfulness and desire to leave the post of duty. Connecticut is as much at a loss now as then to know how to regulate her shad and alewife fisheries. Under a republican form of government, restrictive laws are not popular, and money would never be voted to enforce such laws, which, without an extensive police force, would be powerless. Some one has sagely remarked that the salmon is an aristocratic fish, which can only thrive under the shadow of a throne. Many states now have laws protecting fresh water fish in the breeding season, and numerous game protective associations are laboring with some success for their enforcement. Sales of fish out of season are also successfully prevented in certain city markets.

The attempts to regulate the fisheries at the mouths of rivers, so that spawning fish may be allowed free passage for a few hours, generally from Saturday evening to Monday morning, are meeting with but little success.

Maryland and Virginia attempt to some extent the protection of their oyster beds, and the former state keeps up an expensive police organization. The oyster law is founded in ignorance, however, and the chief effort being to keep away fishermen from other states, for the benefit of their own, there are small results except frequent quarrels and occasional bloodshed.

Connecticut is making the experiment of giving to individuals personal title to submerged land, to be used in oyster culture, and this, perhaps, is the wisest step taken. Oyster production must soon cease to be a free grabbing enterprise, and be placed upon the same footing as agriculture, or the United States will lose its beloved oyster crop, and in this country, as in England, a fresh oyster will be worth as much as a new-laid egg. Great Britain has, at present, two schools of fishery economists, the one headed by Professor Huxley, opposed to legislation, save for the preservation of fish in inland waters, the other, of which Dr. Francis Day is the chief leader, advocating a most strenuous legal regulation of the sea fisheries. Continental Europe is by tradition and belief committed to the last named policy. In the United States, on the contrary, public opinion is generally antagonistic to fishery legislation, and our Commissioner of Fisheries, after carrying on for fourteen years investigations upon this very question, has not yet become satisfied that laws are necessary for the perpetuation of the sea fisheries, nor has he ever recommended to Congress enactment of any description. Just here we meet the test problem in fish culture. Many of the most important commercial fisheries of the world, the cod fishery, the herring fishery, the sardine fishery, the shad and alewife fishery, the mullet fishery, the salmon fishery, the whitefish fishery, the smelt fishery, and many others, owe their existence to the fact that once a year these fishes gather together in closely swimming schools, to spawn in shallow water, on shoals, or in estuaries and rivers. There is a large school of *quasi* economists, who clamor for the complete prohibition of fishing during spawning time. This demand demonstrates their ignorance. Deer, game birds, and other land animals may easily be protected in the breeding season, so may trout and other fishes of strictly local habits. Not so the anadromous and pelagic fishes. If they are not caught in the spawning season, they can not be caught at all. I heard a prominent fish culturist recently advocating before a committee of the United States Senate, the view that shad should not be caught in the rivers, because they came into the rivers to spawn. When asked what would become of our immense shad fisheries if this were done, he said that doubtless some ingenious person would invent a means of catching them at sea.

The fallacy in the argument of these men lies in the supposition that it is more destructive to the progeny of a given fish to kill it when its eggs are nearly ripe, than to kill the same fish eight or ten months earlier.

We must not, however, ignore the counter argument. Such is the mortality among fish that only an infinitesimal percentage attain to maturity. Möbius has shown that for every grown oyster upon the beds of Schleswig-Holstein, 1,045,000 have died. Only a very small percentage, perhaps not greater than this, of the shad or the smelt ever come upon the breeding grounds. Some consideration, then, ought to be shown to the individuals which have escaped from their enemies and have come up to deposit the precious burden of eggs. How much must they be protected?

Here the fish culturist comes in with the proposition "*that it is cheaper to make fish so plenty by artificial means that every fisherman may take all he can catch, than to enforce a code of protective laws*."

The salmon rivers of the Pacific slope, and the shad rivers of the east, and the whitefish fisheries of the lakes, are now so thoroughly under control by the fish culturist, that it is doubtful if any one will venture to contradict his assertion. The question now is, whether he can extend his domain to other species.

Legislation and fish-ways, then, are, as yet, of little practical importance. Actually, they repeat the proverbial act of the clown who locked the stable door after his horse had been stolen. No one makes laws or builds fish-ways until he is of the decided opinion that the fish are pretty nearly gone.

Artificial fish culture seems to offer the only remedy for the evils which have been described.

[TO BE CONTINUED.]

THE LIFE OF GEORGE ELIOT.

There is now and then a biography so written that the reader is able to become intimately acquainted with the subject, to feel after reading that he has had a personal contact, and has formed a friendship which is warm and living. Such “Lives” are rare. Most works of this kind are so biased by the interpretations of the author, so full of facts and opinions that the reader loses all feeling of companionship in reading; he closes the book, knowing much about the subject, but rarely understanding him. A book which presents a man or woman in that personal way which makes a friendship through the medium of the book possible, confers a great gift upon the reading public.

The peculiar fitness of Mr. Cross’s “Life of George Eliot”[B] for giving us a new friend, must be attributed to the really remarkable taste and skill of his editing. The work bears the mark of a reverential hand. It is an *In Memoriam* whose only object has been to lay before the world a memory too strong and precious to be kept secret. But no such a biography could have been given the world had it not been for the peculiar nature of George Eliot herself. The material for this “Life” grew out of two strong elements in her character: the affectionate and persistent friendship which led her to reveal herself so fully to those she loved in her letters, and that constant introspection which made her journal often a mirror of her inner life. These materials make up the book, which is largely a study of her character, and, too, of her character as she understood it. She has verily written her own life. The interpretation remains for the readers.

It would have been possible for Mr. Cross to have given much information about her character which he has withheld, her opinions and much of her conversation; but he has wisely given the world only what she herself chose to reveal to her friends.

The story begins early. The first revelations in character are the strongest; the happiness and misery of the future life are revealed in the childhood traits. The earliest revelations which we find in George Eliot’s life are of affection and ambition; either, if strong enough to become a passion, drives its possessor along a thorny path until it is itself mastered, and where both exist in a nature, continual collision must occur between them. Before each is satisfied there must be a life struggle of the keenest sort. Such a struggle was presaged for Marian Evans very early. She herself tells how, when she was but four years of age, she played on the piano, of which she did not know a note, in order to impress the servant with a proper notion of her acquirements and generally distinguished position. As eager, too, she was for love as for recognition. In her reminiscences of her early life most vividly she portrays her earliest passion—one very common among affectionate girls—her love for her brother. “She used always to be at his heels, insisting on doing everything that he did.” When his first boyish craze took possession of him in shape of a pony, and she found it was separating them, she was nearly heart broken. Impressible, eager for work and ambitious for knowledge, she began life under an emotional pressure, which drove her into incessant distress lest those she loved should fail her, and which brought her devotion to her loved ones in constant collision with her ambition. At twenty-one, writing to a very intimate friend, she said: “I do not mean to be so sinful as to say that I have not friends most undeservedly kind and tender, and disposed to form a far too favorable estimate of me, but I mean I have no one who enters into my pleasures and griefs, no one with whom I can pour out my soul.” Eight years after this, having lost her father, she went abroad for a few months’ residence, and her letters home were full of eager longing for their sympathy and restless fear lest they should forget her. No change in her life diminished this feeling. She became an editor of the *Westminster Review*, and while overwhelmed with manuscripts and proofs she wrote: “You must know that I am not a little desponding now and then, and think that old friends will die off, and I shall be left with no power to make new ones again.” Undoubtedly this feeling tended to make her morbid in her younger days, and consequently dwarf her power. It is an important study of her life to trace the gradual melting of this disposition, and the final growth into a healthy happiness. When quite past the heyday of her life, she wrote a friend of her girlhood: “I am one of those, perhaps, exceptional people whose early childish dreams were much less happy than the outcome of life,” and again, but four years before her death: “I have completely lost my *personal* melancholy. I often, of course, have melancholy thoughts about the destinies of my fellow creatures, but I am never in the mood of sadness, which used to be my frequent visitant, even in the midst of external happiness; and this, notwithstanding a very vivid sense that life is declining, and death close at hand.” This release from morbidness had two causes. She had taught her strong, affectional nature to find satisfaction in that commonplace, but little understood duty, loving her neighbors, and she had learned to enjoy things on their own account. The first article in this creed of happiness became George Eliot’s religion. She had abandoned her belief in the Christian religion when twenty-one years of age. She could not believe fully, and she was too independent and too reliant upon her own mind to conform to a religion she did not believe. The steps she took did not destroy the religious sense in her life. The earnestness which led her to write at nineteen, “May the Lord give me such an insight into what is truly good, that I may not rest contented with making Christianity a mere addendum to my pursuits, or with tacking it as a fringe to my garments;” which induced her to consider the novel and even oratorio as dangerous to spiritual development still remained, though without form. She was intensely in earnest, but it was many years before her love for mankind became a religion to her. A less strong character would have become flippant or scornful under this loss; hers only became more serious. She seems never to have forgotten what she had abandoned.

Though radically differing from most of her friends on religious questions, she never was uncharitable. “Of all intolerance, the intolerance calling itself philosophical is the most odious to me,” she wrote, and she lived out this opinion, in no way allowing the widest diversity to separate her from her friends. Her sympathy and charity indeed seem to increase, even if the breach in their opinion widened.

This habit of thought and feeling resulted in much personal moral benefit. “My own experience and development deepen every day my conviction that our moral progress may be measured by the degree in which we sympathize with individual suffering and individual joy.”

As she grew older, she comforted herself with the thought that “with that renunciation of self which age inevitably brings we get more freedom of soul to enter into the life of others.” She tried for “a religion which must express less care for personal consolation and a more deeply awing sense of responsibility to man, springing from sympathy with that which of all things is most certainly known to us, the difficulty of the human lot." This was the great lesson in her growth toward happiness. A secondary step was her appreciation of the value of things in themselves. It is a serious obstacle to the happiness of women, that in the main they care for exterior life only as it is of value in the personal life. A book is dear because a friend has read or recommended it. This verse is fine from association; this strain of music because they heard it in a certain connection. Take the *personal* out of Art and Nature, and too often women care little for them. George Eliot learned to appreciate and love things for their own value. Music, to which she was from childhood deeply susceptible, she cultivated thoroughly, and no fine rendering of *good* music ever was missed by her. She took the true, high view of life, which declares that from every good all possible enjoyment should be gained. Art was very dear to her, and we find in these quotations from her journal, notes on all the leading galleries of Europe, but in the very midst of her art studies she drops this comment, after noting a sight she had had of the snow covered Alps: "Sight more to me than all the art in Munich, though I love the *art* nevertheless. The great, wide-stretching earth, and the all-embracing sky—the birthright of us all—are what I care most to look at."

But it would have been impossible for even her deep love for mankind, her fine enjoyment of the good and beautiful, to have completed her life. These things satisfied her affections, but there was another quality we have mentioned as prominent in her life: it was her ambition. At twenty she wrote despondently: "I feel that my besetting sin is the one of all others most destroying, as it is the faithful parent of them all—ambition, a desire insatiable for the esteem of my fellow-creatures." Whatever she did, was done with all her might. Her mother died early, and she became housekeeper. Her struggles with the knotty questions of housewifery kept her in a constant worry, but she would do things right—whether it be currant jelly or a German translation. The same perfection marked her novels. Her progress was soon marked; it recommended her to people of standing, and gradually she had a circle of friends—people of strong minds and much culture—about her. Eager to do something, a way opened to her in 1844, when she was twenty-five years old. It was to translate into English a work of the German philosopher, Strauss. She did it, and what was better, did it well. Five years later she writes: “The only ardent hope I have for my future life is to have given me some woman’s duty.” The ambition to excel is already bending to the stronger emotion of affection. For a long time she worked, eager and anxious, but nothing seemed to open. In 1851 she went to London as assistant editor of *Westminster Review*, and here most satisfactory opportunities for culture opened. She formed lasting friendships with Herbert Spenser, Harriet Martineau, Florence Nightingale, and indeed with all the people worth knowing, who filled London in the ’fifties. But her editing left no opportunity to do that special work to which she was looking, and which she did not understand. She wrote many reviews and essays. This writing was a sort of safety valve for her intellect, but it was not until September, 1856, that the new era began in her life. It was when she began to write fiction. The popular idea of fiction as stories which will do to kill time, but which for serious reading are quite useless, was not the idea of George Eliot. A nature so intensely serious, so anxious for noble work, could not content itself with trivial story telling; she did not aim at that, but at studies of life. As she finely writes to Mr. John Blackwood, who became her publisher: “My artistic bent is directed not all to the presentation of irreproachable character, but to the presentation of mixed human beings, in such a way as to call forth tolerant judgment, pity, and sympathy. And I can not step aside from what I *feel* to be *true* in character.” And again: “I should like to touch every heart among my readers with nothing but loving humor, with tenderness, with belief in goodness.” The story of her great success is familiar—her books are well known. In rapid succession she sent out “Scenes of Clerical Life,” “Adam Bede,” “The Mill on the Floss,” “Silas Marner,” “Romola,” “Felix Holt,” “The Spanish Gypsy,” “Middlemarch,” “Daniel Deronda,” and “Theophrastus Such.” Her convictions about how she should work were intense. She wrote and *lived* her story, and once when urged to re-write a tale, replied that she could no more re-write a book than she could live over a year in her own life. Her novels are the embodiment of what she had felt, written that they might strengthen others. The conscientiousness with which she labored made her work sometimes most painful to her. Despondency lest she should fail, fear lest she had misinterpreted a character, depression lest this chapter should fall below a preceding in merit, tortured her in succession, but she worked because she believed she had found her place, and to do her best for mankind was her religion. The slow-growing nature struggling with eager desire for human love, and with a mastering ambition, not often reaches so ripe a stage as did hers. The rigid system of self-culture which she pursued through her life was the outgrowth of her ambition and of her intense interest in things, an interest which we have noticed as being one of the leading elements in her happiness. Reading, study, conversation, observation, writing, travel, were in turns employed in her course of self-discipline. She read incessantly, and *thoroughly*. Notice this list of books, the work of one month, and that when she was nearly fifty years of age: First book of "Lucretius;" sixth book of the "Iliad;" "Samson Agonistes;" Warton's "History of English Poetry;" "Grote," second volume; "Marcus Aurelius;" "Vita Nuova," vol. iv; chapter one of the "Politique Positive;" Guest on "English Rhythms;" Maunce's "Lectures on Casuistry." Few months fell below this in reading, and this, too, while she was writing, seeing people, conversing, and suffering, for she had the misfortune to know, as she says, that "one thing is needful: a good digestion."

As a life of earnest purpose, of continued struggle for a high living, of deepest desire to make the most of everything, and for everybody, there is none more marked than that of George Eliot. Non-conformity to the religion and the law in which we believe, must sadden her life for all, but an honest student of her character must, after reading this "Life," accord to her what she herself never failed to give to the erring—charity.

FOOTNOTES

[B] George Eliot’s Life, as related in her Letters and Journals. Arranged and edited by her husband, J. W. Cross. In three volumes. New York: Harper & Brothers.

ARBOR DAY.

BY THE HON. B. G. NORTHROP, LL.D.

Recent spring floods and the diminished flow of rivers in summer have called public attention to the cause and the remedy as never before. At the opening of the last session of Congress, its attention was called to the subject of Forestry, for the first time in any presidential message. Bills for the protection and extension of forests are now before Congress, and before many state legislatures. The last census presents striking facts which prove this to be a question of both state and national importance. The recent action of the national government shows a new appreciation of forestry. The marvel now is, that the general government did not earlier seek to protect its magnificent forests, once the best and most extensive in the world. Their importance to the nation was little understood. Even after a century of reckless waste, the United States government still owns 85,000,000 acres of timber—a mere fraction of what has been cut, or burned, without a thought of reproduction. The Forestry Division of the United States Department of Agriculture, though organized but six years ago, has already spread much valuable information before the country by its reports and by those of its special agents, commissioned to investigate the forests of the country and the means of their protection and extension. Ex-Governor R. W. Furnas, of Nebraska, for example, investigated the forests of California, Oregon, Washington Territory, and the western slope of the Rocky Mountains. His official reports on the stealing and reckless destruction of those timber lands, and also in regard to the new and extensive timber growing on the treeless plains of Nebraska, were of great public interest. The reports of Dr. F. B. Hough of New York, and F. B. Baker, of Kansas—also agents of the United States Forestry Division—have been extensively circulated and still more widely summarized in the press.

The National Forestry Congress is another index of the growth of popular interest on this subject. A large volume of the proceedings of that association at its meeting in Montreal was officially published by the Dominion of Canada. The best papers given at its three subsequent meetings have been published by the United States Department of Agriculture. The subject has been ably discussed in State Agricultural Reports, and many state and local associations have been formed to further this interest. The passage of the Timber Culture Act has greatly increased the area of planted woodland.

But of all these agencies no one has awakened so general an interest in agriculture as the appointment of Arbor Day, by governors of states, by legislatures, and by state, county and town superintendents of schools. The plan of Arbor Day is simple and inexpensive, and hence the more readily adopted and widely effective. In some states the work has been well done without any legislation. The best results, however, are secured when an act is passed requesting the Governor, each spring, to recommend the observance of Arbor Day, by a special message. The chief magistrate of the state thus most effectually calls the attention of all the people to its importance, and secures general and concerted action. *How* forests conserve the water supplies and lessen floods is aside from the topic of this paper. While the fact of the increase of spring freshets is everywhere admitted, and scientists agree as to the cause, the popular disbelief of the true theory is the great hindrance to remedial action. The bills for the protection of the Adirondack forests, in the legislature of New York, in 1884, failed by reason of the opposition of the lumbermen, and the common doubt and denial of the benefits of forests in the conservation of the rainfall. I often met the same skepticism in the Ohio valley, even among the sufferers from the flood disasters. They were attributed to the extensive use of tile drains. But both in 1883 and 1884, these floods occurred when the ground was frozen deep, and the drains were therefore inoperative.

That so simple a cause as forest denudation should produce such disastrous results seems at first incredible. It is only when the vast areas contributing to a single river are considered, that the proof of the forest theory seems clear. Take the Ohio River, for illustration. The area drained by it is 214,000 square miles, or twenty-two times as much as that which in Vermont, New Hampshire, Massachusetts and Connecticut is drained by the Connecticut River; an area which includes portions of New York, Pennsylvania, Ohio, Indiana, Illinois, West Virginia, Virginia, North

Carolina, Georgia, Alabama, Mississippi, Tennessee, and Kentucky. The length of the Ohio is about 1,000 miles, and that of its ten leading tributaries nearly 4,000, and that of the many minor affluents as much more. The smallest influences working over such immense regions, and ultimately combining in one stream, may enormously swell its volume. As the destruction of forests has been going on for centuries, the remedy must be the work of time, for it must include slow processes and agencies, each *separately* minute, which become important when multiplied by myriads and extended over broad areas. Arbor Day has proved such an agency.

A brief history of Arbor Day will show its aims. The surprising results already accomplished promise a still broader influence in the near future. The plan originated with ex-Governor J. Sterling Morton, the pioneer tree-planter of Nebraska. He secured the coöperation of the State Board of Agriculture, some thirteen years ago, when the Governor was induced to appoint the second Wednesday in April as a day to be devoted to economic tree-planting. By pen and tongue, as editor and lecturer, with arguments from theory and facts from his own practice, Mr. Morton succeeded in awakening popular enthusiasm in this work, in which he was ably seconded by Ex-Governor Furnas, of Nebraska, who has long served as Forest Commissioner for the United States Department of Agriculture.

In Nebraska a remarkable interest was awakened in the observance of her first Arbor Day, and over 12,000,000 trees were planted on that day. That enthusiasm was not a temporary effervescence. Each successive Governor has annually appointed such a day by an official proclamation, and the interest has been sustained and even increased from year to year. The State Board of Agriculture annually awards liberal prizes to encourage tree-planting. Hence Nebraska is the banner state of America in this work, having, according to official reports, as I am informed by ex-Governor Furnas, 244,353 acres of cultivated woodland, or more than twice that of any other state. The originator of Arbor Day is now recognized as a public benefactor, and hence, during the last campaign, as a candidate for Governor, ran some three thousand ahead of his party ticket. Though at first aiming at economic tree-planting, Nebraska now observes Arbor Day in schools. The example of Nebraska was soon followed by Kansas, which had over 120,000 acres of planted woodland. The Governor of that state issues an annual proclamation for Arbor Day, and it is observed by teachers and scholars and parents, in adorning both school and home grounds.

Four years ago the legislature of Michigan requested the Governor to appoint an Arbor Day. Such an appointment has been repeated each succeeding April. For the last three years a similar day has been appointed by the Governor of Ohio. Many schools, especially those of Cincinnati and Columbus, fitly kept the designated day. No man in this country has had a better opportunity of observing the influence of Arbor Day in schools than Superintendent Peaslee, who, after a trial of three years, says: "The observance of Arbor Day is the most impressive means of interesting the young in this subject. Should this celebration become general, such a public sentiment would lead to the beautifying by trees of every city, town, and village, as well as the public highways, church, and school grounds, and the homes of the people. If but the youth of Ohio could be led to plant their two trees each, how, by the children alone, would the state be enriched and beautified within the next fifty years. By our Arbor Day observance the importance of forestry was impressed upon the minds of thousands of children who then learned to care for and protect trees. Not one of those 20,000 children in Eden Park on Arbor Day injured a single tree."

West Virginia furnishes another illustration of the influence of observing Arbor Day in schools. In the face of many difficulties, State School Superintendent Butcher appointed an Arbor Day in schools in April, 1883. Without waiting for any legislative or gubernatorial sanction, solely on his own responsibility, he invited the school officers, teachers, parents, and pupils on the designated day to plant trees on the grounds of their schools and homes. He made the April number of his *School Journal* an "Arbor" number, and circulated it gratuitously over the State. The results exceeded his expectations. It started good influences on *minds* as well as grounds. This great success prompted a similar observance last April, for which greater preparations were made, with still better results. When called to advocate this measure In various parts of West Virginia last spring, I found the people and the press most responsive in encouraging this practical movement. On the day after the celebration, the papers of Wheeling, for example, commended the work in such terms as the following: "Arbor Day was gloriously celebrated yesterday, and was a splendid success. All—the oldest and the youngest—evinced the liveliest interest. Arbor Day will be one of the institutions of our schools."

At the annual meeting of the State Teachers' Association of Indiana, held in December, 1883, a kindred plan was recommended and unanimously adopted, and an efficient committee appointed to carry it out. The State Board of Horticulture heartily endorsed the measure. After a statement of the plan, the State Board of Agriculture invited me to prepare a resolution in its favor, which they promptly adopted. Governor Porter received my suggestions with special interest, and said the measure should have his cordial support. He soon after gave it his official sanction, and issued a proclamation to the teachers and people of the state, in which he predicted that the appointed day would be a memorable one, and "the beginning of a movement for a much more extended system of tree-culture, and the restoration of the varieties of trees, useful and beautiful, which have been so recklessly sacrificed that nature cries aloud for redress," closing by calling on "the teachers to do all in their power to make Arbor Day a day of the most ardent and inspiring interest." State School Superintendent Holcombe gave his personal and official influence heartily to this work. The lectures given by his invitation on this subject were fully reported by the press, for the newspapers of Indiana cordially coöperated in this movement. These combined influences secured the general observance of the appointed day, and the results were most gratifying. Such combined agencies in nearly every state of the Union would promise similar results.

At the last annual Convention of the State Teachers' Association of Wisconsin, the presentation of Arbor Day in schools led to the adoption of a resolution in favor of such an observance, and to the appointment of an efficient committee to carry out the plan. At the National Educational Association held in Madison, Wisconsin, with an attendance of over five thousand, a resolution recommending the observance of Arbor Day in schools in all our states was unanimously adopted. Such a day has been observed with great interest in some of the provinces of Canada.

The American Forestry Congress, which includes the leading arborists of Canada and the United States, adopted, at my suggestion, the following resolution: "In view of the wide-spread results of the observance of Arbor Day in many states, this Congress recommends the appointment of such a day in all our states and in the provinces of the dominion of Canada," and appointed a committee, consisting of the Chief of the Forestry Division of the United States Department of Agriculture, the State Superintendent of Schools of West Virginia, and myself to secure the general adoption of this plan, and especially Arbor Day in schools. As chairman of that committee, I have already presented this subject to the Governors of many states, and the proposition has met a favorable response.

It may be objected to Arbor Day, or to any lessons on forestry in schools, that the course of study is already overcrowded, and this fact I admit. But the requisite talks on trees, their value and beauty, need occupy but two or three hours. In some large cities there may be little or no room for tree-planting, and no call for even a half-holiday for this work, but even there such talks, or the memorizing of suitable selections, on the designated day, would be impressive and useful. The essential thing is to *start* habits of observation and occupation with trees, which will prompt pupils in their walks, or when at work, or at play, to study them. The talks on this subject, which Superintendent Peaslee says were the most interesting and profitable lessons the pupils of Cincinnati ever had in a single day, occupied only the morning of Arbor Day, the afternoon being given to the practical work. Such talks will lead our youth to admire trees, and realize that they are the grandest products of nature, and form the finest drapery that adorns this earth in all lands. Thus taught, they will wish to plant and protect trees, and find in their own happy experience that there is a peculiar pleasure in their parentage, whether forest, fruit, or ornamental—a pleasure which never cloys, but grows with their growth. Like grateful children, trees bring rich filial returns, and compensate a thousand fold for all the care they cost. This love of trees, early implanted in the school, and fostered in the home, will make our youth practical arborists.

They should learn that trees have been the admiration of the greatest and best men of all ages. The ancients understood well the beauty as well as the economic and hygienic value of trees. The Hebrew almost venerated the palm. It was the chosen symbol of Judea on their coins, and was graven on the doors of the Temple as the sacred sign of justice. The Cedar of Lebanon was justly the pride of the Jews, and became to them the emblem of strength and beauty. The Egyptians, Greeks, and Romans were proficients in tree-planting. Hence Thebes, Memphis, Athens, Carthage, Rome, Pompeii and Herculaneum, as their ruins still show, had their shaded streets or parks. Two thousand years ago, the richest Romans maintained a rural home, as does the wealthy Londoner, Viennese, or Berliner to-day, and their ancient villas were lavishly adorned. The Paradise of the Persians was filled with trees and roses. This taste for beautiful gardens was transplanted from Persia to Greece, and the Greek philosophers held their schools in beautiful gardens, or groves. The devastations of parks, the destruction of shade trees, the neglect of public streets and private grounds, the decay of rural tastes, and the utter slight of home adornments, were clearer proofs of the great relapse to barbarism than the vandalism which destroyed the proud monuments of classic art and literature.

Arbor Day has already initiated a movement of vast importance in eight states. In tree-planting, the beginning only is difficult. The obstacles are all met at the outset, because they are usually magnified by the popular ignorance of this subject. It is the first step that costs—at least, it costs effort to set this thing on foot, but that step once taken, others are sure to follow. This very fact that the main tug is at the start, on account of the inertia of ignorance and indifference, shows that such start should be made easy, as is best done by an Arbor Day proclamation of the Governor, which is sure to interest and enlist the youth of an entire state in the good work. When the school children are invited each to plant at least “two trees” on the home or school grounds, the aggregate number planted will be more than twice that of the children enlisted, for parents and the public will participate in the work.

Tree-planting is fitted to give a needful lesson of forethought to the juvenile mind. Living only in the present and for the present, youth are apt to sow only where they can quickly reap. A meager crop soon in hand, outweighs a golden harvest long in maturing. They should learn to forecast the future as the condition of wisdom. Arboriculture is a discipline in foresight—it is always planting for the future. There is nothing more ennobling for youth, than the consciousness of doing something for future generations, something which shall prove a *growing* benefaction in coming years. Tree-planting is an easy way of perpetuating one’s memory long after he has passed away. The poorest can in this way provide himself with a monument grander than the loftiest shaft of chiseled stone which may suggest duty to the living, while it commemorates the dead. Such associations grow in interest from year to year and from generation to generation. By stimulating a general interest in tree-planting among our youth, Arbor Day will yield a rich harvest to future generations. George Peabody originated the motto, so happily illustrated by his munificent gifts to promote education: “Education—the debt of the present to future generations.” We owe it to our children to leave our lands the better for our tillage and tree-planting, and we wrong ourselves and them, if our fields are impoverished by our improvidence.

Arbor Day in school has led youth to adorn the surroundings of their homes, as well as of the schools, and to extensive planting by the wayside. How attractive our roads may become by long avenues of trees! This is beautifully illustrated in many countries of Europe. In France, for example, the government keeps a statistical record of the trees along the roads. The total length of public roads in France is 18,750 miles, of which 7,250 are bordered with trees, while 4,500 are now being, or are soon to be, planted. Growing on lands otherwise running to waste, such trees are grateful to the traveler, but doubly so to the planter.

The influence of Arbor Day in schools in awakening a just appreciation of trees, first among pupils and parents, and then the people at large, is of vast importance in another respect. The frequency of forest fires is the greatest hindrance to practical forestry. But let the *sentiment* of trees be duly cultivated, first among our youth, and then among the people, and they will be regarded as our friends, as is the case in Germany. The public need to learn that the interests of all classes are concerned in the conservation of forests. Through the teaching of their schools this result was long since accomplished in Germany, Switzerland, Sweden, and other European countries. The people everywhere realize the need of protecting trees. An enlightened public sentiment has proved a better guardian of their forests than the national police. A person wantonly setting fire to a forest would there be looked upon as an outlaw, like the miscreant who should poison a public drinking fountain.

HOW TO WORK ALONE.

BY CHANCELLOR J. H. VINCENT, D.D.

Not all members of the "C. L. S. C." can enjoy the benefits of a local circle. Some live in the country, or in remote parts of the city. They can not get out at night, through lack of company, or because the house, the boys, or the baby must not be left alone; the local circle is not under wise direction, and is unprofitable; or it may be that the only accessible local circle is a close corporation, and is "inaccessible." Father or husband objects to the time wasted, or the long walk, or something else. So the student is solitary. Whatever is done must be done alone.

This is not an unmixed evil, because it may develop power in the student, or drive him or her to find associates at home, associates who are not enrolled in Plainfield as "regulars," and some of whom are quite too young to be enrolled at all. No deprivation in this world that does not make a place for some other unsought, unexpected blessing.

I purpose to offer a few hints to these solitary readers, who may, I trust, find much profit out of the restrictions of providence, and do their work well even though it be done alone. On the blank pages of your necessity you may make records of your own, worth more to you than volumes of other people's print.

1. Although alone, remember that you are associated with a great Circle numbering thousands and tens of thousands of members. You are not alone, but one of many. This thought helps you. It sets currents of sympathy in motion. It annihilates distance. It fills the very air about you with companions with whom you are in sweet fellowship, although you have never seen them. They are a great cloud of witnesses. They march under the same banner; put their names on the same great record book in the central office; read the same pages; sing the same songs; answer the same questions; recite the same mottoes; observe the same memorial days; and

turn with tender hearts to the same heavens, under the mystic spell of the vesper hour; experience the same longings after true culture, and have hearts full of sympathy for their fellow-students everywhere. This thought of oneness in work gives feelings of kinship and companionship. The solitary student in the little room—kitchen, sitting room, library, or bed chamber—is surrounded by thousands of fellow-students. They seem to look over the page with you. They seem to whisper words of good will and faith, and some of them, I assure you, are royal people. They would give you such greeting, if they had opportunity, as would make you proud and glad of your connection with the Circle. Indeed, solitariness is impossible to the thoughtful member of the C. L. S. C.

2. This sense of fellowship is increased, and a helpful stimulus given to the solitary worker, by reflecting on the character of the great fraternity of which you are a part. We now enroll more than seventy thousand members. Perhaps twenty-five thousand have practically given up the readings. Only fifty thousand remain with us. Many thousands of readers are connected with local circles who have never joined as “regulars” at the central office in Plainfield, N. J. There are thousands who are reading a part of the course, but who neither belong to the local nor general circle. I believe that these non-recorded and irregular readers make up for the lapsed thousands, so that to-day we have nearly or quite seventy thousand people doing all or a part of the required reading. This, therefore, becomes a great institution. Its territorial extent is as vast as its numerical strength. There are “C. L. S. C.s” in all parts of the world. Our office records contain names from India, China, Japan, the Sandwich Islands, and many other outlying regions, while the list in Canada and on the Pacific coast runs up among the thousands. In every state and territory members are to be found.

And who are these with whom you, my solitary student, are associated? They represent every calling in life, and almost every grade, social and intellectual. Here are lawyers, judges, physicians, clergymen, doctors of divinity, college graduates, literally by the thousand, who seek through our course to review the studies of other and earlier years. Here are seminary and high school graduates, and people old and young who dropped out of the grammar school when they were too young to understand their folly in doing so. Here are business men, mechanics and farmers who have been prospered, and who covet now a measure of culture to fit them for society, that their money may gain for them and their families more than a mere social recognition. Here are mothers good and true, who do not want to part hands with sons and daughters as they enter the higher schools, but who propose by our course of reading to keep in the literary and scientific world where their children are to be at home. Here are people of “low degree,” who toil for bread, with lengthened hours of service, that they may help those who are dependent upon them. They are in shops and kitchens, and have souls that would put honor into palaces. They want outlook as they go weighted down through busy and weary years. They do not expect always to be slaves to society and circumstance. There is blood-royal in every heart-beat, and power to hold princedoms in some near future. So, despised of men who live, they hold converse through books with gifted and kingly souls who, though dead, yet live, and who work in other kingly souls. There are many of these disguised princes and princesses in your Circle.

Here too are sufferers in homes of bereavement and pain, where arms are empty and hearts are full, where love calls but receives no answer, where disease binds the body but leaves the mind free to grieve over its loss; where lack of work gives place to temptation, and renders occupation of some sort a moral and religious necessity; where worldliness, that makes the soul barren, needs thoughtfulness to moisten, beautify and fructify the life.

In such great and gracious companionship you sit down for solitary study. Dismiss the thought, therefore, that this is solitude. Reach out your soul to greet the currents of invisible and loving influence that pour in upon you from every quarter.

3. Select and furnish your Chautauqua corner. Do not be too anxious to have it harmonize with other corners of the room. Put shelves for your books “required” and for “review.” On the lowest shelf pile THE CHAUTAUQUANS. On the wall put up the motto cards, the list of memorial days, the Chautauqua calendar, the photograph or engraving of the Hall of Philosophy, and such other Chautauqua views as you approve. Put up busts or engravings of the great leaders—Homer, Cicero, Dante, Milton, Goethe. Somewhere place a picture of Bryant, the earliest distinguished friend of the C. L. S. C. By gradual additions fill the Chautauqua corner with pictures and bookshelves, busts and mottoes, all in the line of your reading, until the

other corners and the intervening walls shall be filled with reminders of Chautauqua and the world of literature, science and art it represents. And if somehow you can place on the wall that matchless engraving representing the great Master with his two disciples on the way to Emmaus, you will, in a sense, sanctify your room, and set forth most effectively, the aim, scope and spirit of the great Chautauqua movement. In such a room, or in such a corner, can students be solitary?

4. You will greatly increase your power by systematic habits. One may "read up" at any time, but the regular daily reading which renders unnecessary what is called "reading up," is much the better way. It renders the work comparatively light; it makes the C. L. S. C. a help to other less congenial work of the day into which it falls like a refreshing shower. It forces life into a system which always expedites and lightens labor. It schools the will. It brings lower duties into proper subjection to the higher. It is every way better to do each day's work as each day comes. Thus working alone, but systematically, one keeps the hand in and does not lose grasp, taste or delight.

5. Though compelled to work alone, make casual contacts with others afford opportunities for drawing them out, for finding out what they know, or for corroborating your own views. Ask questions. Elicit opinions. Start conversation. Try to tell what you know or think. Tell your children. Tell your neighbors. As you interest them, you set them in search of knowledge, which finding, they will later on report to you, and you thus give them a start in lines of self-improvement.

6. This setting others at work in quest of knowledge for you is a most practicable way of getting knowledge and doing good to the finders thereof.

Write out ten different questions, and give one to each of ten young boys and girls of a high school, for example. They will ransack libraries, consult teachers, find out and report what you want to know, and be immensely helped by the knowledge found and the service rendered. Though alone, you need not work alone.

7. Practice talking to yourself about the things you have read. Put facts and dates into sentences. Now and then write out these sentences, or speak them off. Recite a lesson to yourself every day. Make a speech with

yourself as audience. Put facts into recitative lullabies, by which you sing baby to sleep. Don't do too much of all this, lest it weary you, but do a little of this sentence-framing and solitary speech-making, and nursery-crooning every day. You will then have a local circle of you, and yourself and your own soul. Now one's self makes very good society sometimes; there are so many powers and voices and thoughts and projects in a single soul.

8. Lift your soul up to its height, now and then, and breathe a thought of the heart that may grow into a prayer as you recall the great Circle of which you are a member. Think in silence of their multiplied and varied circumstances, perils, temptations and necessities. Think of the disheartened, the bereaved, the suffering, the doubting; those who have great power, but do not know how to use it; those who are sick of sin and worldliness, and do not know how to get into the path of holiness and peace. Think of all these, and then pray. Let your heart swell toward God in sympathy and longing.

Thus will you find in your solitude the presence of the Spirit invisible and eternal, whose name is love, and whose home is heaven, and whose children are the lowly and meek and devout, who love souls—the world full of souls—and who daily bear them in tender sympathy to the throne.

They who do these things can not be alone.

If it were not for my love of beautiful nature and poetry, my heart would have died within me long ago. I never felt before what immeasurable benefactors these same poets are to their kind, and how large a measure, both of actual happiness and prevention of misery they have imparted to the race. I would willingly give up half my fortune, and some little of the fragments of health and bodily enjoyment that remain to me, rather than that Shakspere should not have lived before me.—*Lord Jeffrey (from a letter to Lord Cockburn, 1833).*

OUTLINE AND PROGRAMS.

OUTLINE OF REQUIRED READINGS FOR APRIL.

First Week (ending April 8).—1. “Chemistry,” chapters XVIII, XIX and XX.

2. “History of the Reformation,” from page 1 to 27.

3. “The Circle of the Sciences,” in The Chautauquan.

4. Sunday Readings for April 5, in The Chautauquan.

Second Week (ending April 15).—1. “Chemistry,” chapters XXI and XXII.

2. “History of the Reformation,” from page 27 to 55.

3. “Home Studies in Chemistry and Physics,” in The Chautauquan.

4. Sunday Readings for April 12, in The Chautauquan.

Third Week (ending April 22).—1. “Chemistry,” chapter XXIII.

2. “History of the Reformation,” from page 55 to 88.

3. “Easy Lessons in Animal Biology,” in The Chautauquan.

4. Readings in *Our Alma Mater*.

5. Sunday Readings for April 19, in The Chautauquan.

Fourth Week (ending April 30).—1. “Chemistry,” chapters XXIV, XXV and XXVI.

2. “History of the Reformation,” from page 88 to 117.

3. “Aristotle,” in The Chautauquan.

4. Readings in *Our Alma Mater*.

5. Sunday Readings for April 26, in The Chautauquan.

PROGRAMS FOR LOCAL CIRCLE WORK.

FIRST WEEK IN APRIL.

1. Essay—Easter.

2. Selection—“All Fool’s Day.” By Addison.

3. A Paper on the Life of Martin Luther.

Music.

4. Fifteen Minutes’ Talk on the Cause of the Present Troubles in the Soudan.

5. Character Sketch—General Gordon.

6. Debate—Resolved, that dynamite is more productive of evil than good.

SECOND WEEK IN APRIL.

1. Selection—“Martin Luther.” From Robertson’s “History of Charles V.” Found also in Chambers’s “Cyclopedia of English Literature.”

2. A Paper on the Inquisition.

3. Recitation—“The Prisoner of Chillon.”—By Byron.

4. Character Sketch—John Knox.

Music.

5. Essay—The Vegetation of the Carboniferous Period.

6. A General Talk on Socialism.

7. Critic’s Report.

THIRD WEEK IN APRIL.

1. Essay—The Massacre of St. Bartholomew.

2. Recitation—“Robinson of Leydon.”—By O. W. Holmes.

3. Character Sketch.—William of Orange.

Music.

4. A Paper on Mount Cenis.

5. Selection—“The Chambered Nautilus.” By O. W. Holmes.

6. Conversation on New Books.

7. Questions and Answers for the month in The Chautauquan.

SHAKSPERE DAY.

Music.

1. Roll call—Quotations from Shakspere.

2. A Paper on the Life and Times of Shakspere.

Music.

3. The Story of “The Tempest.”

4. Recitation—“Perseverance.” Selected from “Troilus and Cressida,” Act III., scene 3; beginning, “Time hath, my lord, a wallet,” etc.; ending, “One touch of Nature makes the whole world kin.”

5. Essay—Characteristics of Shakspere’s Women.

Music.

6. Analysis of “Winter’s Tale.”

7. Court scene in “Merchant of Venice,” Act IV., scene 1; beginning, “Is your name Shylock?” ending with the exit of Shylock.

The plan followed by many Shakspere clubs would afford a fine entertainment. They assign the characters in any one of the plays (that of “Julius Cæsar” being exceptionally fitting for an evening of this kind) to the different members of the circle, who read the parts assigned.

To hold a Shakspere carnival would be a very interesting way in which to commemorate the day. Let each one come dressed in costume to represent any one of Shakspere’s characters and personate that character throughout the evening.

LOCAL CIRCLES.

C. L. S. C. MOTTOES.

“We Study the Word and the Works of God.”—*“Let us keep our Heavenly Father in the Midst.”*—*“Never be Discouraged.”*

C. L. S. C. MEMORIAL DAYS.

1. Opening Day—October 1.

2. Bryant Day—November 3.

3. Special Sunday—November, second Sunday.

4. Milton Day—December 9.

5. College Day—January, last Thursday.

6. Special Sunday—February, second Sunday.

7. Founder’s Day—February 23.

8. Longfellow Day—February 27.

9. Shakspere Day—April 23.

10. Addison Day—May 1.

11. Special Sunday—May, second Sunday.

12. Special Sunday—July, second Sunday.

13. Inauguration Day—August, first Saturday after first Tuesday; anniversary of C. L. S. C. at Chautauqua.

14. St. Paul’s Day—August, second Saturday after first Tuesday; anniversary of the dedication of St. Paul’s Grove at Chautauqua.

15. Commencement Day—August, third Tuesday.

16. Garfield Day—September 19.

The difficulty of holding a circle together is sometimes very great. Not a little thorough study of the needs and natures of the members must tax the leader who would hold a circle which has no interest in its work. At Richmond, Maine, our friends have experienced this difficulty. A circle of fifteen was formed in January, 1884, but did not continue its meetings. The lukewarmness of a few broke the interest of all; but ten of the members did their reading apart. These ten took matters into their own hands last fall, and now Richmond has a “Merry Meeting” circle, of twenty-two members, interested and promising.

Nashua, New Hampshire, has a Chautauqua circle. It has been in existence for two years past, with varying fortunes. Last fall, when reorganized for the season, it consisted of ten ladies, but now numbers fourteen. Though this number is less than one half that of the last year, the interest and enthusiasm are much greater. The weekly meetings are occasions of great interest and instruction. They follow, with frequent modifications, the program arranged in The Chautauquan, making the roll call and question box regular features. The only difficulty with which they meet is that they are all so busy that they can scarcely prepare for each program. They also derive much pleasure and profit in observing the memorial days. The circle is called the “Raymond” circle, in honor of the Rev. B. P. Raymond, president of Lawrence University, Appleton, Wis., founder of this branch.

The “Athenian Chautauqua Literary and Scientific Circle” of West Enosburgh, Vt., has entered upon its first year in the Chautauqua course. Although in its infancy, it shows a great deal of interest and enthusiasm. The circle was organized September 29, 1884. The officers are president, vice president and secretary. The circle began with eight members and has increased to thirteen. One of the most interesting exercises of this circle is the pronouncing match, each person being allowed to try once; if he misses he sits down. The words for the next match are the names of the sixty-six elements in chemistry.

Our travels through MASSACHUSETTS this month furnish much interesting circle news. The "Star" circle, in FOXBORO, reorganized in October with twenty-eight members, which includes all the graduates with one exception. They believe there in once a C. L. S. C. always a C. L. S. C. The weekly meetings are reported in the local paper, and more are inquiring about the work than in previous years. One reason may be that they are but eighteen miles from the "Hall on the Hill," which is in process of erection in South Framingham.——The "Henry M. King" circle, connected with the Dudley Street church, BOSTON, was organized in November, and has twenty-five members. Of these the larger part are gentlemen, not of leisure, but business men, who bring with them into the bi-weekly meetings the same energy and perseverance that characterize a successful business enterprise. These are certainly the ones who might with a good show of reason say: "No time." But on the contrary they *have* time, not only for the regular work, but for the preparation of papers requiring much time and research.——At NORTH ATTLEBORO the new "Bryant" circle is four months old, and numbers twenty-six members. They open the meetings with reading Scripture lessons and singing Chautauqua songs. Roll call is responded to by quotations from a standard author, followed by essays, recitations, blackboard exercises, questions, discussions, etc., as the committee of instruction has arranged. The secretary writes: "If we are not great, our hopes are."——"Profit as well as enjoyment we are getting from our studies," says a member of the circle at NORTH WEYMOUTH. This organization is a circle of '83, and has had time to thoroughly test the course. They have had recently a pleasant memorial service, and have been favored with chemical experiments by a chemist.——Pleasant notes of the work at WEST MADFORD have been sent us by the secretary: "Through the influence of one sturdy little lady, six or eight people met together last October and talked up the feasibility of the C. L. S. C. They elected a president and secretary, drew up a few by-laws, and are now in good running order. They meet once in two weeks. Their membership was limited to twenty, which was quickly reached. The opinion of these members seems to be that this circle is as good, if not better, than any reported in your magazine. We all work with a will, cull the best from the programs given for the local circles, and add original ideas. Each member, in the order of his enrollment, makes out the program. This gives each one an opportunity to do his share, as well as to add his own ideas. We think this feature much superior to the general mode of allowing the ‘chair’ to prepare all programs.”——Amesbury has a circle of unusual strength. We have been so fortunate as to receive a letter which gives an account of a delightful entertainment held by them in December. Our friend says: “Thinking perhaps you might like to hear from us once again, we are glad to write you of our pleasant and prosperous winter of literary work, brought about by the grand C. L. S. C. movement. Our meetings are held on the second and fourth Tuesday of each month, the programs comprise essays, music, readings and conversation, and are social and very delightful, showing a marked improvement on our ‘feeble beginning’ a year ago. Two new circles have been formed this winter, one, the ‘Delphic,’ having forty or more members. On the 18th of December we held our first public meeting in honor of ‘Our Poet’s’ (Mr. John G. Whittier) birthday, to which we invited the ‘Delphic’ circle, also the ‘Thursday Evening Club,’ an older literary society of Amesbury, and other friends, about three hundred in all. Members from the three circles took part in the program, which had been carefully prepared. We were greatly pleased to receive from Mr. W. C. Wilkinson a paper entitled ‘Whittier at the Receipt of Customs,’ which was read to us by his friend, the Rev. P. S. Evans, of Amesbury. As Mr. Whittier, owing to a previous engagement, could not be present with us, resolutions were drawn up and sent to him, as follows:

> “‘Dear Mr. Whittier;—The three literary circles, together with a goodly company of the citizens of Amesbury as their invited guests, are met to celebrate the return of your birthday. We have talked together of all that you have done and suffered in the cause of freedom and of truth. We have listened to many of your words, rendered by living voices. We have looked at your ‘counterfeit presentment’ as it has hung before us covered with evergreen—our New England laurel. Because you were not with us in person, to receive them, we desire to send you our most hearty congratulations on the completion of your seventy-seventh year. We rejoice that after your “Thirty Years’ War” you have been spared to enjoy so many years of peace, and that in the prolonged “Indian Summer,” the “Halcyon” days of your

life, you are receiving a well deserved tribute of reverence and affection. We think ourselves happy to have known you, not merely as a poet, but as a citizen, a neighbor, and a friend.

"'We feel we can not better voice our sentiments than by retaliating upon you the words you once so fitly spoke of one who has been a co-laborer with you in the cause of humanity—the mild "Autocrat of the Breakfast Table."

"'"The world may keep his honored name,
The wealth of all his varied powers;
A stronger claim has love than fame,
And he himself is only ours."'

"'In the name and by the request of three hundred citizens of your own village.'

"To which Mr. Whittier responded with the following charming letter:

"'OAK KNOLL, DANVERS, 12 MO., 23, 1884.

"'MY DEAR FRIEND:—Thy kind letter in behalf of the literary associations of Amesbury and Salisbury has just been received, and I hasten to express my thanks for the generous appreciation of my life work by "mine own people," who know the man as well as the writer. That I am neither a prophet myself, nor the son of a prophet, may account perhaps for the rather remarkable fact that I am not without honor in my own country. I scarcely need say that among the many kind testimonials of regard which, on the occasion of my birthday, have reached me from both sides of the water, none have been more welcome than that conveyed in thy letter. If the praise awarded me is vastly beyond my due, I am none the less grateful for it.

"'I know too well my own deficiencies and limitations, but my heart is warm with thankfulness to the Divine Providence which so early led me to consecrate the ability given me to the cause of heaven, freedom, and the welfare of my fellowmen.

> The measure of literary reputation which has come to me is as far beyond my expectation as my desert, and I am glad to share the benefit of it with my home friends and neighbors. With thanks to thyself personally, and to those whom thee represents, I am, very truly, thy friend,
>
> “‘John G. Whittier.’”

The “Crescent” circle, of Wakefield, grew out of a meeting held last September, and addressed by Mr. Fairchild, of Malden, in the interest of the Chautauqua movement. A circle was formed as a result of their meeting. About twenty members are now recorded on the books, although more than that proposed at first to join. The meetings are quite interesting, the programs being varied.——The “Alpha,” of Uxbridge, is a new name on the books. This cutting from a recent letter is suggestive of their spirit: “We start with six members only, but all are *very* enthusiastic. We propose to do thorough work. Our object is improvement and genuine culture. We shall use the best means to bring in others to reap with us the golden harvest, and not be selfishly content with ‘our set.’”——There are in Fall River about sixty members of the C. L. S. C., but the “Amity” circle is the first organization in the city. It at present numbers only thirteen members. A larger number certainly ought to be in the organization. The “Amity” will undoubtedly soon bring them in.——From Pittsfield a friend writes: “I am happy to report to you a constantly increasing interest in the C. L. S. C. work in Pittsfield. Our circle reorganized in October for another year’s work. To the leadership of our efficient president, the Rev. Geo. Skene, we owe our present prosperity. We have now sixty-four members, twenty-three of whom belong to Class ’88. We have one graduate, our president, who took his diploma at Chautauqua last summer. We also have one member of Class ’85, making five classes represented in our circle. Our meetings are full of interest, and the attendance is excellent, the smallest number present at any meeting this year being twenty-five. Programs are arranged by a board of seven managers, who serve for three months. Singing, prayer, roll call, with responses by quotations and reading of minutes of last meeting, always form the opening exercises. We have also used the Chautauqua vesper service, and enjoyed it. Our pastor has had the Sunday vesper service several times, and we have found it very enjoyable in both church and circle. We have had, too, experiments in chemistry, illustrating some of the articles on that subject in The Chautauquan. As another specialty we have had ‘pronunciation of Greek names,’ conducted as the old fashioned spelling matches. This proved highly entertaining, as well as instructive. We have recently changed our name to ‘Bryant Chautauqua Circle.’ We think it particularly appropriate, as Cummington, the birthplace of Mr. Bryant, and where he spent much of his life, is situated only twenty miles from this town. Another circle has been formed here since November, taking as a president one of the members of our circle. They have at present thirty members. On Monday evening, February 2d, Dr. Vincent gave a lecture, both circles attending, and after the lecture a joint reception was given him. It is expected that arrangements will soon be made for occasionally holding union meetings. Thus the C. L. S. C. prospers in Pittsfield. We find that here, as elsewhere, the C. L. S. C. is promoting the best interests of the people.”——For several years the two or three members of the C. L. S. C. in Marshfield have been accustomed to meet weekly for reading, study and conversation, but they never dignified the gathering by the name of a local circle. Within a few months they have organized under the name of the “Webster” circle, inasmuch as they are the nearest members of the C. L. S. C. to the home and burial place of that great statesman. They meet once in three weeks, and have a membership of eight or ten, including representatives of nearly every class.

The “Phelps” local circle, of New Haven, Connecticut, of the C. L. S. C., started in November with five Chautauquans, and now numbers twenty-six, with a number of others who are reading. So far they have kept very closely to the Greek part of the course, and in the meetings have had a number of map exercises, which they find very interesting.——West Winsted, of the same state, has a year-old circle, from which we have had our first letter: “Our local circle numbers sixty, thirty of whom are regular members of the central Circle. Nearly all of these members belong to the ‘Pansy’ class, and are loyal to it. We have never labored under great difficulties, always having had good meetings. We have a most efficient lady president, to whom, in a large degree, the success of our circle is due. Early in the fall of 1883 a few enterprising men and women sent for the books for the year and commenced reading, hardly daring to hope that a circle would be formed. Our village is not lacking in literary circles, having an almost countless number of different kinds, and for this very reason it seemed that another one would not meet with success, but at the first call nearly forty responded. We organized our circle that night and continued the meetings during the year, taking up the work in essays, questions and readings, and observing, as far as possible, the memorial days, by appropriate exercises. This year we reorganized in October, and, if possible, have had more interesting meetings than last year. Some of our members who have a long distance to walk in order to attend have proved themselves filled with the Chautauqua enthusiasm by their regular attendance, whatever the condition of the weather. At our last meeting we had chemistry for the topic, and devoted the evening to experiments, having twenty or more, nearly all of which are given in THE CHAUTAUQUAN. We have had sometimes, in addition to the regular literary work of the evening, a personation of some author given by a member, the remaining members guessing the author personated. One feature of our program for January 20th was a match, similar to an old-fashioned spelling match, upon the questions on 'Preparatory Greek Course' in THE CHAUTAUQUAN for October and November. From the fact that new members join our ranks at almost every meeting, we are encouraged in the feeling that though popularity is not the winning feature, the good 'Idea' has taken deep root."

A RHODE ISLAND friend writes from WARREN: "To the numerous reports from local organizations, I am pleased to add a few lines from the 'Delta' circle, organized last October, in this part of 'Little Rhody.' It consists of nineteen 'regular' and four 'local' members, assembling on the second and fourth Monday evenings of each month. Our president and vice president are enthusiastic Chautauquans, respectively of the classes of '86 and '87, the remainder belonging to the class of '88. Our programs are arranged by 'the committee of instruction' during the intermission, and reported to the circle before its adjournment each evening. In the arrangement of these great help is rendered by those published in THE CHAUTAUQUAN. Our memorial days have been pleasantly observed, and we shall shortly have a Sunday evening vesper service. We also intend to have a supper, the cooking of which is to be '*à la* CHAUTAUQUAN.' While waiting for the Chautauqua songs our president has carefully prepared by hektograph, for our use, both notes and words of several selected from his copy, and we are delighted with the harmonies. Should we discover any new departure that would be helpful to local circles, we shall write again."

Almost as numerous reports reach us this month from the “Empire State” as we received last; several are of circles hitherto unknown to our columns. The “DeKalb” circle, of Brooklyn, is one of these. It was organized in the fall of 1883, with fifteen members. Since that time the membership has increased to twenty-six.——At Batavia a local circle was formed in October last, and consists of about fifty members. These are mostly beginners in the Chautauqua course, with a few who will finish next year. They have done some good work in the way of essays, readings and experiments, and hope to do more. The work upon Greece has been made particularly interesting, from the fact that the leader, the Rev. C. A. Johnson, has described so faithfully many of these landmarks of the past as seen by him in recent years.——In October, 1884, a new C. L. S. C. was organized at Whitestown. It is called the “Hestia” circle, and has fifteen enthusiastic members, all ladies. At one meeting leaders are appointed to conduct the exercises on the various readings at the next meeting, having as many different leaders as there are different subjects in the readings for the week. The leaders are appointed in alphabetical order, so each member is required to lead in some exercise as often as once in every three or four weeks.——The “Lakeside” circle, of Fair Haven, is to be counted “one of us.” Many readers have been at Lakeside, but the circle is a new organization. Thus far the work has been, most of it, on the Greek course; they take the questions in The Chautauquan, have essays on the leading characters, selections, questions, discussions, etc. The president drew for them a large map of Greece, which was a great help in fixing the position of the different places in their minds—an admirable plan, which more presidents would do well to follow.——A delightful circle of seventeen exists in the pleasant city of Rome. Unfortunately, they have recently lost their president, a gentleman of scholarly taste, to whom the success of the first two years of their life was largely due.——At Little Genesee there is an enthusiastic circle of sixteen members. At each circle one of the members presents a program for the next session, every member taking his turn in the order in which his name stands on the secretary’s book. Although not formally made a rule, it is understood that no member shall refuse to undertake any work assigned on the program. Chautauqua songs, roll call, and “Questions and Answers” from The Chautauquan are the standard features of the programs. Essays, discussions, select reading, questions, etc., furnish variety, and conversation is always in order. At the last circle the responses were to be from “Kitchen

Science.” The responses assumed form, as well as expression, and a bountifully spread table gave opportunity for practical tests of kitchen science.

At Latrobe, Pennsylvania, a C. L. S. C. was properly organized, and went earnestly to work October 1, 1884, with twenty-five members. It being the first Chautauqua circle in the place some difficulties had to be overcome before getting rightly started. The circle is now under good progress, and doing a good work. They have enjoyable monthly meetings, where a regular program is carried out, consisting of readings, recitations, music, etc. The benefit gained by the members is far beyond expression. Both old and young are alike profited and pleased with the readings. October 22, 1884, the circle was called to mourn the sad death of Miss Alice Newcomer, one of their most beloved members.——A very interesting variation from the usual response by roll call has been introduced into a program of the Harrisburg circle. It is that each person respond by mentioning some one important event which has occurred in the past month. This circle sends a very skilfully prepared program.——At Berwick the C. L. S. C. pursues the plan of study laid down in The Chautauquan, finding it admirably adapted to complete the required reading in the given time. A friend telling of their prosperity says: “We have lost a few members by removals, and one or two have withdrawn, after a year’s study, but the backbone and sinew of the circle remain, and the body is growing vigorous and symmetrical. At the dawn of the Chautauqua year we were compelled to part with our learned and valued preceptor, Prof. L. H. Bower, who was called to the Dickinson College Preparatory School. The circle, with appropriate ceremony, presented him with a copy of ‘Knight’s Illuminated Pictorial Shakspere,’ in eight volumes, as a token of their appreciation of his services. His talented brother, Prof. A. V. Bower, was elected to succeed him as president of the circle, and the change was made without any friction whatever. We congratulate ourselves upon being members of the Class of ’86.”

The outlook which a friend from Maryland sends of the new circle at Frederick is very encouraging: “Through the energy of a lady of the Methodist church we have organized a C. L. S. C. local circle under the name of ‘Mountain City.’ We organized November 24, 1884, with nine members, elected a president, vice president, secretary and treasurer. We are glad to say we now have thirteen members, and hope soon to increase this number. We have enthusiastic meetings every week at the homes of the members; read in the circle some of The Chautauquan required readings, and carry out as far as practicable the programs for local circles, and expect to observe all memorial days.”

We have just received a very encouraging report from the Madisonville, Ohio, circle which was organized last year. They have twenty-five members, all of whom take a great interest in the circle. The committee of instruction, composed of the officers, has a full program prepared for each meeting. Two ministers of the town belong and take an active part. Miscellaneous questions have been introduced, and beside a question on the lesson, each member is required to bring one on outside matters. All questions remaining unanswered are distributed, to be answered at the following meeting. There is no doubt that if the interest in the circle still continues there will be a second circle started in the town next year.——At Defiance a local circle was organized October 1st, with a membership of twenty, all of whom belong to the general Circle of the C. L. S. C. The president is the Rev. B. W. Slagle, pastor of the Presbyterian church in the town. They have prepared special programs for the memorial days, which have proved very delightful, as well as instructive. There is a good prospect of doubling the membership by next year.

The annual report of the work of Calvary church, Detroit, Michigan, for last year, includes an account of the work done by the “Calvary” circle, a society which has been made a part of the church organization. From it we learn that the society has thirty-three active members. They have held twenty-two meetings; the programs have included—essays, 36; select readings, 28; music—instrumental pieces 21, vocal pieces 17; general talks, 4; debates, 2. The regular Chautauquan review questions have been taken up at each meeting. There has been a great deal of interest manifested in the meetings and a disposition on the part of officers and members to make them a success; every one who has attended them has been benefited, not only in the improvement of his or her mind, but also in some degree morally.

Indiana reports two circles: the “Wide Awakes,” of Moscow, a circle of four, and the “Laconia,” of Gumfield. Some five years ago, when the “Chautauqua wave” was moving westward, it reached Gumfield in a modified form. Eight persons began taking THE CHAUTAUQUAN, but did not perfect an organization; only one of the number matriculated and kept up the required reading. In the fall of 1882 they began the work vigorously, organizing a promising circle. As time advanced their influence gradually widened and extended, until this year there are over twenty enthusiastic Chautauquans enrolled at the Plainfield office. The "Laconia" meets weekly, and has endeavored to make thoroughness one of the characteristics of its work. It is composed entirely of housekeepers, but they feel more than compensated for sacrifice of time by inspiration received from the reading and study. Most memorial days have been observed. By this means the public has become interested in the C. L. S. C., and a similar society has been organized among the young people.

One of the most enthusiastic circles of ILLINOIS is a quartette of "Irrepressibles," at NOKOMIO. The circle had the novel experience of graduating in a body at Chautauqua last August. Now they are working more vigorously than ever, trying to cover their diplomas with seals.——ELGIN has four large circles, the result of the "Alpha" circle, an organization formed in December, 1883, with six members. Last fall this society increased its numbers to nine, and most zealous has been their work. A sad loss recently befell them in the death of one of the charter members, Miss Mary Warde.——The circle at SULLIVAN, was organized in October, with a membership of eleven—one "Progressive" and ten "Plymouth Rocks." They meet once a week at the homes of the members. The president appoints the members in turn to act as leaders, and the circle is composed of enthusiastic workers. Seven members visited New Orleans in the holidays, and two are spending this month in the "Crescent City."——From PROPHETSTOWN a friend writes: "We are a modest bouquet of 'Pansies,' counting only seven, but we feel the charm of the Chautauqua Idea, and propose to 'Neglect not the gift that is in us.' One of our number, Mrs. Amelia K. Seely, passed 'beyond the gates' December 15, 1884. We sadly miss her cheery presence and unfailing interest in the work."——Wednesday, January 21st, was a "red-letter" day for the Chautauquans of HINSDALE. Their usual enthusiasm was raised to a high key by the long-looked-for visit to their suburb of Chancellor Vincent, who made a stop of two hours on his way to Aurora. He was received by the class, who were out in full force, at the residence of the secretary. A lunch was served, and the time was most agreeably and profitably spent in conversation upon topics of interest connected with the C. L. S. C.

The "Oak Branch" circle was organized at Oakfield, Wisconsin, in November. There are only seven members, and all are busy people, but they are zealous and interested in the work, and thankful that they may enjoy the benefits of the C. L. S. C. They meet once in two weeks, their circle being conducted similarly to others which have been reported in The Chautauquan.

The "Centenary" local circle, of Minneapolis, Minnesota, writes us: "Our city boasts no less than twelve circles, but Centenary, the pioneer circle, still lives, and while our members are about one half what they were when ours was the only one in the city, we are going on quietly and promptly with all our work, and expect to furnish ten graduates for the class of '85. We have our cottage engaged for the coming Assembly at Chautauqua, and hope to send a good delegation next summer. We have some eight or nine members of the class of '88, and several representatives of classes of '86 and '87."——At Spring Valley, a circle of seventeen members organized last fall, the president being from the class of '84, but the members from '88. The interest in the circle is decidedly increasing.

The friends of The Chautauquan in Iowa have been unusually kind this month. The following brief clippings from their letters give an excellent outlook on the work there: "A circle was organized at Afton, in October last, consisting of eleven regular and fifteen local members. Although nearly a month behind in organizing, we intend continuing our society through July, so as to be able to commence the next year at the regular time. In making out our program for local circle work we usually follow the one given in The Chautauquan, and find it a great help, but occasionally vary our exercises to adapt it to peculiar circumstances. The average attendance is good, and most of the members seem to take quite an interest. We hope the society will prove of lasting benefit to each member."——"Through the energetic efforts of our village doctor, there was started last October a C. L. S. C. circle at Le Grand, and we feel worthy of mention in your columns. The circle consists of eleven members of the great Circle, and four or five local members. We appoint a new teacher for each book. We are learning much, and very much enjoy the circles. We have chosen for our name

‘Philohellemon.’”——“The ‘Ladies’ Chautauqua Reading Circle,’ of Sioux City, Iowa, has seventeen members. We organized in October, 1884. Our society is full of earnest enthusiasm. We meet once a week, following with slight variations the programs suggested in The Chautauquan. Chemistry is a favorite study, made specially interesting by the fact that a gentleman familiar with the subject gives us lectures with illustrative experiments.”——The “Kelly Humboldt” circle, of Humboldt, was reorganized last fall with renewed energy and vigor. About fourteen new members were admitted. “Our circle being now so large (numbering about twenty-six) as to almost require dividing, next season we intend organizing one in the adjoining town, just half a mile from here; then those living in that vicinity can withdraw from our circle to their own, leaving room for more to join us. To say that we enjoy our study, would be saying but very little; we can hardly wait for Monday evening to come, so anxious are we to meet and discuss the topics prepared for us. The programs arranged in The Chautauquan are a great help to us, although we vary them a little, generally opening by prayer and music; then, as a sentiment, we each give a current event of the week. We observe all the memorial days, and are now making extensive preparations to hold a public meeting in the church on Longfellow’s day. So that we may not be confused with the other ‘Humboldt’ circle, we have, in honor of the originator, Miss Mary Kelly, named our circle the ‘Kelly Humboldt’ C. L. S. C.”——Wapello has the “Qui Vive” circle, which enjoys the work. It was organized in September, 1884, and is composed almost entirely of members of the class of ’88.——In a recent letter from Burlington, we find some entertaining news from still another Iowa friend: “You always have something in the local circle column from Iowa. You know Iowa has two great staples, corn and Chautauquans, and we think you would surely be glad to hear of our flourishing circle, as well as others of the thousands of Chautauquans. Our circle was organized for the year’s work on Garfield day. We have the best circle we ever had, and are conceited enough to think there are no better ones anywhere. Our president is a busy lawyer. Indeed, our circle is composed of the busiest people in the town. We meet *regularly* and *promptly* every Monday evening. Burlington is a city of seven hills. Then you understand what regular meetings are here, for the circle is comprehensive and takes in all the hills. Our chemistry lessons are taught by a practicing physician who is a thorough chemist and teaches intelligently and enthusiastically. We have the willing coöperation of many of the educated people of the city, and when necessary for either our own advancement, or more perfect instruction on a topic, we find them ready to give us an address or essay. Our most enthusiastic members are graduates of colleges, or advanced academies. We recognize each memorial day. One of our daily papers freely makes any announcement we have to make, and aids us all it can. I can not undertake to tell you the good our circle is doing for us individually. Some of us, deprived of early advantages, can not be too thankful for the C. L. S. C. It is an influence for good that enters into our everyday life, and overbalances and counteracts some of the *other* influences that every soul must encounter."

With an excellent program of a regular meeting has come to us a notice of a circle at Hatboro, Tennessee. The secretary says: "With great pleasure I report a local circle in our little town. We started with two members; we now enroll thirteen. We all are deeply interested, and think the Chautauqua Idea a grand one. We call ourselves 'Golden Flower' (Chrysanthemum) local circle, and our badges are clusters of chrysanthemums."

From Greenville, South Carolina, come very cheering reports: "Our circle was organized in the fall of 1883, and we are therefore of the 'Pansy' order. We have twelve members, six young ladies and six young men. Most of the members are college graduates, and take the course to keep bright in their studies. We adhere, with occasional changes, to the following order of business: First, roll call and reading of minutes; second, examination of question box, in which each member is required to deposit at least three questions, bearing directly on the subjects for the time in the regular course; third, an essay; fourth, reading by two members appointed by the president; fifth, twenty minutes allowed for informal discussion of the lessons. We of course celebrate the memorial days with appropriate ceremonies. Some additional interest is given by having some extra literary entertainment. A Dickens party we had recently was very enjoyable. The book we selected was 'Our Mutual Friend.' Each member represented one of the leading characters in the book. Besides we acted several scenes, which added much to the enjoyment. We are all enthusiastic in our interest in Chautauqua, and fully determined to finish the course."

At ATLANTA, GEORGIA, there is a circle of fifteen in West End, the largest suburb of Atlanta. The Rev. H. C. Crumley, a pastor of the city, deserves the credit of founding this organization.

A very kindly and graceful courtesy has been extended to those Chautauquans visiting New Orleans, by the “Longfellow” circle, of that city. It is an invitation prettily framed, which has been hung in the Chautauqua alcove. The placard reads:

C. L. S. C.

GREETING OF THE LONGFELLOW CIRCLE OF NEW ORLEANS.

To any and all Fellow-Chautauquans who may be visiting The World’s Industrial and Cotton Centennial Exposition, we offer a cordial invitation to attend the meetings of our Local Circle, which are held every Tuesday afternoon, at five o’clock, at No. 393 South Rampart Street, corner of Erato Street.

Also, we extend a like invitation to all Resident Chautauquans to join our Circle, wishing to awaken renewed interest in the Great Movement.

O. F. GROAT, Secretary.
J. HASAM, Cor. Sec’y.
K. L. RIGGS, President.

NEW ORLEANS, January 26, 1885.

A very encouraging report of the circle at EUREKA SPRINGS, ARKANSAS, has reached us: “We organized the Eureka Springs Chautauqua Literary and Scientific Circle October 1st, 1884. Our circle has about thirty members, half of whom are reading the books. We follow the programs given in THE CHAUTAUQUAN. A great many spectators attend. Everybody is interested in our circle. We are talking of establishing a lecture course at this place for the summer months, probably in July, in the interest of the Chautauqua Circle. We always have between 4,000 and 6,000 people here, in the summer many more. We have very suitable grounds, near the purest water in the place. Our town is easy of access from Missouri and Kansas, as well as from other parts of this State. So far as known, we are the only organized Chautauquans in this State. Probably many persons are reading the course at different places, but we know of no circle.”

From CLARKSVILLE, MISSOURI, a lady writes: “This Pansy bed by the ‘Father of Waters’ has much for which to be thankful: Fifteen earnest workers compose our number. We are all teachers and scholars, by turns. We attempt as much thoroughness as practicable in the readings, brought out by recitations and conversation. We carry out some parts of the programs in THE CHAUTAUQUAN. Some of the Pansies hope to be transplanted for a time to Chautauqua in ’87.”

OTTAWA, KANSAS, circle was organized in time for the October work, with a membership of fourteen. “Our circle has increased, until now we are twenty-eight in number. Our meetings, held twice a month, are both pleasant and profitable, each member faithfully doing his part. We respond to roll call by quotations or class mottoes. We find the programs in THE CHAUTAUQUAN quite beneficial. The essays, recitations and music form a pleasing variety. We adopted the question match, also the question box, and find these not only amusing but profitable. This month we will try some of the chemical experiments in connection with a lecture. We are all looking forward to the Sunday-school Assembly, which meets here in June, and to the meetings of the circle conducted by the Rev. Hurlbut. The spirit of the C. L. S. C. is spreading, and we hope to report a large circle to you next year.”

A friend writes from SEATTLE, WASHINGTON TERRITORY: “I notice in your January number a communication from Mr. K. A. Burnell, in which he states that at Seattle and Tacoma he found but a single reader and one family reading the Chautauqua course, a statement from which one might infer that he was indeed so much under the ‘shadow of Mount Tacoma’ as to obscure his vision. There are at Seattle as many as forty readers, at least, who have been pursuing the Chautauqua course of study since October last. There are three regularly organized circles in this place, holding weekly meetings, and a general semi-monthly meeting in which the members of all the circles join. One of the circles, named ‘Alki,’ has a membership of sixteen. This circle has the honor and advantage of numbering among its members a noted linguist and scientist in the person of Dr. John C. Sundberg. Considerable interest is being awakened throughout the whole of the Puget Sound country in the Chautauqua readings, and it would not be surprising if, in another year, the regular Chautauquans in this section of country are numbered by hundreds.”

The “Washakie” circle, of Evanston, Wyoming, was organized on the 10th of last October. The names of twenty-six members have been enrolled. Starting late, they were behind with their studies until lately, consequently the program for each week as laid down in The Chautauquan was not followed. The meetings, however, have been very interesting. The leaders appointed for the different subjects on each evening came well prepared. Essays on Milton, Burns, and others, have been read. Prof. Halleck, of the public schools, has delivered short lectures on the scientific subjects. Prof. Capen has given experiments in chemistry. Music, and recitations from the classic authors by a fine elocutionist, have rendered the meetings more entertaining. The enthusiasm has grown with the year.

The first circle that was regularly organized in Portland, Oregon, was that established by the Y. M. C. A., last October. This circle is composed of about twenty members. The other two circles which have joined the class of ’88 are those connected with the Taylor Street and Grace Methodist Episcopal Churches. The latter was organized during the month of December, and is composed of about twenty-five members, who seem to be now deeply interested in their work. The former is the largest circle in the State, composed of about forty active and progressive young men and women, who are now deeply interested in their studies, and a notable fact of this circle is that there is no restraint in thought by the members, as is often the case where freedom of opinion is withheld, thus repelling the progress of the meeting. The able secretary of their circle deserves great credit for the time and trouble he has exercised performing that office, and volunteering to assume all responsibility with regard to books, dues, and pamphlets. The Rev. G. W. Chandler, the efficient president, is the originator of this circle. Their efforts and untiring energies have made this circle most interesting, and have brought into it some of the best scholars in the State. By perseverance and thorough study, with the watchword “Forward,” they are determined to ever press onward and upward in this grand work, and receive their reward.

THE C. L. S. C. CLASSES.

CLASS OF 1885.—“THE INVINCIBLES.”

“Press on, reaching after those things which are before.”

OFFICERS.

President—J. B. Underwood, Meriden, Conn.

Vice President—C. M. Nichols, Springfield, Ohio.

Treasurer—Miss Carrie Hart, Aurora, Ind.

Secretary—Miss M. M. Canfield, Washington, D. C.

Executive Committee—Officers of the class.

Class badges may be procured of either President or Treasurer.

The members of the Chautauqua circles have now a third of a year only in which to finish their readings and fill out their papers for the current year. So far as we have been able to learn, a much larger number of persons have been pursuing the C. L. S. C. course this year than have been in the ranks during any previous corresponding period. Those connected with journalism, in looking over their exchanges, rarely pick up a local paper that does not have some reference to the doings of a local Chautauqua circle. Then it has been discovered that those who read the Chautauqua books and periodicals have been led to go beyond the lines, and to search for intellectual treasures in “pastures new”—in books, reviews, public journals of character and excellence, and, also, to seek association with people of culture. Indeed, it is pleasantly and encouragingly apparent that the Chautauqua system is becoming, from month to month, broader, deeper, more far-reaching in its wholesome and really powerful influence, in promoting moral as well as intellectual culture.

The members of the Class of 1885 should bear these facts in mind, and accept the special degree of responsibility involved. Let this class be not only the best, but the largest that has ever passed within the Golden Gate on Commencement day! Why should it not be three thousand strong? If we begin now, in April, to make our plans and preparations, perhaps we can all “get there,” and present a solid phalanx of honest, thorough, intelligent and aggressive Chautauquans, marching toward and through the Gate and into the Hall, with banners and songs, that will promise largely and grandly for the moral and mental improvement of thousands of communities throughout the land.

“What would be the result if we report to Miss Canfield our intentions to be at Chautauqua to receive our diplomas, and something should happen to prevent?”

The only result would be that those who expected you would be as sadly disappointed as you would be in not being able to come. The fact that you intended to come and were detained by good cause would be accepted, and you would “stand excused,” and would receive your diploma in good time.

Missouri.—As one of the “Invincibles,” I would add my testimony with others of Class ’85 as having received pleasure and benefit beyond computation in pursuing the C. L. S. C. course. I commenced alone, but after a few months succeeded in organizing a circle for ’86, which keeps up a large membership, persistent and thorough in study, with rigid class drill; also remembrance of memorial days.

Pennsylvania.—What a well-spring of joy is the C. L. S. C. in the homes of those who have not enjoyed the advantages of a liberal education! The students born of this great movement are rising up all over this great land with blessings for the founder of this happy Circle. I am reading alone, as there are no members near me, but at some little distance I have interested some bright young friends of mine in the work, and I am glad to know that they are so much pleased with it.

CLASS OF 1886.—“THE PROGRESSIVES.”

“We study for light, to bless with light.”

CLASS ORGANIZATION.

President—The Rev. B. P. Snow, Biddeford, Maine.

Vice Presidents—The Rev. J. C. Whitley, Salisbury, Maryland; Mr. L. F. Houghton, Peoria, Illinois; Mr. Walter Y. Morgan, Cleveland, Ohio; Mrs. Delia Browne, Louisville, Kentucky; Miss Florence Finch, Palestine, Texas.

Secretary—The Rev. W. L. Austin, New Albany, Ind.

The new badge, bearing the motto and emblem of the class, is now ready to be sent out. The design meets hearty approval. The cost, including postage, will be 15 cents. For badges, address the president or the secretary.

The New England branch of the class will have superior headquarters at the Framingham Assembly in July. This important section of ’86 have plans and arrangements in view that will insure a most pleasant and successful class gathering at the Assembly.

CLASS OF 1887.—“THE PANSIES.”

“Neglect not the gift that is in thee.”

OFFICERS.

President—The Rev. Frank Russell, Mansfield, Ohio.

Western Secretary—K. A. Burnell, Esq., 150 Madison Street, Chicago, Ill.

Eastern Secretary—J. A. Steven, M.D., 164 High Street, Hartford, Conn.

Treasurer—Either Secretary, from either of whom badges may be procured.

Executive Committee—The officers of the class.

Class paper may be procured from Mr. Henry Hart, Atlanta, Ga.

The attention of the members of ’87 is called to the letter by Mrs. Alden in the March number of THE CHAUTAUQUAN, page 353.

President Russell had charge of the Sunday-school Normal Department at the Florida Chautauqua, Lake de Funiak, and is one of the Board of Managers.

At Milton, Mass., recently, the representatives of the Class of ’87 had a table at a church fair and cleared over $100.

It is our painful duty to record the death of two members of the Class of 1887: Miss Mary Dayton, of Binghamton, N. Y., and Mrs. Lou L. Dunn, of Bonham, Texas. The deepest sympathy not only of the class, but of all members of the C. L. S. C. is with the sorrowing friends.

TO NEW ENGLAND ’87S.—The second mid-year reunion of New England ’87s will be held on Friday, April 3d, in Union Congregational Chapel, Stewart Street, Providence, R. I. The business meeting will be held at half-past one o’clock p. m.; the literary and musical entertainment at two o’clock. A social reunion will precede and follow the regular exercises. Will all New England members of ’87 please make a special effort to attend this reunion? Providence Chautauquans are enthusiastic, and will doubtless strive to make this meeting thoroughly enjoyable. Let us, by our presence, show our appreciation of their efforts. Our Providence classmates have kindly offered to meet at the station any strangers who will communicate the hour of their arrival to Miss Nellie F. Crocker, 6 Kepler Street, Providence.

CLASS OF 1888.—“THE PLYMOUTH ROCKS.”

“*Let us be seen by our deeds.*”

CLASS ORGANIZATION.

President—The Rev. A. E. Dunning, D.D., Boston, Mass.

Vice Presidents—Prof. W. N. Ellis, 108 Gates Avenue, Brooklyn, N. Y.; the Rev. Wm. G. Roberts, Bellevue, Ohio.

Secretary—Miss M. E. Taylor, Cleveland, Ohio.

Treasurer—Miss M. E. Taylor, Cleveland, Ohio.

All items for this column should be sent, in condensed form, to the Rev. C. C. McLean, St. Augustine, Florida.

The “Chautauqua Quartette,” Avon, Indiana, organized December 5, 1884, writes: “We are four country girls, living two to three miles apart, but hold weekly meetings, alternately, at our homes.”

In Harlem, N. Y., is a class of seven, organized October 1, 1884. The secretary writes: “Each member in turn takes charge, assigning lessons and questioning the class.” In addition to the required study they take some prominent author, giving biography and quoting from works.

From Portland, Maine, we learn that they have a large and interesting circle, meeting semi-monthly.

The “Castalian,” of Philadelphia, ten members, was organized October, 1884. This circle thinks too many members make each other timid, and therefore advocates many circles of few members. They are fortunate in having a president who makes chemical experiments.

A flourishing circle of fifty members was organized in Batavia, New York, October, 1884.

The Rev. J. D. Gillilan, of Toocle, Utah, writes that “here among the Mormons a class of three is formed; one of the number was a Mormon when he joined the circle, but has since united himself with the M. E. Church.” There is a flourishing circle in Salt Lake City.

The “Wilkesbarre” circle, of Wilkesbarre, Pa., was organized October, 1884, with sixty members. This circle meets every alternate week, each member responding to roll call with a quotation from the “readings.” A physician makes fine experiments in chemistry.

A circle has been organized in Topeka, Kan., with thirty members. The secretary says: “Most of us are busy girls, figuring as teachers, office and store clerks, but find time to take the reading course thoroughly, and hope to graduate with the 88s.”

Kansas.—“I am well pleased with our class motto and name. I am a sculptor by profession and wish a higher aim, a sculptor of life, for I have caught that angel vision. I am pursuing my studies with energy and enthusiasm, and life to me is more pleasant since I have taken up the course. Whenever I feel vexed and comfortless I only need to read over Chancellor Vincent’s articles in The Chautauquan for encouragement.”

From Buffalo, Pa., a friend says that “*all* dislike the Class name, and desire it changed.”

Toronto, Canada, raises a protesting voice against our name, saying, “I am well aware of the fact that the name stands on history’s page as a synonym for grand and noble qualities, but I am forced nevertheless to object to it on account of its ‘fowl’ association. Could we not have a name *unwinged*, *unplumed*, and of no marketable value.”

One of the ’88s, who is reading alone, tells us, “In the study of the past four months I have received more instruction and enjoyment than in any amount of the general reading done in the same number of years.”

“Vincent” circle, of Portland, Maine, sends us an interesting program of a meeting held January 16th. A most exquisite Plymouth Rock engraving graces its first page.

“Longfellow” circle, of Eastern Promontory, Portland, Maine, sends us their constitution and by-laws, including the names of its 103 members.

Kansas.—I am pursuing the course alone, and feel that I need the stimulus of outside aid and correspondence. Since my school days were over my reading has been of too miscellaneous a character to result in the profit it should have done. I am enjoying the Greek History and the Preparatory Course very much. My husband has been brushing up his knowledge of the Greek language, and comes to my assistance occasionally, so it is a source of profit to him as well. Even my eleven-year-old boy has caught the spirit, and begs me to mark all the battles for him to read, and is learning the Greek alphabet. I am pleased with the name of our class—“The Plymouth Rocks.” My ancestors were among those that landed on the bleak old Rock, and I know something of the sturdy perseverance and uprightness of their character. I can only hope that the “mantle” of those old pilgrims will fall upon us as “Plymouth Rocks,” and that, like them, we may grow strong in wisdom and goodness.

QUESTIONS AND ANSWERS.

BY A. M. MARTIN,
General Secretary C. L. S. C.

I.—SEVENTY-FIVE QUESTIONS AND ANSWERS ON "SHORT HISTORY OF THE REFORMATION."

1. Q. What is the Reformation? A. It is that great religious and intellectual revolution which marks the boundary line between the Middle Ages and the Modern Period.

2. Q. What was the first aim of the reformers, and which proved a total failure? A. The purification of the church within itself, and by its own servants.

3. Q. What was the next step, and one which succeeded? A. To withdraw from the fold, and establish an independent confession, and a separate ecclesiastical structure.

4. Q. Who planted the first seeds of Protestantism in France? A. The Paris reformers.

5. Q. Who were three prominent Paris reformers? A. D'Ailly, Gerson, and Clémanges.

6. Q. What was the most obvious cause of the failure of the Paris theologians? A. They never withdrew from the Roman Catholic Church, or took steps to establish a separate ecclesiastical organization.

7. Q. How did the Mystics of the fourteenth and fifteenth centuries arise? A. As a spiritual reaction against the supremacy of the scholastic philosophy.

8. Q. What was the central scene and native country of the most notable reformatory Mystics? A. Germany.

9. Q. What four names are prominent among the early Mystics of Germany? A. Eckart, Ruysbroek, Suso, and Tauler.

10. Q. Who were the two most notable members of the school of St. Victor? A. Hugo and Richard.

11. Q. What was the chief of important general movements, without connection with prominent characters, in progress to hasten the approach of reform? A. In the field of intellectual progress, was the revival of literature, which took the name of Humanism.

12. Q. In this revival, what were the studies, as distinguished from the theological themes which had long held sway in all the universities and learned circles of Europe? A. They were purely human and literary.

13. Q. Who were three prominent champions of the new Humanism? A. John Reuchlin, of Germany, Erasmus, of Rotterdam, and Thomas More, of England.

14. Q. What three councils were formal acknowledgments, on the part of the Roman Catholic Church, of the evils within its pale, and the necessity of relief from them? A. The councils of Pisa, Kostnitz, and Basel.

15. Q. With what bitter controversy did the fourteenth century open? A. A controversy between the church and the leading civil rulers. It was the old question of authority—whether pope or king was the supreme head.

16. Q. Why was the Avignon papacy popularly called by the Romanists "The Babylonian Captivity?" A. From the light in which it was held as an ecclesiastical calamity, and from its continuance of nearly seventy years—from 1309 to 1377.

17. Q. Although the three councils failed of their prime object, what fact did they reveal to the world? A. The fact that no prospect for reform could exist in any new council.

18. Q. What way was it now clear was the only one open for improvement? A. The independence of the individual reformer.

19. Q. What now became the theater for the Reformation? A. Central Germany.

20. Q. Who responded to the universal aspiration for a leader to guide into new and safe paths? A. Martin Luther.

21. Q. When and where was Luther born? A. In Eisleben, Saxony, November 11, 1483.

22. Q. What wealthy lady befriended Luther in youth, and gave him the advantages of an excellent teacher? A. Ursula Cotta.

23. Q. After finishing his course at the University of Erfurt, what did Luther then do? A. He bade the world farewell, and in 1505 entered the Augustinian cloister as a monk.

24. Q. In 1508 to what place was Luther called as professor? A. To Wittenberg.

25. Q. After two years in Wittenberg to what city did he make a visit? A. Rome.

26. Q. What effect did this visit have upon Luther? A. He took with him, when he left Rome, an abhorrence of the superstition and immorality of the church at its fountainhead, which never left him.

27. Q. In what bill of charges did Luther subsequently arraign the church? A. His ninety-five theses, directed principally against the sale of indulgences.

28. Q. In an “Address to the Nobles of the German People” what did Luther declare which led to his excommunication by the pope? A. That the time had come when Germany ought to cast off allegiance to Rome, to start out on an independent religious and national life, and take care of its own interests.

29. Q. Before what body was Luther summoned, where his doctrines were condemned, and the sentence of ban and double ban pronounced against him? A. The Diet of Worms.

30. Q. To what place was Luther taken for safety after leaving Worms? A. To the Wartburg Castle, where he remained for eight months.

31. Q. About how many separate writings appeared from the pen of Luther? A. About one hundred and twenty, among them a translation of the Bible.

32. Q. To whom did Luther commit the task of formulating a systematic treatment of doctrine? A. To his nearest friend, Melancthon.

33. Q. Of what do the annals of literature and theology not furnish a more beautiful illustration than we find in the case of Luther and Melancthon? A. Of the manner in which a great work can be performed by the combined action of two men.

34. Q. To what were the labors of Melancthon directed, in the great cause of reform? A. To the improvement of the methods of study in the university of Wittenberg. He urged the students to the fountain-heads of truth, and placed before them the Bible as the only source of real knowledge.

35. Q. What five princes of Saxony were devoted friends of the new movement for the liberation of the conscience? A. George, Maurice, Frederick the Wise, John, and John Frederick.

36. Q. Who was the leader of the new movement in Switzerland? A. Ulric Zwingli.

37. Q. Into what did the religious conflict in the eastern cantons of Switzerland grow? A. Into an appeal to arms, that resulted in civil war.

38. Q. What followed the battle of Cappel, where Zwingli was killed? A. The peace of Cappel, which declared that each canton should decide its religion for itself.

39. Q. What name is most prominent in connection with the Reformation in French Switzerland? A. John Calvin.

40. Q. What work did Calvin publish in 1536, which became the doctrinal standard for all the Reformed Churches of the Continent and Great Britain? A. “The Institutes of the Christian Religion.”

41. Q. By what great reformer was the work, left unfinished by Calvin at his death, taken up? A. Beza.

42. Q. In the history of the Reformation, what honor belongs to England? A. That of having discovered the need of a universal religious regeneration in Europe.

43. Q. In whom did the beginnings of reform in England center? A. Wyckliffe, who was born about 1324.

44. Q. What were Wyckliffe’s greatest services to the coming Reformation? A. First, his translation of the New Testament, and afterward the whole Bible, into English.

45. Q. What was a striking feature of the English Reformation, from the outside? A. Its political character.

46. Q. What three names are prominent in the first period of the English Reformation? A. Colet, Sir Thomas More, and Cranmer.

47. Q. What was the most powerful single agency in bringing about the English Reformation? A. The publication of the Bible in the language of the people.

48. Q. What followed the ascension of Mary to the throne of England? A. A violent persecution of the Protestants, during which, it is estimated, about eight hundred persons were burned at the stake.

49. Q. What faith did Elizabeth, the successor of Mary, recognize as national? A. Protestantism.

50. Q. From what sect did the puritan Pilgrims of America come? A. The Brownist sect.

51. Q. Who was the first Protestant leader in Scotland? A. Patrick Hamilton. He suffered martyrdom.

52. Q. Who was the natural successor to Hamilton? A. John Knox. By the time of his death the triumph of the Scotch Reformation was complete.

53. Q. What was the chief aim of the Brothers of the Common Life, a society of the Netherlands, founded in 1384? A. To improve the morals of the people, and looked intently upon a thorough reform.

54. Q. What preparation was there for the Reformation in the Netherlands? A. In no land was there such a complete and popular preparation for the Reformation as in the Netherlands.

55. Q. What character did the Reformation assume in the Netherlands? A. A political character.

56. Q. What order against all sympathy with the Protestant cause was made binding upon the Netherlands? A. The Edict of Worms.

57. Q. Who, of Rotterdam, belongs to the front rank of reformers? A. Erasmus.

58. Q. How alone was Erasmus important as a Reformer? A. As a profound and versatile scholar.

59. Q. What is one of the most unpleasant chapters in the history of the Reformation? A. The unfraternal relationship between Erasmus and Luther.

60. Q. From what did the real danger to the French Protestants come? A. From a firm alliance between the authorities at Rome and the French throne.

61. Q. What were the Protestants in France called? A. Huguenots.

62. Q. What great massacre of the Protestants took place in France on the 24th of August, 1572? A. The Massacre of St. Bartholomew.

63. Q. By whom were the Italians prepared to give hearty credence to the new doctrines of the Reformation? A. Savonarola.

64. Q. What causes led to the failure of the Reformation in the Spanish Peninsula? A. Protestantism was largely a measure of scholars and thinkers, while the persistent energy of the Spanish authorities, reinforced from Rome, made thorough work of suppression.

65. Q. In what was the groundwork of Protestantism in the three Scandinavian countries—Sweden, Denmark, and Norway—already laid? A. In the dissatisfaction of the people with the prevailing order of civil and ecclesiastical government.

66. Q. Into what two Scandinavian countries was the Reformation introduced and formally adopted? A. Sweden and Norway.

67. Q. Who was the great reformer of Bohemia? A. John Huss.

68. Q. As what did his followers afterward become known, under Zinzendorf? A. As the United Brethren.

69. Q. What was the political effect of the Reformation? A. To elevate the people to a thirst for liberty, and a higher and purer citizenship.

70. Q. Of what did the Reformation become the mother? A. Of republics.

71. Q. To what does the American Union owe a large measure of its genesis? A. To the European struggle for reform.

72. Q. What was one of not the least benefits conferred upon the world by the Reformation? A. The promotion of learning.

73. Q. What sprang up throughout Germany, as an immediate fruit of the Reformation? A. Universities.

74. Q. By what celebration have the memories of the Reformation been recently renewed? A. By the celebration on November 11, 1883, of the four hundredth anniversary of the birth of Luther.

75. Q. How was the day observed? A. With becoming festivities in all the Protestant countries of the world.

II.—TWENTY-FIVE QUESTIONS AND ANSWERS ON “CHEMISTRY,” FROM PAGE 157 TO THE END OF THE BOOK.

76. Q. What are some of the most important uses of borax? A. In the manufacture of porcelain, and in other of the industrial arts, and as a

remedial agency in medicine.

77. Q. In addition to the well known substances sodium and oxygen, what element does borax contain? A. A special and peculiar element, called boron.

78. Q. What are two of the most important sources of borax? A. Borax Lake, in California, and the borax lagos in Tuscany.

79. Q. What element constitutes about eighty per cent. of our atmospheric air? A. Nitrogen.

80. Q. As a simple and uncombined substance, by what is nitrogen characterized? A. By extreme inactivity. It does not burn; it does not support combustion; it can not be made to enter into chemical union with other substances, except by specially devised and circuitous processes.

81. Q. Of what is nitrogen a constituent? A. Of a very large number of compounds, which are themselves often characterized by a high degree of activity.

82. Q. What are two important compounds of nitrogen? A. Ammonia gas and nitric acid.

83. Q. In addition to oxygen and nitrogen what are some of the other substances always present in atmospheric air? A. Vapor of water, carbon dioxide, and ammonia gas; minute quantities of a vast multitude of other gaseous substances; and it is likewise charged most of the time with still more minute quantities of solid dust materials of various kinds.

84. Q. To what do the principal explosives owe their activity to a very large degree? A. To the presence of nitrogen in them.

85. Q. What are the four explosives of chief importance? A. Gunpowder, the fulminates, gun cotton, and nitro-glycerine.

86. Q. What are the three principal constituents of gunpowder? A. Potassic nitrate, charcoal, and sulphur.

87. Q. Why is phosphorus a most interesting chemical element? A. Because of its exceptional chemical properties, the very important part it

plays in the chemistry of animal and vegetable life, and its employment in the friction match.

88. Q. In what country is the manufacture of friction matches carried on to a very large extent? A. In Sweden; and that country, it is now stated, produces about seventy-five per cent. of all the matches made in the world.

89. Q. What is probably the most familiar and representative form of carbon? A. That known as charcoal.

90. Q. How is lamp-black produced? A. It is a product of the imperfect combustion of substances like oil, tar, resin, and the like, which are very rich in carbon.

91. Q. What are two well known compounds of carbon? A. Anthracite coal and bituminous coal.

92. Q. Of what origin do both of these combustibles, when carefully studied, show distinct evidences? A. Of their vegetable origin.

93. Q. What is the diamond? A. It is nearly pure carbon, crystallized.

94. Q. What are some of the other natural forms in which carbon is found in large quantities? A. In petroleum, marble, and limestone.

95. Q. When combined with oxygen alone, what two compounds only does carbon form? A. Carbon mon-oxide and carbon di-oxide.

96. Q. What is the material on which the manufacture of illuminating gas is based? A. Bituminous coal.

97. Q. In the distillation of coal for the manufacture of gas, what three distinct classes of substances are produced? A. Solids, which are left in the retorts; liquids, which are condensed in the various coolers; and gases, which pass on to the gas holder.

98. Q. What coloring matters are obtained from the liquids produced by these processes? A. Alzorine, affording Turkey red and other colors, and the well known analine colors.

99. Q. To what quantity does silicon exist in our globe? A. In a quantity equal to about one fourth its entire weight, including its atmospheres and its oceans.

100. Q. What is the principal earthy matter of our planet? A. The compound of silicon and oxygen, existing either alone in the form of sand, quartz crystal, and similar minerals, or else in combination with other well known abundant earth materials, such as oxides of calcium, magnesium, and aluminum.

EDITOR’S OUTLOOK.

PUBLIC MEN IN LITERATURE.

Until recently Americans have had good grounds for complaining that their public servants were almost a minus quantity in literature. The complaint had an especially sharp edge in view of the fact that at an earlier period our Washington, Jefferson, Hamilton, Adams, and others, had been among the foremost writers of the country; and it was still further aggravated by the contrast we seemed to present to France, England, and Germany, where a public man is usually also a literary man. The rule in France is that an eminent politician is an author, and the most distinguished statesmen and princes have written books. Even Louis Napoleon wrote a book on Cæsar, and one of the best accounts of our late war is the stately volumes of Count de Paris. In England, the rule is the same. The queen herself takes pride in the books she has produced. John Bright is almost alone in having no literary tastes, but his speeches will long survive in the volumes they will fill. Disraeli and Gladstone, Bulwer and Macaulay, Fawcett and Dilke, are only a few contemporary names in along list of distinguished statesmen who have excelled as writers for periodicals and as producers of books. In this country, from about 1830 to 1880, our public men wrote little. Benton’s “Thirty Years,” Webster’s speeches, and Sumner’s orations, and some other less famous works, do indeed redeem the half century; but when we have said all that can be said in praise of exceptions, the rule seems to have been that an American politician was not a writer, and a phrase of contempt attributed to an eminent Senator expresses the feelings of our politicians against “them literary fellers” in a form which is full of a significance from which we prefer to turn away our ears. Too many of our public men have despised literature, and justified literature in returning the sentiment with interest.

We are entering upon a happier period. The American statesman is returning to authorship. It is a wholesome change. Mr. Blaine’s history will occur to many readers as an illustration. It is hardly less noteworthy that his late associate on the Republican ticket has written for The Chautauquan able papers on a public question which is a living issue. A very long list might be made of public men who are in good fame as writers. The witty S. S. Cox will at once occur to all our memories, and another eminent Democrat is said to be writing a history of his times. General Grant finds relief from the terrible strain of his financial misfortunes in writing the history of his battles. We have employed some of our most gifted authors as diplomats; as, for example, Motley, the historian, James Russell Lowell, the poet and literary critic, and George P. Marsh, the man of universal knowledge; but it may, probably will, come to pass that some of their stamp will more and more appear in our public life at home. We have kept poets, philosophers, and novelists alive by giving them clerkships in Custom Houses. Nathaniel Hawthorne, Howells, "Ik Marvel" (Donald G. Mitchell), and others, rose to the dignity of consulships. Francis Lieber was tolerated in a Custom House clerkship in New York. We are probably coming to the time when such men may be members of Congress and shape the legislation of the country. Literary men are usually the most practical of men; that they are dreamers of impossibilities is the strangest of our popular delusions. A few exceptions have been carelessly considered as making the rule for the class. The sort of practicality—tempered by philosophy—which the literary man brings to affairs is what our public life most needs. All clean knowledge is a light where it abides, and the value of unclean knowledge (such as some practical politicians boast themselves in), is a forlorn minus quantity.

The advantages to be anticipated from the increase of the literary spirit in our public men are too numerous to be here set forth in detail. A few suggestions must suffice for our present purpose. In the first place, public men are experts, and have therefore valuable knowledge to impart. We are all well aware that General Grant knows important things about his battles which other men do not know. It is equally true that any clerk in a department, or any member of Congress has an intimate acquaintance with many concernments of considerable moment. A man who has served ten years in Congress could instruct and please us all if he had the art of describing the methods of law making. It is not a pleasant fact that the writing of a book on "Congressional Government," which is at once philosophical and entertaining, should have been left to a college professor; nor is it pleasant to feel that the author of this book, Professor Woodrow Wilson, is probably the last man whom Baltimore will think of sending to Congress. The men who see the meaning of things and connect them with principles, and align them with historical precedents, are needed in Congress to give it dignity and character. In short, we ought to send our best men to Congress, and we are approaching an era when “best men” will generally be possessed of literary tastes and habits. Our public life is rich in materials for useful books and entertaining novels. Most of these materials lie neglected because we send inferior men to our public work. Another distinct advantage will be found in the preservation of many bright men whom we send to Congress from rusting out of intellectual brightness and becoming mere political workers. The majority of men sent to Congress are college men; they have had some literary tastes and habits. They have often been journalists. The public opinion which hedges them in converts them into office hunters and office peddlers, and consumes their lives in routine and political anxiety, to the detriment of all generous and aspiring manhood. The man whose brain work, in periodicals and books, will secure his position before his constituents, is a man saved.

The change which is going on is mainly the work of the enterprising managers of periodicals. Most good literary work in our day first reaches the public in periodicals. Much excellent work is found only in periodicals. The editors have discovered that there is valuable matter to be had by encouraging public men to write. Our articles by General Logan, for example, contain a view of a great question which is best seen in all its aspects by a public man who has seen all sides of it in a Congressional committee. Many similar articles have in recent years appeared in literary periodicals. An invitation to present his views to the public through such a periodical as The Chautauquan is a challenge to candor and a stimulus to thoroughness. The work done educates the statesman while it informs the people. It creates an intelligent sympathy between public servants and those whom they serve. It carries on that form of education in which light and wisdom are put into the first place, while turgid bombast and self-seeking buncombe are rendered odious to the people whom they have deceived.

THE DECLINE OF SPIRITUALITY IN THE CHURCH.

Among the unpleasant reflections which the reading of Bishop Hurst's "History of the Reformation" will be apt to awaken in many minds is that there has been a great decline in the spirituality of the church. In those days religious earnestness was at its maximum; we seem to be passing through a period when it is at a minimum. How far the seeming is accurate, it may not be easy to determine; but appearances are against the modern church. All our religious services lack in spirituality. The lack is in the sermon, the song, the prayer. Family religion has *apparently* little of the intense power of the former days. The conversation of Christians is less frequently on religious subjects. We are carefully weeding out of familiar speech the old references to Providence, Death and Judgment. We fall into silence when one among us introduces such themes. Religious feeling and expression have disappeared *from the surface* of our life in a most astonishing way. We are not made, the unconverted are not made, to feel the force and warmth of religious conviction. The sermons are logical, literary and cold; if there be warmth, it seems to be rather intellectual than religious. The more able religious editors complain that they can not get written for them articles which are at once readable and spiritual; while other editors condemn any articles of that type as savoring of a "dreary religiosity;" and others say that the expression of religious experience has "hopelessly gone into the keeping of cranks and weak-headed and morally-unsound persons." One man says: "I can imagine nothing sweeter to hear than religious experience ought to be; but when I listen to it I hear either out-worn phrases or senseless fanaticism; and these have been driven from the respectable churches and are monopolized by ignorant egotists in the out-of-the-way corners of the country."

A partial explanation of the facts lies in the statement just quoted. But it is very partial. Why should fanatical zeal kill genuine earnestness? If we think and feel earnestly in religion, why do we not talk of what is burning in our hearts, as the fathers did, in language of our own? A round of set phrases does denote vacancy of spirit, but the earnest spirit is not banished from our heart by the formalism of another's speech. It may be pleaded for us that we are in a transition state; that the Reformation did develop a form of earnestness, and that our earnestness can not work in that harness and is reverently silent because appropriate speech is wanting. But why do not hundreds of ministers who have all gifts of intellect utter spiritual thought and emotion? Why are they forever dealing rather with opinions than facts of the spiritual life? We ask such questions in no censorious spirit; they are pressed home to many anxious hearts, and the wonder grows whether modern Christianity is tongueless respecting its experience because it is backslidden and even skeptical. We could frame, as has often been done, explanations; but we still doubt whether they really explain. The spiritual activity is of all inner motions the one least likely to lose all power to express itself.

It is true that a vast body of believers have the spirit of giving and of work. They make noble offerings, they teach the children in Sunday-schools, they make sacrifices of time and ease and money to carry on churches. In these things no former generation had so glorious a record. It is probably true that this vast body of believers contains as large a proportion as any Reformation body of persons who would die for their faith. It can not be said of such a body of persons that faith is not in it. Making all allowances for conventionality and religious fashion, there remains proof enough that the modern church believes. Nor can we doubt its spiritual poverty. It is poor in the divine life. This state of things can not last. We are in a condition where faith must fail if love does not come to the rescue. The greatest of all revivals may be at the door. The church wants nothing but vital godliness—experience of divine things. It has so much of zeal, benevolence, self-sacrifice, philanthropy, that we can not so much as hint at despair. Is it possible that some of our philanthropies are too consuming and exhaust us? If we will stop to think and take account of ourselves, we shall probably find that we lack spirituality because we do not want it. That discovery may be the one thing needed to arouse us to strenuous spiritual endeavor.

THE SHAKSPEREAN ANNIVERSARY.

The fourth century of Shakspere will be remembered either as the century of Shaksperean skepticism or as the one in which the play-actor was stripped of Bacon’s clothes and reduced to his proper condition of play-actor. That we can so much as entertain this latter thought proves that the skepticism has made considerable progress. We do not believe that Bacon wrote the Shaksperean plays; but we are obliged to pay to those who do believe it such respect as is paid to Strauss with his theory that Jesus is a mythical person. Another Shaksperean year is completed on the 23d of April, and its most significant event is an increase of skeptics. We are doubtless to have a thorough sifting of the facts and a large debate. No lover of the great dramas need regret the discussion. It will provoke the study of them and enlarge their fame. They are the great dramas of the world. No others equal them in breadth and fervor. Whatever stimulates the study of them must be useful to the higher forms of literature. One way of looking at the subject of the authorship of these plays is to regard the question as of no absolute importance. The plays are what they are, whoever wrote them; just as the Homeric poetry does not lose a line through the Homeric skepticism. It is an audacious thing to attack Shakspere as a wearer of another man's clothes, after three centuries of his renown. He lived in the public eye. All London knew him. Some envied and sneered, but none doubted him until some three hundred years after his birth; if there were doubts they were so feeble that nothing came of them. Is it the function of the press and the reporter—making great and small seem alike—which has made Shaksperean skepticism almost respectable, if not entirely so? Whatever be the cause, "the news" spreads that Bacon wrote our Shaksperean works, and the debate is growing into bulk, if not into a serious concernment. We are not a bit touched with the skepticism; it seems to us unreasonable, beyond ordinary measure in unreason; and yet we must recognize the growth of the new theory of the authorship of our glorious drama.

The change next to the foregoing in importance which marks the fourth Shaksperean century is the new way in which the great mass of his admirers come to know and enjoy him. He has passed from the stage to the study, the parlor, the school-room. He is acted a little; he is read a great deal. In his first and second centuries he was known almost exclusively through the stage; in his third, the stage and the book divided about equally the office of making him known; in the fourth, Shaksperean acting has become insignificant in comparison with the general reading and teaching of Shakspere. His works are coming to be studied in all high schools, academies and colleges. Shakspere is in nearly all libraries, be they large or small. One may almost say that he is at home in nearly every house where English is read. There is hardly a town in the country which does not boast at least one well-established "Shakspere Club." Year after year the members meet weekly to read and talk over the merits of the one writer who never tires them. The scholars of all lands know him in the printed page; all the great tongues have books of criticism in which he occupies a conspicuous place. One view of this transition from the stage to the study and the school is that Shakspere was always too large for the theater. It was in the largest sense impossible to act his plays. All acting narrowed and misrepresented him. The larger field of the book is his proper home. He gains by the liberty and healthfulness of the modern environment. The two changes which we note will bear on each other. Too many persons are coming to know what and how Shakspere wrote to permit any star-chamber of criticism to settle the authorship of these plays in darkness and secrecy; the power to form a judgment is being created in the minds of the great jury whose verdict will probably kill off the Shaksperean skepticism. We do not believe it will survive to 1964, the end of Shakspere's fourth century.

ART IN THE UNITED STATES.

If it is *possible* for this nation to become artistic in tastes and habits, we shall not fail. There is no branch of special education more enthusiastically advocated and patronized. Of course the end in view will require more advocacy and more patronage—a great deal more—but we are doing so much that the necessary more will doubtless be done. It should be remembered too, that if blood tells in the matter of art culture we have no lack of blood drawn from the artistic nations. Flemings, Germans, Italians, Spaniards, and even Greeks, come to New York in large numbers; and if Anglo-Saxon blood were condemned as unartistic by inevitable natural incapacity, we should still be able to produce great artists in abundance—if method, zeal, and patronage could do this thing. We will not prophesy; let us wait and see. It is understood, of course, that much, perhaps most of our art, is industrial. We need to educate a large corps of designers for useful goods, which are also ornamental; and this type of artist is so well rewarded when he displays inventive faculty that we are likely to surpass all other peoples in this department of work. It is not easy to separate completely in our thinking this branch of art from that which aims only at artistic pleasure. A design for goods may be perfect art, and satisfy all the requirements of the æsthetic sense. But it is obvious that decorative art is very close to industrial art in nature and purpose. And the purpose has always been condemned by high art, for it looks straight at the sale of the goods at good prices.

It is complained every year in New York, when the art exhibitions come on for criticism, that the pecuniary motive for work, and the avidity of artists for good sales, depress the imagination of the lovers of good work. In substance, then, our trouble as to art—that we are a commercial race—seems to get into the schools and infect their atmosphere. The evil is not that success is rewarded; but that success is not possible to an artist who thinks always of his reward. Art, like religion, requires a spirit of self-renunciation. Success in art is not possible to one who consumes his energy in thinking about the sale of his pictures. To become rich by art one must be first willing to starve to death in the service of art truth. We are not demanding such sacrifices; we are only suggesting that without the spirit of them the pure art of this country will not attain the eminence which our enthusiasm seeks to reach.

"Sordid treatment" of themes is inevitable in the sordid atmosphere which, we are told, is breathed in all our circles of art. Besides, the museums are founded by good natured people who are poor guides and directors and yet must control, because they are patrons. One art journal declares that enough energy has in the last five years been expended in behalf of art to have given it a firm establishment. It adds that most of this energy has been wasted. Art students and art teachers and art institutions and publications multiply, but they do not give us high art. We read this complaint and recall the story of the oil-king in western Pennsylvania who ordered the teacher of his daughter to "buy her a capacity, without regard to expense." If art comes to us to stay it will come by a slow change of thought, feeling, and aspiration. It is probable that this change has begun; let us hope that it will ripen to a gracious and mellowed maturity. The art-life will find ample room in our hospitable civilization, if it can acquire the courage to live its own life and escape being a parasite on the robust body of our commercial life.

EDITOR’S NOTE-BOOK.

The force of the “Chautauqua Idea” is not abating; on the contrary it works its way into new homes and distant fields—for instance, we have the C. L. S. C. Class of 1888, which commenced to form last July, and now numbers about 20,000 members. The “Florida Chautauqua,” in Florida, is a new plant, and now our C. L. S. C. friends in Canada are raising a fund of $50,000 with which to purchase and furnish grounds near Niagara Falls, for an Assembly after the Chautauqua fashion.

Many prominent Chautauqua workers are at the Florida Chautauqua now in session at Lake de Funiak. Among them are the Rev. Frank Russell, Mrs. G. R. Alden (Pansy), the Rev. A. A. Wright, Dean of the Chautauqua School of Theology, Prof. W. D. Bridge, Prof. W. F. Sherwin, Mrs. Juvia C. Hull, the Rev. S. G. Smith, D.D., the Meigs-Underhill Combination, Prof. C. E. Bolton, Mrs. Emily Huntington Miller, Dr. O. P. Fitzgerald, Prof. R. L. Cumnock, Wallace Bruce, Hon. John N. Stearns, Col. G. W. Bain, Mrs. Emma P. Ewing, Hon. Lewis Miller, etc. Many prominent lecturers, singers and readers as yet not known publicly at Chautauqua, are at this Southern Chautauqua, or on the program for the closing week. Dr. Gillet, in preparing the royal feast of four weeks’ continuance, has subsidized the country generally for his purposes, and all prominent denominations are tributary thereto. Nearly or quite all the departments (save the C. S. L.), known at Chautauqua, are in successful operation. The Assembly already takes high rank in design and desire, and professors, lecturers, readers, singers, helpers, are among the very best. No Assembly in the land starts off with a more brilliant outlook, or with such strong financial backing.

In our December issue we called attention to the effort being made to establish an Assembly in Canada, at Niagara. The plan is developing very satisfactorily. The proposition involves the acquirement from the Dominion Government of the piece of land known as Paradise Grove, containing about eighty acres, situated upon the bank of the Niagara River just outside the town of Niagara. The company which holds the lease has signified its willingness to consent to a transfer. Toronto is also thoroughly aroused to the importance of the movement, so much so that at a very largely attended public meeting called in February to discuss the matter the citizens pledged themselves almost unanimously to give a bonus of ten thousand dollars to the company. In addition to this promises of stock subscriptions have been made of at least as much more. It is easy to see that, if carried out, this project will prove a great boon to the old town. Already a large number of persons on both sides of the line have signified their intention to erect cottages and make it their summer home.

The Chautauqua Circle has just added a new and important branch to the many into which it is already divided. This is an art "circle," to be called the Chautauqua Society of Fine Arts, in which it is proposed to give lessons in drawing and painting by correspondence. Every branch of art will be taught, from elementary drawing to oil-painting. The plan is a thoroughly practical one, and will be carried out in the best interests of the fine arts. Mr. Frank Fowler has been appointed director, and Messrs. R. Swain Gifford, Thomas Moran and Will H. Low will act as a committee of award. The course of study will extend over two years, at the end of which time diplomas will be given and prizes awarded for the best work in the different classes. The membership fee is fifty cents a year. Application for circulars and further information should be made to Miss K. F. Kimball, Plainfield, N. J.

With the fall of Khartoum, the death of General Gordon, the Irish dynamiteurs and their explosions in London, together with the land troubles in Ireland, a growing dissatisfaction with the Gladstone ministry, and the threatening aspect of Russia, England has enough of perplexing questions on hand to keep her Queen, Ministry and Parliament employed for an indefinite period of time. To be an English politician to-day is to have unrivaled opportunities for strong and vigorous action. Apropos of the Soudan trouble our readers will find the article by Dr. Wheeler, on England and Islam, in this impression, both spirited and profitable reading.

Roller skating is now claiming the attention of, first, physicians, who seem to be divided in their verdict as to the injurious physical effects of the exercise; second, of clergymen and laymen in the churches, who object to the “rink” on account of the associations, quite as much as the doctors do to the skating; third, of economists. In a railroad car bound west recently, we overheard a conversation between two cattle drovers on the “Roller Rink,” one of whom held up a paper named the *Rink and Roller*, the organ of the new sport. These two men discussed the financial side of “roller skating,” one insisting thus: “A boy will chop wood for seventy-five cents a day, or work at the bench for that amount, and then spend fifty cents in the evening for himself and girl to attend the rink; they keep it up; what’s the good; it is a craze.” Rinks are being built in all our towns and cities, but it will come to an end like every craze. Some will be injured physically—perhaps some will date a moral lapse to an unfortunate acquaintance made in the promiscuous company; while all who go will spend their money. *What is the profit?*

The venerable Mr. George Bancroft, having passed his eightieth birthday, still preserves his physical vigor and looks like one of the patriarchs of Washington. His mind is active and retains its strength, though now enjoying a much needed respite from literary work. Mr. Bancroft has finished his “History of the United States,” which has been a long and laborious task. Some new historian must appear, who can live in the midst of political changes, and like this great man, preserve an impartial judgment, as a historian, to continue Bancroft’s standard history of the United States.

The closing act of President Arthur’s term of office was one of simple justice to a worthy man. The following note explains it all:

> To the Senate of the United States—I nominate Ulysses S. Grant, formerly General Commanding the armies of the United States, to be General on the retired list of the army, with full pay of such rank.
>
> Chester A. Arthur.

Executive Mansion, March 4, 1885.

Congress had passed an act which made it possible that General Grant could be placed on the retired list. The Senate by a unanimous vote confirmed the President’s nomination.

The number of books and periodicals supported on a given subject, is a good sign of its interest to the public. Following this indication we conclude that the public interest in sports and amusements is fully double what it was a year ago. A tabulated statement of the publications of 1884, compared with the books issued in 1883, gives the works on sports in the two years as twenty-two in 1883 to fifty-one in 1884. This suggestive comparison is but one of many signs that we are awakening to the absolute necessity of healthful exercise, if we would lead useful lives ourselves, and would propagate a sturdy race.

Every summer many ladies and gentlemen engaged in educational work make a vacation tour to the Old World. Those having such intentions for the coming summer will perhaps accept a few words of advice. In order to economize your time and derive the full benefit of your trip abroad, the best thing to be done is to join a party, the management of which is in the hands of an experienced traveler. The question naturally arises, Where is there a party formed in which we will find most advantages for the money expended? We do not hesitate in saying that we can recommend no better than Professor de Potter’s parties, organized each year in Albany, New York, and which have the reputation of being ably conducted.

There is going on in the newspapers just now a very suggestive contest over the spelling of a word. Shall it be dynami*teur* or *ter*? Both forms have reliable followings, though no reasons have been advanced for either termination. The word is a good example of several interesting features of word-making. It illustrates how each new development in history requires a vocabulary, and how the vocabulary is formed from the facts involved. Further, the difference in the termination shows how each word must have its period of instability before usage selects the form which shall be permanent. This Irish agitation has, by the way, introduced several new words into the language.

We Americans believe very firmly in ourselves. But sometimes we can not help wondering if this vigorous, athletic government of ours, and these growing institutions, seem to others a success. It will be gratifying to read Mr. Matthew Arnold’s opinion of us: “A people homogeneous, a people which had to constitute itself in a modern age, an epoch of expansion, and which has given to itself institutions entirely fitted for such an age and epoch, and which suit it perfectly—a people not in danger of war from without, not in danger of revolution from within—such is the people of the United States. The political and social problem we must surely allow that they solve successfully.”

Last year women were for the first time admitted to the Oxford University examinations. Since they have been allowed to hear certain college lectures, and are now finally admitted to the classes. It is a surprising concession, but it is the course of the future. Women in England have proven conclusively their ability to cope with university studies. They have zealously and quietly improved each added liberty. This last recognition comes as the inevitable effect of a law which works through all human affairs, viz.: a demand creates a supply.

President Arthur closed his term of service with the confidence and respect of the American people. He performed the difficult task of filling the highest office in the government with prudence and ability, when, in fact, he was not the choice of the people for the place, but it fell to his lot in the order of a mysterious providence. Among the Vice Presidents who have succeeded to the presidency Chester A. Arthur will be honored in history as a wise statesman, faithful to the people whom he served. President

Cleveland's administration is the dawn of a new political era in the country, but we believe that he will make a safe President.

There has been recently organized in New York State a State Forestry Association. President White, of Cornell University, has accepted the presidency. The society proposes to make a vigorous effort to arouse the people to the necessity of laws which shall preserve their forests from the lawless destruction which has robbed thousands upon thousands of acres in the Adirondacks of their wealth of timber. Such a society is, without doubt, the only means by which a proper sentiment can be aroused. The cause of the wholesale depredations has been lack of thought. As one of the lumbermen put it: "It all comes to this—it was because there was nobody to think about it, or do anything about it. We were all busy, and all to blame. But I could do nothing alone, and my neighbor could do nothing alone, and there was nobody to set us to work together on a plan to have things better; nobody to represent the common object. Why did not you come along to talk to us about it years and years ago?"

A look through a railroad guide shows a list of names which are a sad criticism on our refinement. Think of going down to posterity as born in such a place as You Bet, Red Hot, Fair Play, Muddy Creek, Looking Glass, Lone Star, or Saw Tooth. These undignified, ill-sounding names are very common, and in the new portion of the country it seems to be a matter of pride to invent absurd names. A gentleman who had the misfortune to reside in a town which bore one of these unmelodious names recently said to us: "I am actually ashamed to register myself when traveling, as from 'Goose Creek,' and for years I have had my mail sent to a town three miles away rather than endure the sight of that odious name on my letters." There are ways of changing these names, and in the interest of good taste it should be done.

In spite of the difficulties in his way, the Rev. Sheldon Jackson has succeeded in getting into his Industrial Training School for Indian Boys and Girls at Sitka, Alaska. Not the least of these difficulties has been getting lumber for the building. Here is the story as he writes us: "Since coming here last August, I sent a crew of three Indians in a canoe a round trip of

400 miles along the coast, with a letter to a saw mill. The trees were felled, the logs were sawed up, a schooner chartered to bring the lumber, and in due time 100,000 feet of lumber was rafted from the schooner on the beach, through the surf, carried on men's shoulders to the building site, a three story building 130x50 feet in size erected, and so far completed that we were able to move into it the first week in January. I have also in the same time organized a church of seventy members, of whom sixty are natives, received on confession of faith and baptism. These converts are largely the fruit of the work of Mr. Austin, one of our teachers."

The Rev. H. M. Bacon, D.D., pastor of the Central Congregational Church, Toledo, Ohio, in a recent article on "Our English Tongue," in which he quotes Richard Grant White's statement in THE CHAUTAUQUAN for December, that "This modern English, which is the youngest, is also the greatest language ever spoken," mentions several valuable confirmations of this opinion. Among them is the tribute of Jacob Grimm, the learned German lexicographer, who says: "In wealth, good sense, and closeness of structure, no other language at this day spoken, not even our German, deserves to be compared to it" (the English). He also calls attention to similar opinions expressed by the late Baron Humboldt, and by Guizot, and recalls the fact that once when the Academy of Berlin offered a prize for the best essay on a comparison of fourteen of the ancient and modern tongues, the prize was awarded to a writer who had given the first place to the English.

On the 23d of February the Washington Monument was formally dedicated at Washington, D. C. Thirty-six years have elapsed since its corner stone was laid. Of the Senate which attended the ceremonies on that occasion but nine are still living, and since that date the most trying years of our national life have been passed. Though the delays in completing the work have been annoying, now that it is complete, it is gratifying to know that the monument is in every way worthy of its object. Indeed, we have no hesitancy in pronouncing it the most beautiful structure in the nation's capital. An obelisk of light gray stone, at a distance it looks like a clearly defined cloud lying against the sky. Its great height (555 feet) is not realized, so perfect is the proportion. The location of the monument has been criticised. It stands on a Government reservation, adjacent to the Potomac River, and directly facing the Capitol. The land is low, and many believe it was a serious mistake not to have placed the obelisk on Capitol Hill. We can not agree with them. The advantage of having the monument on public grounds, where the view of the entire shaft will never be obstructed, is much greater than a higher location with an obstructed view would have been. Then, too, this site was one approved by Washington himself for a monument which, in 1783, the Continental Congress voted to be erected to him. Of course "going to the top" is, and will be, one of the chief features of sight seeing in Washington. Every half hour the steam elevator in the monument carries a crowded load to the top, allowing them ten minutes for looking around before the descent. The stairway is not yet open to the public, and even if it were most people would hesitate before undertaking to mount its 900 steps. The interior of the shaft is lighted by incandescent electric lights. Not the least interesting feature of the monument is the number of marble tablets presented by different states and institutions, and which are being inserted in the inside walls. Several of these have considerable artistic and historic value.

C. L. S. C. NOTES ON REQUIRED READINGS FOR APRIL.

SHORT HISTORY OF THE REFORMATION.

There is so much reading on the Reformation, and it is so well known and easily accessible that it seems almost unnecessary to give a list of supplementary readings. But among so many books it is hard to choose, so we append the names of a few, thinking we may perhaps help some to decide what to read. In order to enjoy this little "History of the Reformation" in the required course, one ought to read many larger ones. "History of the Reformation." By G. P. Fisher. $3.00; D'Aubigne's "History of the Reformation;" Burnet's "Reformation in England;" "History of the Christian Church." By W. W. Blackburn. $2.50; Motley's "Rise of the Dutch Republic." $6.00; "Protestantism." By De Quincy. "Short Studies." By J. A. Froude. "History of the Rise of the Huguenots." By Henry M. Baird. $3.50; "John Knox." By Thomas McCrie. $2.00; "Martin Luther and his Work." By J. H. Treadwell. $1.00; "The Massacre of St. Bartholomew." By Henry White. $1.75; "Schönberg-Cotta Family." By Mrs. Charles. $1.00; "The Martyrs of Spain." By Mrs. Charles. $1.00; "Savonarola." By W. R. Clarke. $1.50; "Romola." By George Eliot. (Treats of the times of Savonarola); "Christians and Moors of Spain." By Miss Yonge. $1.25.

P. 3.—"Council of Constance." A council of the Roman Catholic Church, opened in 1414, and closed in 1418. In its earlier sessions the doctrines of Wycliffe were examined and condemned. John Huss was also condemned and executed, as was Jerome, of Prague. The council was called to consider measures to remedy the division arising in the church from the long residence of the popes at Avignon, and the consequent desire on the part of the French for a national church. See page 89 in the "Short History."

"Julian, the Apostate." (331-363.) A Roman emperor, the nephew of Constantine the Great. Immediately upon his accession he openly avowed his abandonment of Christianity, but he published an edict which granted perfect liberty to all sects and all religions. He, however, excluded Christians from civil and military offices, and compelled them to contribute toward sustaining pagan temples. He permitted the Jews to rebuild their temple at Jerusalem, and published a large volume against Christianity.

P. 4.—“Medici,” māˈde-che. A distinguished Florentine family appearing in history since the close of the thirteenth century.

P. 5.—“d’Ailly,” dāˈye; “John Chartier Gerson,” shär-te-ā zhair-so^{ng}.

P. 6.—“Nicholas Clémanges,” clā-manj; “Gallican Church.” The name given to the Catholic Church in France.

“Father Hyacinthe.” Charles Loyson, a French pulpit orator, born in 1827. At the age of twenty-two he was ordained a priest. He was highly educated. Suspicions as to his doctrines were awakened, and he was summoned to appear before the pope, but cleared himself. Shortly after some speeches of his gave offense, and he was ordered to change his manner or be quiet, but he paid no heed. He was soon forbidden to preach, and threatened with excommunication. In 1869 he visited America, where he was warmly welcomed by many Protestants, but he declared he had no intention of leaving the Catholic Church. He protested against the doctrine of the pope’s infallibility, and defended the right of the clergy to marry. In 1870, on his return to France, the pope relieved him of his monastic vows, and he became a secular priest. In 1872 he was married to an American lady. He is now pastor of a church in Paris, a sort of independent Catholic church.

P. 7.—“Huguenots,” hūˈgē-nots. The name applied to the French Reformers. Its origin is uncertain, some asserting that it was derived from one of the gates of the city of Tours, named Hugons, where the Protestants held their first assemblies. Others say it came from the name of their first leader, Hugues.

P. 8.—“Dominican Order.” An order founded by St. Dominic, in 1216; “John Ruysbroek,” roisˈbrek.

P. 12.—“Wittenberg,” vitˈten-bairg. A town in Prussia, in which there is an immense bronze statue of Luther, and not far from it one of Melancthon.

It is the seat of a great university.

“St. Victor.” A monastery in Paris.

P. 13.—“Origen.” (185-253.) One of the fathers of the church, noted for his unwearied diligence and life of self-denial. For two years, during the persecution under Maximin, he lay concealed in a friend’s house, and here wrote his “Hexapla.” In the Decian persecution he was imprisoned and subjected to extreme torture. Many of his valuable writings have been lost.

“Alexandrian school.” A name applied to the philosophers of Alexandria in the second century. It aimed to harmonize all philosophy and all religion.

P. 14.—“Thomas à Kempis. (1379-1471.) A German writer, a prior in the monastery of Mount St. Agnes.

“Kaisersheim,” kīˈzers-hīmeˌ; “Rheinfeld,” rīneˈfelt; “Pfaffenheim,” päf ˈfen-hime.

P. 17.—“Boccaccio,” bok-katˈcho. (1313-1375.) An Italian novelist, and friend of Petrarch; “Chrysoloras,” kris-o-loˈras.

P. 18.—“Pa-læ-olˈo-gus;” “Bes-sāˈri-on.”

P. 19.—“Argyropylus,” ar-ghe-ropˈoo-los; “Lasˈca-ris;” “Chalkondylas,” kal-konˈde-las; “Gemistus Pletho,” je-misˈtus pleˈtho; “Moschopylus,” mos-kopˈy-lus; “Gasperinus,” gäs-pä-reeˈnus; “Aurispa,” ow-rēsˈpä; “Poggius,” pojˈus; “Perothes,” perˈō-tēs; “Politianus,” po-lishˈā-nus.

P. 20.—“Hierarch.” One who rules or has authority in sacred things.

P. 21.—“Vulgate Bible.” One of the oldest Latin versions of the Scriptures. So called from its common use in the church. The Catholic Church claims this to be the only authentic translation.

“Guizot,” gēˌzōˈ. (1787-1874.) A French historian.

“Reuchlin,” roikˈlin. (1455-1522.)

P. 23.—“Bordeaux,” bor-dōˈ; “Avignon,” ă-vē-nyo^{ng}.

P. 27.—“Eisleben,” iceˈla-ben.

P. 28.—“Eisenach,” īˈzen-näk.

P. 30.—“Scala Santa,” sacred staircase. A staircase in the church and palace of the Lateran, so called because Christ was said to have ascended and descended it. This magnificent building was used as the residence of the popes, from 312 till their removal to Avignon in 1309. The staircase, according to tradition, belonged to the house of Pilate, and was brought to Rome by the mother of Constantine. It is composed of twenty-eight marble steps, which have been covered by order of the popes with a casing of wood. The wood has several times had to be replaced, having been worn through by the knees of ascending pilgrims. This staircase was preserved from the fire which destroyed the building in 1308. The Lateran was rebuilt, to be again burned in 1360. It was restored in 1364, and completely modernized in 1559. This church has always been the cathedral of the bishops of Rome, and takes precedence of all other churches in the Catholic world.

P. 32.—“Schlosskirche,” schlusˈkeer-ka. The church belonging to a castle; “Mos-celˈla-nus.”

P. 33.—“Bull.” An edict of the pope, sent to the churches over which he is head, containing some decree or decision.

“Hapsburg.” Originally a castle in Switzerland. It gave its name to the imperial house of Austria.

P. 35.—“Frederick the Wise.” Frederick III., elector of Saxony.

P. 37.—“Zwickaw,” tswikˈkow. A city in Saxony.

P. 43.—“Augsburg Confession.” The first Protestant confession of faith.

“Convention at Smalcald,” smälˈkält. A confederation of the Protestants held in 1531, in which they were secretly aided by England and France.

P. 45.—“Melancthon,” me-lankˈthon; “Pforzheim,” pfortsˈhime; “Tüˈbing-en;” “Œcolampadius,” ĕkˌo-lăm-pāˈdĭ-us.

“Terence.” (B. C. 195-159.) A Roman comic poet.

P. 50.—“Ulrich von Hutten,” oolˈrik fon hootˈen; “Si-kingˈen;” “Cranach,” kräˈnäk.

P. 51.—“Zwingle,” tswinˈgle.

P. 52.—“Wittenbach,” vitˈten-bäk; “Glarus,” gläˈroos. A canton of Switzerland; “Einsiedeln,” īneˈze-deln.

P. 53.—“Mariolatry,” mā-rí-olˈa-try. The worship of the Virgin Mary.

P. 54.—“Helvetic Confession.” This differed materially from the Lutheran only in holding that Christ was not bodily present in the eucharist.

P. 57.—“Viret,” vē-rā; “Froment,” frō-mo^{ng}; “Farel,” fä-rel.

P. 58.—“Bourges,” boorzh; “Angoulême,” a^{ng}-goo-laim.

“Psychopannychia,” sī-kō-pan-nikˈi-a.

P. 59.—“Tillet,” til-lā; “Martianus Lucanius,” mar-she-āˈnus lu-caˈni-us; “Courault,” coo-rō.

P. 61.—“Neuenburg,” noiˈen-boorg. A town in Germany.

P. 62.—“Bucer,” booˈtser. (1491-1551.)

P. 64.—“Lausanne,” lō-zanˈ.

P. 66.—“Archbishop of Canterbury.” This archbishop is the primate or ruling officer in the national Church of England, the first peer of the realm, and member of the privy council. It is he who places the crown upon the king.

P. 67.—“Lambeth Palace.” The town residence of the Archbishop of Canterbury. It stands on the Thames River, and is surrounded by gardens twelve acres in extent.

P. 68.—“Ochino,” o-kīˈno; “Fagius,” fäˈge-ŏos; “Anne Boleyn,” ann bul ˈlen.

P. 72.—“Froschover,” froshˈo-vair.

P. 78.—“Act of Uniformity.” An act enforcing observance of the English Church service. Severe penalties were enforced against any one who should conduct religious service in any other way than that prescribed by the Book of Common Prayer.

P. 80.—“Cardinal Beatoun,” bēˈtun. Usually written Beaton. (1494-1546.) A persecutor of the Protestants. On the death of King James, he conceived the idea of seizing the government, and forged a will of the king’s, naming himself as successor, but he was prevented from carrying out his plan and was imprisoned for a time. He was shortly afterward reëstablished in his ecclesiastical administration. His enemies seeing no release from his terrible persecutions put him to death.

P. 84.—“Gerard Groot,” jĕ-rardˈ grōt; “Florentius Radewin,” flo-ron ˈshe-us räˈde-win; “Herzogenbusch,” hairts-ōˈgen-boosh.

P. 85.—“Yuste,” yoosˈtā.

“Inquisition.” This was a court established for the purpose of examining and punishing heretics.

P. 87.—Luther’s doctrine concerning the will was that it has no “positive ability in the work of salvation, but has the negative ability of ceasing its resistance under the general influence of the Spirit in the Word and Sacraments.”

P. 88.—“Momus.” In Greek mythology the god of mockery and censure. He is represented as raising a mask from his face.

P. 89.—“Vaudois,” vo-dwä.

P. 90.—“Sorbonne,” sor-bun. A school of theology in Paris, founded in 1253, by Robert de Sorbonne, whence its name. The members were divided into fellows and commoners. The former were selected for their eminent learning, and took the position of teachers. The commoners were chosen from among those receiving instruction, after a severe ordeal, and were supported by the college, but had no voice in its government. They ceased to be members when they graduated as doctors. No member of any religious body was allowed to enter this order. The large lecture halls of the institution were opened free of all charges, to all poor students, and the professors were directed never to refuse instruction to such. Students who had money were required to pay regular fees. The school was without a rival all through the Middle Ages. Its controlling power was felt everywhere. It was frequently appealed to in disputes between the civil power and the papacy. It opposed the claims of Henry VIII. for a divorce from Catharine; condemned the doctrines of Luther and other reformers, and declared that Henry III. had forfeited his crown. It was suppressed in 1789, and its buildings are now used by the University of France.

"Meaux," mō; "Angers," â^{ng}-zhā; "Poictiers," pwä-tyā.

P. 91.—"Gallic Confession." This was essentially Calvinistic in its import, as were also the system of government and method of discipline adopted. They were, however, modified somewhat, to suit a church—not like that at Geneva, in union with the state, but antagonistic to it.

"Bourbons." This line of kings in France began with Henry IV. Six of his descendants in direct line occupied the throne after him. The Louises XIII., XIV., XV., XVI., XVIII., and Charles X. The last representative of this line was the Count de Chambord, who died in 1883. There is a younger branch known as the Orleans branch.

"Guises," gheez. A branch of the ducal family of Loraine, which took a prominent part in the civil and religious wars in France.

P. 95.—"Sä-vo-nä-roˈlä," "Brescia," brāˈsha.

P. 98.—"Chardon de la Rocette," shar-do^{ng} dĕ lä rŏh-shĕt; "Brucioli," broo-choˈlee; "Marmochini," mar-mo-keeˈnee; "Teofilo," tā-o-feeˈlo.

P. 99.—"Mauricha," mä-rēˈka; "Della Rovere," delˈlä rō-vāˈrā; "Cherbina," sher-beeˈna; "Gonzago," gon-zäˈgō; "Ca-rafˈfa;" "Paschali," pas-caˈlēe.

P. 100.—"Paolo di Colli," pä-oˈlo dē colˈlee; "Gratarole," grät-ä-rōˈlee; "Cor-räˈdo;" "Teglio," täˈglē-o.

P. 103.—"Vives," vēˈvace; "Ponce de la Fuente," pōnˈthā dā lā fwenˈtā; "Enzinas," en-zēˈnas; "Valladolid," väl-yä-dō-leedˈ; "Varelo," vä-rāˈlo; "Ægidius," ē-gidˈē-us.

P. 104.—“Hernandez,” her-nanˈdā; “Boborguez,” bō-borˈgā.

P. 110.—“Cyriace,” si-rēˈä-see.

P. 116.—“Dollinger,” dolˈling-er. A learned Catholic theologian, born at Bamberg, in 1799. He has published a church history, and several other works.

CHEMISTRY.

P. 169.—The formula (N_2O_2) shows that two atoms of nitrogen have united with two atoms of oxygen to form a molecule of nitrogen di-oxide. The formula $Cu\ (NO_3)_2$ shows that one atom of copper has united with two molecules, each composed of one atom of nitrogen and three atoms of oxygen, to form one molecule of cupric nitrate. In like manner $Fe_2\ (NO_3)_6$ indicates that two atoms of iron have united with six molecules, each composed of one atom of nitrogen and three atoms of oxygen, to form one molecule of ferric nitrate.

P. 173.—“Refractive power” of water. When a ray of light strikes the surface of a new medium, a portion of it is turned out of its original course or refracted. This gives rise to some well-known effects. When any object is placed in water and viewed obliquely it looks to be nearer the surface than it is, because the light in passing from the denser medium takes a direction more inclined to the horizontal, and an object always appears directly in line with the ray of light entering the eye.

P. 178.—“Crécy,” kresˈse. This battle took place August 26, 1346, between the English under Edward III. and the French under Philip VI. It is said that Edward had six pieces of artillery. Artillery had probably not been used in the field before this time.

P. 182.—“Trinˌi-tro-cĕlˈlu-lose.”

P. 185.—“Mont Cenis,” mōng sŭh-nē. This tunnel under the Alps is in reality some sixteen miles from Mont Cenis, whose name it bears. The first mine was fired in 1857, and for four years the piercing was done by hand; the need of a quicker method led to the invention of a machine drill—a perforating machine worked by compressed air. The work was carried on by day and night, from both sides of the mountains, until the two bodies of workmen met, December 26, 1870. The tunnel was opened for railway travel September 17, 1871. Its length is nearly eight miles, and the cost of the tunnel was $15,000,000.

“St. Gotˈhard.” This tunnel is also through the Alps. The length is nine and one fourth miles. Its construction was begun in 1872, and it was completed in eight years.

P. 189.—“Phosphorus ne-croˈsis.” The latter term is derived from a Greek word, meaning to make dead, to mortify, and is a disease which attacks bony tissues, as gangrene effects the soft parts. “The acid fumes thrown off from phosphorus in the various processes of making matches, frequently cause among the people employed a terrible disease, which attacks the teeth and jaws.… Its natural course is to rot the entire jaw bone away.”

P. 190.—“Al-lŏtˈro-pĭsm.” Dana says allotropism is “the property of existing in two or more conditions which are distinct in their physical or chemical relations. Thus, carbon occurs crystallized in octahedron, and other related forms, in a state of extreme hardness, in the diamond; it occurs in hexagonal forms, and of little hardness, in black lead; and again occurs in a third form, with entire softness, in lamp-black and charcoal. In some cases one of these is peculiarly an active state, and the others a passive one. Thus, ozone is an active state of oxygen, and is distinct from ordinary oxygen, which is the element in its passive state.”

P. 194.—“Chemicking,” kemˈik-ing.

P. 203.—Translation of French sentence: “This last virtue I believe it still to possess, if the husband is rich enough to buy the jewel which his wife is ambitious to own.”

P. 217.—“Boussingault,” booˌsănˈgoˌ; Jean Baptiste. A noted French chemist of this century.

NOTES ON REQUIRED READINGS IN “THE CHAUTAUQUAN.”

ARISTOTLE.

1. “Schoolmen.” Philosophers and divines who in the middle ages adopted the principles of Aristotle and dwelt much upon abstract speculation. Scholasticism was a philosophy of dogmas. “Its elements were doctrines which the authority of the church made indisputable,” and which were looked upon as absolute truth. Facts in nature were set aside and an artificial, logical scheme developed. Scholiasts thought experiment only fit to follow and illustrate theories.

2. “Haroun-al-Raschid,” Aaron the Just. (765-809.) The caliph who raised Bagdad to its greatest splendor, and whose reign was looked upon as the golden era of the Mohammedan nation. He reigned twenty-three years and performed the pilgrimage to Mecca nine times. He is famous as the hero of the Arabian tales. Tennyson wrote of him in his “Recollections of Arabian Nights.”

3. “Ex post facto.” A Latin expression, meaning an after act or thing done afterward. An *ex post facto* law is a law enacted after the commission of a crime, for the purpose of being enforced upon the person having committed the crime, who could not be held a criminal, or at least a criminal in the same degree, until after the enaction of the law. All such laws are forbidden by the constitution of the United States.

4. Transcriber’s Note: This note (presumably “Hypolais.”) was omitted in the original. *Hippolais* is a scientific genus of tree warblers.

CHEMISTRY.

[ERRATA.—A few typographical errors in articles of this series have escaped correction. Page 254, change "300,000,000" to 3,000,000; for "alcohol," thirty-third line, page 325, substitute paraffine; same page, eighth line, second column, use not for "but;" in "experiment," same column, use heat for "sensation," and in next to the last line of the article change "topics" to optics.—*Prof. J. T. Edwards.*]

1. "Geysers," gī˙sers. Intermittent hot springs, found in different parts of the world. Those of Iceland are the best known. More than one hundred of these springs are there found, within a space of two miles. The geysers of the Yellowstone are the most wonderful ever discovered. The country lying between latitude 43° and 47° north, and longitude 110° and 114° west, is dotted with groups of springs. Some of them, when in action, send up columns of water to a height of 200 feet.

2. "Hugh Miller." (1802-1856.) A British geologist. He was by trade a stone mason, but he devoted all his leisure hours to study. Soon "detecting the wonders of the fossil world" in the quarries in which he worked, he made them the special subject of his thought, and soon became an eminent geologist. He published many works, most of them bearing on this subject. He worked so incessantly, taking little sleep or exercise, that his mind was on the verge of giving way. Realizing this with terror, he took his own life. A note left for his wife read as follows: "A fearful dream rises upon me. I can not bear the horrible thought."

The old red sandstone is the name given to the rock in Great Britain formed in the Devonian age, or age of fishes. Its thickness is in some parts 8,000 to 10,000 feet. It includes sandstone, marlytes of red and other colors, and some limestone.

The sandstone of the Triassic period, which includes the latest formations of the earth's crust, is also characterized by fossils, and is often red in color; hence the name, new red sandstone, has been applied to it.

3. "Dr. Hitchcock." (1793-1864.) An American geologist and author.

4. "Minute animals." The carbonate of lime which is found in rocks is most of it formed directly of shells, corals, and other animal remains. These little creatures take their stony-like structures from the water or from their

food through the power of secretion, just as man forms his bones, and after their death they are given over to be made into rocks. The great extent and thickness of the limestone rocks of the earth give some idea of the amount of life that flourished there in past time.

5. “Anhydride.” For definition see “Chemistry,” page 151.

6. “Old Stone Mill.” It is asserted by some antiquaries that this structure was built by the Northmen, 500 years before Columbus landed on these shores. Its purpose, as well as its origin, has been a theme of much discussion. Its present appearance is that of a large round tower overgrown with vines.

7. “The Stone age.” One of the divisions of prehistoric time. In this age men were not acquainted with the use of metal and fashioned their rude implements exclusively out of stone.

8. “Ceramic,” se-ramˈic.

9. “Cesnola collection.” Cesnola was an American soldier and archæological explorer, born in Italy in 1832. He served in the Crimean war, and in the civil war, was for a long time in Libby Prison. At the close of the war he was sent as consul to Cyprus. Having his attention attracted by some fragments of terra cotta and some coins, he began making excavations in search of relics. He met with such rewards that he continued his work for three years, employing hundreds of men. Among his discoveries were statues, lamps, vases, coins, glassware, gold ornaments, bronzes, and inscriptions, in all about 13,000 articles. This remarkable collection is now in the Metropolitan Museum of Art in New York.

10. “The mound builders.” The race of people found in America by its first settlers had clearly been preceded by a race of higher type and attainments. Relics proving this have been discovered throughout the Mississippi valley. Earthworks are their principal testimony, of which many thousands have been found in Ohio alone. These mounds vary in size and shape, but are always regularly formed, sometimes being square, sometimes round, hexagonal, octagonal, or truncated. They are ascended by spiral paths, and frequently contain skeletons. Sometimes the earthworks are

thrown up so as to represent in outline men and animals, and appear as huge "bas-reliefs on the surface of the ground."

11. "Humboldt," Baron von. (1769-1859.) A German naturalist, the most distinguished scholar of the nineteenth century. After a thorough education, under the best masters in different universities, he determined to devote himself to finance as a business, and familiarized himself with everything pertaining to this calling. He changed his career and wished to engage in practical mining. And again he went through with a full preparation for this work. He was sent to explore several mining districts, and made many experiments to discover the nature of fire-damp. Later he made a great scientific expedition which only led the way to others, until he had visited as a scientist almost every land. He is distinguished for the comprehensiveness of his researches. During his travels he made astronomical, botanical and magnetic researches, measured elevations, investigated the nature of the soil, and the thermometrical relations; he also collected herbariums, and founded the new science of the geography of plants. Of his numerous published works, "Kosmos" has perhaps attracted public attention most widely. It has been without an equal in giving an impulse to natural studies.

12. "Lord Lytton," Sir Edward George. Earle Lytton, son of General William Earle Bulwer, born in 1805. Upon his succeeding to the vast estates of his mother, the heiress of the Lyttons, he by royal license assumed this name, writing it after his own. He is the author of several works, mostly of fiction.

THE CIRCLE OF THE SCIENCES.

1. "Syllogism." Every argument, to be valid, must be placed in regular logical form, which consists of three propositions, two called the *premises*, and the third the *conclusion*. The conclusion follows from the premises, so that if the former are true, the conclusion must be. For example: Major premise—It was not lawful to scourge a Roman citizen; minor premise—Paul was a Roman citizen; conclusion—Therefore, it was not lawful to scourge Paul.

2. “Dr. Porter.” An American scholar and author, born in 1811. The eleventh president of Yale College.

3. “Ruskin,” John. (1819- .) An English author. He has given much attention to the study of art, many of his numerous books being written on that subject.

4. “Utopia.” See The Chautauquan for February, 1885.

5. “Campanella.” (1568-1639.) An Italian philosopher. He was suspected of joining a conspiracy against the Spanish government, was put to the rack, and finally imprisoned in Spain. Later he was transferred to the inquisition at Rome. On gaining his liberty he went to France. He was famous for undermining other systems of philosophy rather than for establishing one of his own.

6. “Owen,” Robert. (1771-1858.) An English social reformer. He lived for a few years in Scotland, where he advocated his theory of communism, an absolute equality in all rights and duties. By the aid of his large fortune he was enabled to distribute great numbers of tracts explaining his views, and these soon won him a large following. He was, however, opposed and attacked on all sides. In 1823 he came to the United States, bought 20,000 acres of land in Indiana, intending to found his community there, but the scheme proved a failure and he returned to England. He spent the rest of his life as a traveler, advocating his views both by his books and public lecturing.

7. “Fourier,” (foo-ri-ā) Charles. (1772-1837.) A French writer on social science.

SUNDAY READINGS.

1. “Thebes and Luxor.” Thebes was a celebrated Egyptian city, formerly the capital of Upper Egypt. Its ruins are among the most magnificent in the world, and comprise what form now nine villages, among which Luxor is one. The large and costly palaces of the Luxor quarter were founded by Amenophis III., and from here was taken the obelisk which stands in Place de la Concorde in Paris.

2. “Archbishop Whately.” (1787-1863.) An English prelate. He was for some years a professor at Oxford, and in 1831 was consecrated archbishop of Dublin. He was the author of many important works.

3. “Paˌlæ-o-cosˈmic.” Pertaining to the ancient universe.

4. “Old Man of Cromagnon.” “In the *Reliquiæ Aquitanicæ,* by Messrs. Lartet and Christy, there is a full account of the archæology of the old Stone age, as exhibited in the south of France, especially in the caves in the valley of Cro-Magnon. … Bones of the reindeer are abundant, and the co-existence of man with this animal in latitudes so much lower than its present habitat implies a certain degree of elevation above savages, as not only food, clothing and implements, but materials for ornamentation were obtained from it. The domestic economy of these earlier races is shown by their hearths, boiling stones, rough hammers, and hollow, dish-like pebbles. … M. Pruner-Bey, from the examination of skeletons found in the cave of Cro-Magnon maintains that the crania of the reindeer age belong to a double series, one approaching the Lapp and the other the Finn of the present day. He concludes that they had massive bones, long, flat feet, comparatively short arms and long forearms, with powerful muscles, greatly developed jaws, widely opened nostrils, and were of unbridled passions.”

ANIMAL BIOLOGY.

PRONUNCIATION OF NAMES IN TABLE.

Pro-to-zoˈa, mo-neˈra, greg-a-rinˈi-da, rhiz-opˈo-da, in-fu-sōˈri-a, spon-gĭˈda, cœ(kē)lenˈte-rā-ta, hy-dro-zoˈa, an-tho-zoˈa, mad-re-poˈra, poˈrites, tuˌbi-pōˈra, cor-ralˈle-um, ruˈbrum, cte(te)-nophˈo-ra, e-chinˌ(kin)o-dermˈa-ta, crī-noidˈe-a, as-ter-oidˈe-a, e-chen(ken)-oidˈe-a, holˈo-thu-roidˌe-a, ver ˈmēs, ro-tifˈe-ra, pol-y-zoˈa, brachˌ(brak)-i-opˈo-da, an-nelˈi-dæ, mol-lus ˈca, la-melˌle-bran(g)-chi(ki)-āˈta, gas-ter-opˈo-da, ceph(sef)-a-lopˈo-da, ar-ticˈu-lāˈta, crus-taˈcē(se)-a, a-rachˈ(rak)-ni-da, myr-i-opˈo-da, tu-ni-cāˈta, ver-te-brāˈta, pisˈ-cēs(sēs), aˈvēs.

1. “Amœba.” This little animal is known to microscopists under the name of proteus, from the rapid and continuous changes of shapes which it

presents to their notice.

2. “Tentacles.” Processes usually slender and thread-like, proceeding from the head of invertebrate animals, such as insects, snails and crabs, being used for the purpose of feeling, prehension or motion.

3. “Oviparous.” An adjective applied to all animals which produce eggs, as distinguished from *viviparous*, producing young in the living state.

4. “Ganglia.” Collections of nerve cells, from which nerve fibers are given off in different directions. They are thought to be the organs in which all action originates.

5. “Ventral surface.” The surface of the body opposite the back. The back is called the dorsal surface.

6. “Medˈul-la-ry.” Consisting of marrow. The fibrous nervous matter of the brain contains nerve tubes, within which is a layer of thick, fluid, highly refractive matter, called the medullary layer.

PARAGRAPHS FROM NEW BOOKS.

Portraits from Carlyle.—If Carlyle had taken to the brush instead of to the pen he would probably have left a gallery of portraits such as this century has not seen. In his letters and journals, reminiscences, etc., for him to mention a man is to describe his face, and with what graphic pen and ink sketches they abound. Let me extract a few of them. Here is Rousseau's face, from "Heroes and Hero Worship:" "A high but narrow contracted intensity in it; bony brows; deep, straight-set eyes, in which there is something bewildered looking—bewildered, peering with lynx-eagerness—a face full of misery, even ignoble misery, and also of an antagonism against that; something mean, plebeian there, redeemed only by *intensity*: the face of what is called a fanatic—a sadly *contracted* hero!…"

Here we have Dickens in 1840: "Clear-blue, intelligent eyes; eyebrows that he arches amazingly; large, protrusive, rather loose mouth; a face of most extreme *mobility*, which he shuttles about—eyebrows, eyes, mouth, and all—in a very singular manner while speaking. Surmount this with a loose coil of common-colored hair, and set it on a small, compact figure, very small, and dressed à la D'Orsay rather than well—this is Pickwick."

Here is a glimpse of Grote, the historian of Greece: "A man with straight upper lip, large chin, and open mouth (spout mouth); for the rest, a tall man, with dull, thoughtful brows and lank, disheveled hair, greatly the look of a prosperous Dissenting minister."

In telling Emerson whom he shall see in London, he says: "Southey's complexion is still healthy mahogany brown, with a fleece of white hair, and eyes that seem running at full gallop; old Rogers, with his pale head, white, bare, and cold as snow, with those large blue eyes, cruel, sorrowful, and that sardonic shelf chin."

In another letter he draws this portrait of Webster: "As a logic-fencer, advocate, or parliamentary Hercules, one would incline to back him, at first sight, against all the extant world. The tanned complexion; that amorphous, crag-like face; the dull black eyes under their precipice of brows, like dull anthracite furnaces, needing only to be *blown*; the mastiff mouth, accurately closed; I have not traced as much of silent *Berserkir rage*, that I remember of, in any other man."—*From John Burroughs's "Fresh Fields."*

SCOTT AT WORK.—I never can forget the description Sir Adam Fergusson gave me of a morning he had passed with Scott at Abbottsford, which at that time was still unfinished, and swarming with carpenters, painters, masons, and bricklayers, was surrounded with all the dirt and disorderly discomfort inseparable from the process of house-building. The room they sat in was in the roughest condition which admitted of their occupying it at all; the raw, new chimney smoked intolerably. Out-of-doors the whole place was one chaos of bricks, mortar, scaffolding, tiles, and slates. A heavy mist shrouded the whole landscape of lovely Tweed side, and distilled in a cold, persistent and dumb drizzle.

Maida, the well beloved stag-hound, kept fidgeting in and out of the room. Walter Scott every five minutes exclaiming, "Eh, Adam! the puir beast's just wearying to get out;" or, "Eh, Adam! the puir creature's just crying to come in;" when Sir Adam would open the door to the raw, chilly air for the wet, muddy hound's exit or entrance, while Scott with his face swollen with a grievous toothache, and one hand pressed hard to his cheek, with the other was writing the inimitably humorous opening chapters of "The Antiquary," which he passed across the table, sheet by sheet, to his friend, saying, "Now, Adam, d'ye think that will do?" Such a picture of mental triumph over outward circumstances has surely seldom been surpassed. House-builders, smoky chimney, damp draughts, restless, dripping dog, and toothache form what our friend Miss Masson called a "concatenation of exteriorities," little favorable to literary composition of any sort; but considered as accompaniments or inspiration of that delightfully comical beginning of "The Antiquary," they are all but incredible.—*From Mason's "Traits of British Authors."*

PARADISE FOUND.—Could it once be proven that the Arctic terminus of the earth has always been the ice-bound region that it is, and which for thousands of years it has been, it would of course be useless to entertain for a moment the hypothesis that the cradle of the human race was there located. Probably the popular impression that from the beginning of the world the far North has been the region of unendurable cold has been one of the chief reasons why our hypothesis is so late in claiming attention. At the present time, however, so far as this difficulty is concerned, scientific studies have abundantly prepared the way for the new theory.

That the earth is a slowly cooling body is a doctrine now all but universally accepted. In saying this we say nothing for or against the so-called nebular hypothesis of the origin of the world, for both friends and foes of this unproven hypothesis believe in what is termed the secular cooling or refrigeration of the earth. All authorities in this field hold and teach that the time was when the slowly solidifying planet was too hot to support any form of life, and that only in some particular time in the cooling process was there a temperature reached which was adapted to the necessities of living things. On what portion of the earth's surface, now, would this temperature first be reached? Or would it everywhere be reached at the same time? … We asked the geologist this question: "Is the hypothesis of a primeval polar Eden admissible?" Looking at the slowly cooling earth alone, he replies: "Eden conditions have probably at one time or another been found everywhere upon the surface of the earth. Paradise may have been anywhere." Looking at the cosmic environment, however, he adds: "But while Paradise may have been anywhere, the *first* portions of the earth's surface sufficiently cool to present the conditions of Eden life were assuredly at the Poles."—*From Warren's "Paradise Found."*

SEPARATION OF DE LONG AND MELVILLE.—De Long verbally directed both of us to keep, if possible, within hail, and reiterated his orders in case of separation: "Make the best of your way," said he, "to Cape Barkin, which is eighty or ninety miles off, southwest true. Don't wait for me, but get a pilot from the natives, and proceed up the river to a place of safety as quick as you can; and be sure that you and your parties are all right before you trouble yourselves about any one else. If you reach Cape Barkin you will be safe, for there are plenty of natives there winter and summer." Then addressing me particularly, he continued: "Melville, you will have no trouble in keeping up with me, but if anything should happen to separate us, you can find your way in without any difficulty by the trend of the coast-line; and you know as much about the natives and their settlements as any one else.” This was our last conversation in a body.

So when De Long waved me permission to leave him, I hoisted sail, shook out one reef, and as we gathered way the boat shot forward like an arrow, and the spray flew about us like feathers. Heretofore we had been running dead before the wind on our southwest course for the land, but the heavy sea and lively motion of the boat caused the sail to jibe and fill on the other tack, whereupon we would broach to and ship water. For this reason I hauled up the boat several points, or closer to the wind, and our condition at once improved. Now that we were separated I resolved to concern myself directly with the safety of my own boat; so that when one of the men said that De Long was signaling us, I told him he must be wrong, and further directed that no one should see any signals now that we were cast upon our own resources.

When last seen, the second cutter was about one thousand yards astern of us, the first cutter probably midway between, and there is no doubt in my mind that she then foundered. A conversation with the only two surviving members of the first cutter (Nindemann and Noras) has confirmed me in this belief; for they witnessed the scene as I have described it, and state that it was the general opinion of DeLong’s crew that I had shared the same fate simultaneously with Chipp.—*Melville’s “In the Lena Delta.”*

TALK ABOUT BOOKS.

There can scarcely be a sadder story than that of the loss of the Jeannette and the subsequent search for DeLong and his party. No event of recent years has caused more horror in the public mind and led to more urgent expostulation against further Arctic exploration. We believe, however, that as a better knowledge of the aim and value of these undertakings grows on the public, censure will be removed. Certainly a careful reading of Melville's "In the Lena Delta"[C] produces this result. The book is perhaps more full of horror than most readers imagine, but after reading there can be but one opinion; terrible as it all is, it has been worth the suffering. It is worth while to have died bravely in carrying out orders. The unflinching resolution of those men places them among the heroes of modern history. You can not help feeling that there is a wonderful amount of unusual heroism in the story. The Jeannette expedition has furnished a much needed lesson on the nobility of endurance. The results to our knowledge of these regions have been considerable, the people of Siberia, the Russian exile, the homes and customs of various tribes are more fully explained to us in Melville's notes than elsewhere; again, no future exploration will be open to equal dangers.

We have never experienced a greater shock in our book reviewing than that which came to us when, on turning from Melville's description of the Arctic regions, we were told that Paradise had been found[D]—at the North Pole. It will be a long time before the public mind with its present ideas of the Pole will be willing to consent to this conclusion, even if President Warren is able to advance still more skillful arguments than he yet has in proof of his theory. And the arguments are skillful. He has quoted high authority to prove that Eden conditions once existed on every portion of the earth, and first of all at the Pole; he has done his best to remove our prejudices against a night of six months by showing that not more than "four fortnights" is the probable length of the night there; he shows us that palæontology teaches that life first began at the polar regions, and quotes the mythical lore of Egyptian, Assyrian, Babylonian, Buddhist, Greek, and

Roman, to support his theory. The hypothesis is certainly entertaining, and this attempt at demonstrating it contains some very probable arguments.

A book bearing the title “Personal Traits of British Authors”[E] is sufficient of itself to win the attention and awaken the interest of all book lovers, but when, on turning its leaves, it is found that these traits have been noted and given to the world by other authors, the desire to know what they are is doubled. What great men think and say about other great men is a matter of interest to all well informed persons. That pardonable, commendable curiosity to know “what about” earth’s gifted ones, that lurks in human hearts, has a sort of double chance to satisfy itself with such an arrangement as this. The persons of whom this book treats are seven in number: Scott, Hogg, Campbell, Chalmers, Wilson, DeQuincey, and Jeffrey. In a tabulated form all the leading events of their lives are given. Several pages are devoted to sketches of each one, all filled with exquisite little pen pictures, drawn by master hands at widely differing periods, and from widely differing scenes in life, giving the greatest contrasts in attitudes, words, and expression. Indeed, one has hard work sometimes to make himself believe they were intended to represent the same person. It would be difficult to find a finer collection of character studies than Mr. Mason has given in this volume.

There is considerable probability in the suggestion which Mr. Lang makes in his “Custom and Myth.”[F] He has attempted to find the key to myths in customs which have prevailed among early tribes, in opposition to those scholars who find their solutions in the names which they claim were once applied to objects, and of which the original meaning has been lost. The essays are made valuable by a great deal of material gathered evidently by much research in the lore of remote tribes, but they are singularly unsatisfactory. The work is loosely done. The solutions are mere suggestions, although such interesting suggestions that one feels loath to give up the search without more work. In some cases the myths presented simply show that similar tales exist in many nations. Until Mr. Lang does more work on this entertaining theory he must not expect a very wide following.

A good work on English cathedrals has for a long time been in demand. The interest in architecture which recent years has developed, the increase in travel and the large scale on which the English people have carried on the restoration of their cathedrals have made such a work necessary. “The Cathedral Churches of England and Wales”[G] quite fills the demand. A book of superior make up, a very handsome parlor book, indeed, it is yet full of information. Thirty-five cathedrals are described, and so fully illustrated as to give very satisfactory ideas of the leading features of each. The articles, written in an encyclopedic style that seems slightly out of place in the company of such illustrations, paper, and letter press, are historic and architectural, concerning themselves very little with poetic associations and fine descriptions. They are, however, all the more useful for that.

Two little books, helpful to all persons and bearing comfort for stricken hearts, and for those weary with the burdens of life, are to be found in “Meditations on Life, Death and Eternity.”[H] They are like friends to whom one would turn for companionship. The books were translated and compiled from the larger work of a distinguished German writer, and were arranged in their present form at the request of Queen Victoria, who prizes them very highly, as the original was a great favorite of the Prince Consort.

That Mr. Barnes fully accomplished what he set out to do when he produced the “Hand-Book of Bible Biography,”[I] a brief examination of the work will satisfy any one. His aim was to produce a book that would be complete as to names, that should contain all the facts, and that should be within the means of all Bible students. Each biography is a story complete in itself, with many illustrations and maps.

FOOTNOTES

[C] In the Lena Delta. A narrative of the search for Lieutenant Commander De Long and his companions. Followed by an account of the Greely Relief Expedition and a Proposed Method of Reaching the North Pole. By George W. Melville. Boston: Houghton, Mifflin & Co. 1885.

[D] Paradise Found. The Cradle of the Human Race at the North Pole. By William F. Warren, S. T. D., LL D. Boston: Houghton, Mifflin & Co. 1885.

[E] Personal Traits of British Authors. By Edward T. Mason. New York: Charles Scribner’s Sons. 1885.

[F] Custom and Myth. By Andrew Lang, M.A. New York: Harper & Brothers. 1885.

[G] The Cathedral Churches of England and Wales. Cassell & Company. New York: 1884. Price, $5.00.

[H] Meditations on Life, Death and Eternity. By Johann Heinrich Daniel Zschokkè. Translated by Frederika Rowan. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe.

[I] Hand-Book of Bible Biography. By the Rev. C. R. Barnes, A.B. New York: Phillips & Hunt. Cincinnati: Cranston & Stowe. Price, $2.25.

THE CHAUTAUQUA UNIVERSITY.

WHAT ARE ITS CLAIMS?

BY PROF. R. S. HOLMES, A.M.

We shall be careful in what we say to make no claim for the correspondence system of teaching, as against any other. We claim for it simply a place as a co-laborer in the work of education. Lest any one should be misled by any utterances we may have made, or may hereafter make, and think that here was cast up a royal road over which one could pass with flying feet to the goal of educational culture, and enter it, to find only a narrow path, rough, stony, and filled with difficulties, we wish to plainly state what we claim for this system of instruction. Lest any one should conceive that the need for university and college has passed, and that results can be obtained by a home correspondence-university course, as good or better than can be obtained from actual college residence, we wish to plainly state what we do not claim. It may place our positive claims in a stronger light, if we set them forth against what we do not claim, as a background. Accordingly, our first statements will be negatives, as follows:

1. We do not claim that the correspondence system of teaching is the superior of oral teaching;

2. Nor that it is destined to supersede oral teaching;

3. Nor that it has wrought or will work any revolution in educational methods;

4. Nor that it can compete with oral teaching, on anything like equal terms;

5. Nor that by this method, years of study in classroom, under able, living teachers are made unnecessary;

6. Nor that it uses newer and better methods of instruction than are used in the classroom;

7. Nor that it is freer from defects than other existing systems;

8. Nor that a class, school, college, or university, dependent for its entire work upon pen, paper and post, should be sought by the student in preference to established resident institutions;

9. Nor that it is without serious disadvantages, even to the student most favorably circumstanced;

10. Nor, finally, that it is able to teach all branches of study without other than postal facilities.

We might carry this line of disclaimer farther, but are persuaded that enough has been said to enable us to make our claims for the correspondence system, without danger of being misunderstood. Still further, we desire the power of voice and pen, as far as it may reach, to be felt on the side of the college and university. To all who can go to college, our word is most emphatically—go; and having gone, stay; let nothing come between you and the completion of the course. Still further, we will say to such as are so limited by circumstances as to feel unable to devote the requisite time, means, and presence, to a college course, “If possible, let not circumstance compel you, but do you compel circumstance, till the desired way shall open; and this though years be occupied in the struggle. The goal is worth the race.”

Here, then, we present what we claim for the correspondence system of teaching:

1. We claim that the majority of those who are likely to avail themselves of this system, are men and women of mature mind, and hence are able to make the very best use of whatever advantages are offered them;

2. That the majority of those who are likely to avail themselves of the advantages we offer, are actuated by an earnest purpose to obtain an advanced education, by *any* means which are available to them;

3. That wise direction through correspondence, by competent and experienced teachers, is calculated to produce better results than can be expected ordinarily from unaided individual effort;

4. That teaching by correspondence can be successfully applied to a course of study so wide and comprehensive that one who masters it will secure a culture that would be rightly called liberal;

5. That this system of teaching is therefore entitled to a place, as associate, in the ranks of the teaching systems of the age;

6. That as a system, it is no untried experiment, but has been so tested that it can point to tangible results with no fear of discomfiture if these results be examined;

7. That it requires determined effort, and calls for rigid self-discipline, to insure success;

8. That it tends to form critical habits of study;

9. That it tends to produce self-reliance, and to develop individuality in methods of study;

10. That it affords marked opportunity for deliberation, and so fosters the judicial habit in study;

11. That it tends to systematize and render methodical all habits, whether of study or of life;

12. That opportunities for *mal-application* are reduced to a minimum;

13. That its possibilities are such as to warrant corporate effort to extend its advantages to those who would be otherwise deprived of any advanced educational opportunities;

14. That such a corporation is entitled to be called a School of Liberal Arts;

15. That it allows tests of the student’s acquirement, as rigid as can be desired by the highest standard of educational excellence;

16. That the student who has submitted to such tests, and successfully borne them, is entitled to the reward of a diploma and a degree;

17. Finally, that the corporation or institution which can prepare the student for such an ordeal is entitled to confer such diploma and degree.

The claims which we have now presented are sufficient to show the spirit and belief which have led to the incorporation of the Chautauqua University. We have attempted to state them logically, clearly, and forcibly. There is in them no element of disputation.

We appeal to a vast, an eager and earnest constituency. To know, only to know, is the earnest cry of multitudes of our fellows. Lament for lack of early opportunities, and consequent self-depreciation, is the undertow that sweeps to ruin the possibilities of many a life. High purposes and noble ambitions have been thwarted on life’s threshold by the cruel limitations of circumstance. Mistaken views of life’s best aims, in days when opportunities were possible, have been dispelled when the opportunities have long been left behind. To each of these classes the Chautauqua University brings the correspondence system of teaching, and says: for you, it is possible to supplement the lack of early years; for you, to realize your ambitions, even within the bond by which circumstance has bound you; and for you, in the new light which experience has given, to see other opportunities for obtaining that culture which, years ago, you neglected and passed by.

SPECIAL NOTES.

THE ACADEMY OF LATIN AND GREEK,
Summer Term of Six Weeks.

TO THE CHANCELLOR OF CHAUTAUQUA UNIVERSITY:

My Dear Doctor Vincent:

It gives me great pleasure to be able to offer this summer, at Chautauqua, a course in Latin and Greek of unusual merit. Of the assistant teachers, Mr. Otto is already favorably known to our pupils of last summer, and to many correspondence students as an energetic and thorough teacher. Dr. Bevier will be a great acquisition for Chautauqua. He was graduated from Rutgers with first honors, having also during his course won honors in Latin and Greek at the inter-collegiate contest. After graduation he studied at Johns Hopkins University (which conferred on him the degree Ph.D.), and then continued his studies in Europe. He was a student at the American School at Athens, Greece, and is now an enthusiastic and successful teacher.

Although our session in Latin last year began a week late, and we suffered from other disadvantages, I believe our numbers in Latin reached a total unparalleled in the history of Chautauqua.

What was, however, especially gratifying, was the improved quality of scholarship manifested by students.

For this summer we offer the following course:

1. ROMAN LAW (using the Institutes of Justinian) with information. Not only every lawyer, but every teacher of Latin to-day should familiarize "*thon*"self with Roman law, lying, as it does, *at the base of Roman civilization*.

2. The Latin of the early Church Fathers.—Recent publication and discussion have rendered so prominent the influence of the early Latin Fathers on church doctrine that *every clergyman*, present or prospective, will do well to examine this question for himself.

3. Comparative Philology.—(Every student preparing to enter either of these three classes should at once communicate with the principal, that there may be no delay at the opening of the session, in securing apparatus.)

4. Plato.—Apology and Crito, Tyler's Ed. (Appletons.)

5. Cicero.—*De Natura Deorum*, Stickney's Ed. (Ginn, Heath & Co.)

6. Homer.—Odyssey.

7. Virgil.—Æneid.

8. Horace.—Chase's Ed. (Eldridge & Bro's.)

9. Cicero.—Orations.

10. Xenophon.—Anabasis.

11. Cæsar.—*De Bello Gallico* (two hours per day).

12. Beginners in Greek. Harkness's Text-Book, last ed. (Appletons.)

13. Beginners in Latin (three hours per day by the induction method).

Latin students must have the "Hand-Book of Latin Synonyms." (Ginn, Heath & Co.)

I hope you will give us at Chautauqua zealous students, who will concentrate their work on Latin and Greek, but especially two classes: Teachers of Latin and Greek, and those who are absolutely beginners. A clear-headed student who doesn't know a word of Latin, can, by devoting six weeks to it, secure five

HOURS per day (*Beginners* and *Cæsar*) or ONE HUNDRED AND FIFTY HOURS in six weeks—quite as much time as the average school gives in one year.

It is thought that teachers of Latin and Greek will find not only the method of value, but also the inspiration which indubitably does arise when teachers gather.

Your ob't servant,

EDGAR S. SHUMWAY, Principal.

RUTGERS COLLEGE, February 23, 1885.

The C. L. S. C. Correspondence Department, though not often heard from publicly, is doing an important work. Several hundred students are enrolled upon its books, and the work is being prosecuted this year with renewed vigor. An Illinois lady writes: "Having enjoyed and been benefited by the letters of my C. L. S. C. correspondent, I very much wish to continue that branch of the work this year. We followed no special plan, but the letters I received encouraged and strengthened me, and kept me from falling by the wayside. I love the C. L. S. C. and am proud to say I have gained for it some members. In my judgment the Correspondence Circle is grand, good and beneficial." From New Hampshire comes the following: "I tender hearty thanks to the originator of the Correspondence Circle. The frequent letters of my two correspondents are a continual stimulus. The sympathetic words, the exchange of essays, the comparing of work done, I find very helpful, while the questions of my bright girl correspondent have led me to search for and find many items of information I should have otherwise neglected." These and many similar letters received from members of the Correspondence Department show how helpful this work is proving to many isolated members of the Circle, shut out from all other means of communication with their fellow students.

From a circle in Connecticut numbering five members comes the suggestion that Local Circles be put in communication with each other, the correspondence to be carried on, of course, through the respective secretaries. There is no reason why a correspondence of this sort should not prove both interesting and valuable, as it will serve to increase the feeling of fraternity among local circles, give opportunities for the exchange of programs, the discussion of difficulties, and in other ways make the circles of practical benefit to each other.

Members of the C. L. S. C. or local circles wishing to join the Correspondence Department should report to the office of the C. L. S. C. at Plainfield, N. J.

The list of C. L. S. C. graduates in the Class of ’84 has been lengthened by the following names:

Daniels, Mrs. Margaret P. S.	New York.
Longee, Mrs. Mary P.	New Hampshire.
McConnell, Edward B.	Pennsylvania.
Smith, Miss Anna	Michigan.
Van Ingen, M. Gertrude	New York.

Communications intended for the “Local Circles” of The Chautauquan should be sent directly to our office. Any circle which has not reported this year we should be glad to have do so at once.

Transcriber’s Notes:

Obvious punctuation errors repaired.

Page 388, “II” changed to “IV” (Class IV.)

Page 389, “carniverous” changed to “carnivorous” (they have all the five senses, and are carnivorous)

Page 398, “Fate” changed to “Gate” (Traitor’s Gate)

Page 398, “Tewksbury” changed to “Tewkesbury” (in the field at Tewkesbury)

Page 403, “ahd” changed to “and” (and have bought bonds)

Page 405, “extirminated” changed to “exterminated” in two places (and their kind exterminated / the native fish are actually exterminated)

Page 406, “extirmination” changed to “extermination” (extermination so recklessly begun)

Page 406, “their” changed to “the” (the narrow, tortuous defiles)

Page 407, “neans” changed to “means” (by artificial means)

Page 408, “Mariner” changed to “Marner” (Silas Marner)

Page 413, “Easer” changed to “Easter” (1. Essay—Easter.)

Page 424, “make” changed to “made” (which has made Shaksperean skepticism almost respectable)

Page 429, “with” added (two atoms of nitrogen have united with two atoms of oxygen)

Page 429, “hydrogen” changed to “oxygen” (The formula $Cu(NO_3)_2$ … three atoms of oxygen)

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